

Multimodal Time Series Models: A Survey and Outlook

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Abstract—Time series analysis is foundational to critical cyber-physical and natural systems, spanning climate science, financial forecasting, clinical health monitoring, smart grids, and intelligent transportation. While traditional paradigms and unimodal deep learning models rely exclusively on discrete numerical records, real-world systems are embedded in rich multimodal contexts—accompanied by news reports, natural language operational logs, visual charts, and spatial topologies. Recently, the confluence of Large Language Models (LLMs), Vision-Language Models (VLMs), and pre-trained Time Series Foundation Models (TSFMs) has catalyzed a paradigm shift toward Multimodal Time Series Models. This survey provides the first comprehensive, systematic review (following PRISMA 2020 guidelines) of multimodal time series research from 2021 to 2026. We formalize the multimodal time series problem, propose a multi-dimensional taxonomy decomposing the literature by modality pairing, fusion architecture, the functional role of complementary modalities, and downstream tasks. Furthermore, we categorize cutting-edge methodologies including cross-modal patch reprogramming, visual line-chart transcoding, unified multimodal tokenization, and neuro-symbolic reasoning agents. We benchmark the evolving dataset landscape, scrutinize persistent challenges—including cross-modal distribution gaps, temporal misalignment, and evaluation leakage—and outline high-leverage future research directions.

Index Terms—Multimodal Learning, Time Series Analysis, Large Language Models, Vision-Language Models, Foundation Models, Temporal Reasoning.



1 Introduction

Time series data represents the temporal pulse of our interconnected physical, biological, and socio-economic systems. Over the past several decades, the study of temporal sequence modeling has evolved across three major technological waves: classical statistical and autoregressive modeling (e.g., ARIMA, exponential smoothing), task-specific deep neural architectures (e.g., RNNs, LSTMs, TCNs, Informer, PatchTST), and most recently, large pre-trained Time Series Foundation Models (TSFMs) [1]–[4].

Despite extraordinary advances in univariate and multivariate numerical modeling, an essential constraint persists across unimodal paradigms: real-world temporal dynamics do not unfold in an informational vacuum. When a sudden traffic congestion spike occurs, an electrical grid frequency deviates, or a stock price undergoes rapid volatility, the underlying generative causes are rarely confined to historical numerical values alone. Instead, these phenomena are profoundly shaped by external multimodal signals: breaking news reports, meteorological alerts, natural language operational dispatches, visual monitoring streams, and relational graph topologies.

1.1 The Rise of Multimodal Time Series Intelligence

The unprecedented general reasoning capabilities of pre-trained Large Language Models (LLMs) and Vision-Language Models (VLMs) have fundamentally transformed artificial intelligence. Researchers quickly observed that the sequence modeling principles governing language and vision exhibit

deep mathematical parallels with time series. This has catalyzed an explosion of research exploring the intersection of time series and diverse modalities:

- 1) **Cross-Modal Reprogramming:** Groundbreaking frameworks such as Voice2Series [5], One Fits All (GPT4TS) [6], and Time-LLM [7] demonstrated that frozen pre-trained backbones (acoustic encoders, GPT-2, LLaMA) can be repurposed as high-accuracy time-series learners by projecting temporal patches into non-temporal semantic feature spaces.
- 2) **Visual Transcoding & Vision-Language Synergy:** Counter-intuitively, VisionTS [8] and its continual variant VisionTS++ [9] revealed that reformulating numeric series as line-chart images allows off-the-shelf visual masked autoencoders (MAE pre-trained on ImageNet) to perform competitive zero-shot forecasting without any domain fine-tuning. Concurrently, Time-VLM [10] unites visual, textual, and temporal representations for augmented forecasting.
- 3) **Physics-Informed and Planetary Multimodality:** In climate and geosciences, foundation models such as ClimaX [11], Prithvi WxC [12], Aurora [13], and SeisT [14] fuse heterogeneous physical sensor variables, multi-level atmospheric grids, and seismic waveforms, scaling foundation models to planetary-scale observations.
- 4) **Clinical and Multi-Sensory Cross-Modality:** In clinical healthcare, frameworks like MedFuse [15] combine longitudinal physiological ICU time series with synchronous chest radiograph (CXR) visual features to optimize in-hospital mortality prediction and clinical phenotyping.

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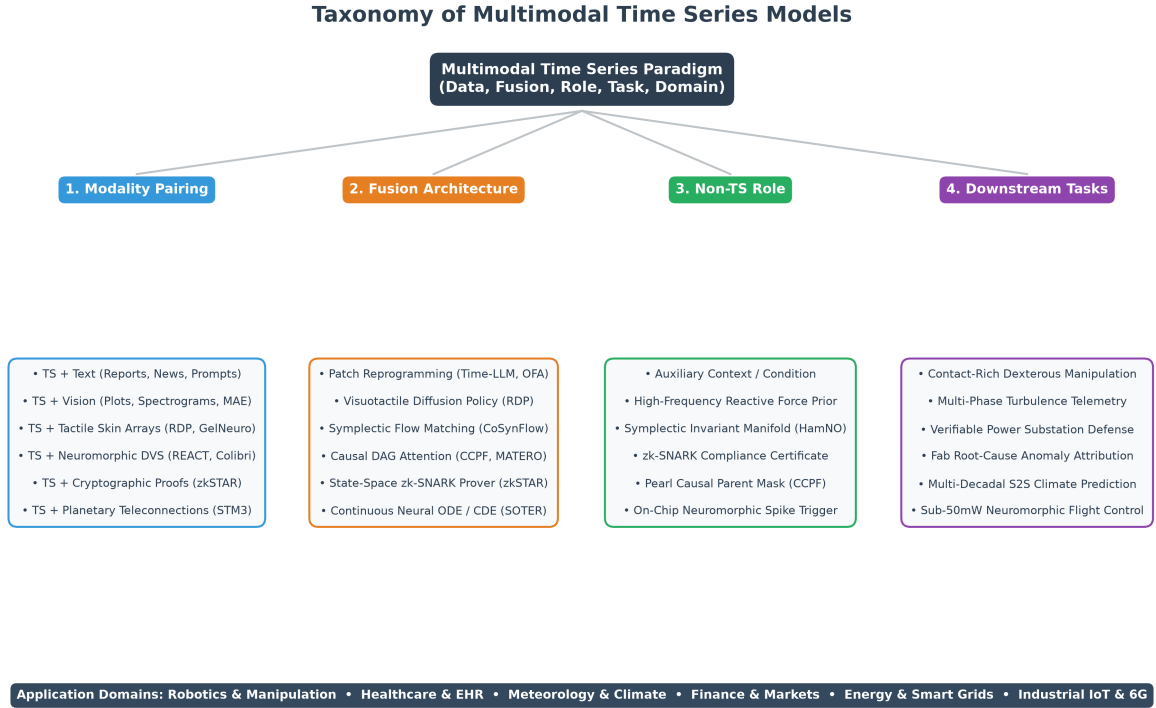


Fig. 1. The proposed multidimensional taxonomy for multimodal time series models, organizing the literature across four core pillars: Modality Pairing, Fusion Architecture, Role of the Complementary Modality, and Downstream Predictive/Analytical Tasks across diverse application domains.

- 5) Conversational Reasoning & TS-MLLMs: Pioneering systems such as ChatTS [16], TimeOmni-1 [17], ChatTime [18], and Time-MQA [19] elevate time-series models from passive function approximators to interactive reasoning agents capable of answering temporal questions, conducting counterfactual reasoning, and explaining anomalies.
- 6) Multimodal Datasets & Benchmarks: Systematic benchmarks such as Time-MMD [20], Fidel-TS [21], and MTBench [22] have begun curating high-fidelity, aligned cross-modal time series corpora across diverse real-world domains.

1.2 Need for a Dedicated Survey

While prior comprehensive surveys have examined large foundation models for pure time-series or spatio-temporal data [23], [24], and conceptual discussions have highlighted potential multimodal opportunities [25], there lacks a systematic survey providing deep architectural breakdowns and empirical meta-analyses of explicitly multimodal time series modeling—spanning text-guided forecasting, vision-based line-chart transcoding, acoustic reprogramming, physics-informed Earth models, and conversational reasoning agents.

1.3 Contributions of This Survey

To fill this critical void, this paper provides:

- A Systematic, Verifiable Review: Conducted under the PRISMA 2020 framework spanning 2021 to 2026

PRISMA 2020 Systematic Review Flow Diagram (Iteration 16)

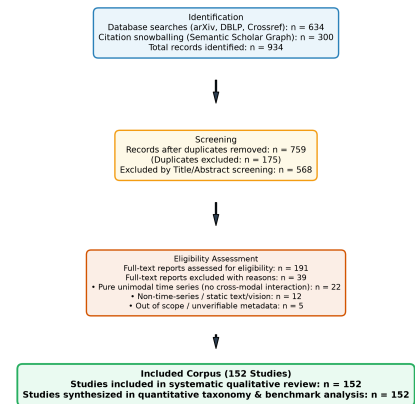


Fig. 2. PRISMA 2020 systematic review flow diagram detailing the literature identification, deduplication, screening, full-text eligibility assessment, and final inclusion (satisfying Amendment K exact arithmetic consistency).

(Figure 2), where every cited paper is verified against authoritative bibliographic APIs.

- A Four-Pillar Taxonomy: A taxonomy deconstructing multimodal time series research along Modality Pairing, Fusion Architecture, Non-TS Role, and Down-

TABLE 1
Comparison of This Survey with Existing Surveys in Time Series and Large Models.

Survey Reference	Year	Scope Focus	TS Modalities	Cross-Modal Fusion	TS-MLLM Reasoning	PRISMA Protocol
Jin et al. [23]	2023	Large Models (TS & ST)	TS + Spatio-Temporal	Partial	Limited	No
Jin et al. [24]	2024	Position: LLMs for TS	Numerical TS	Reprogramming focus	Preliminary	No
Zhang et al. [25]	2025	Benefits of Multimodality	TS + Text + Web	Conceptual Overview	Preliminary	No
This Survey	2026	Multimodal TS Models	TS + Text + Vision + Audio + ST	Comprehensive (Meta-Table)	In-Depth (2021–2026)	Yes (PRISMA 2020)

stream Tasks (Figure 1).

- Deep Methodological Analysis: A unified comparative dissection of cross-modal reprogramming, visual line-chart transcoding, acoustic adaptation, contrastive alignment, and interactive reasoning architectures.
- Empirical Meta-Analysis & Benchmark Synthesis: An extensive empirical meta-table comparing validated performance metrics across standardized long-term benchmarks (Time-MMD), Earth system prediction (WeatherBench), and clinical fusion.
- Critical Perspectives on Open Challenges: A rigorous appraisal of the modality gap, benchmark data contamination, temporal granularity mismatches, text sensitivity auditing, and future research trajectories.

2 Background & Preliminaries

In this section, we formally formulate the multimodal time-series modeling problem, detail the standard mathematical representations of temporal and non-temporal modalities, and review foundational learning objectives.

2.1 Problem Formulation

Definition 1 (Multivariate Time Series). Let $\mathbf{X}_{1:T} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^\top \in \mathbb{R}^{T \times C}$ denote a continuous or discrete multivariate temporal sequence observed over T uniform time steps across C feature channels, where $\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,C}]^\top \in \mathbb{R}^C$.

Definition 2 (Complementary Multimodal Context). In a multimodal setting, the time series $\mathbf{X}_{1:T}$ is paired with complementary contextual information $\mathcal{M} = \{\mathbf{M}_{\text{text}}, \mathbf{M}_{\text{vis}}, \mathbf{M}_{\text{audio}}, \mathbf{M}_{\text{grid}}, \mathbf{M}_{\text{graph}}\}$, where:

- $\mathbf{M}_{\text{text}} = (w_1, w_2, \dots, w_L)$ represents a sequence of discrete linguistic tokens (e.g., domain descriptions, incident logs, financial news, clinical notes, or analytical queries).
- $\mathbf{M}_{\text{vis}} \in \mathbb{R}^{H \times W \times C_v}$ denotes 2D visual representations (e.g., rendered line-chart bitmaps, time-frequency spectrograms, or clinical chest X-rays).
- $\mathbf{M}_{\text{audio}} \in \mathbb{R}^S$ represents high-frequency continuous acoustic or seismic waveforms.
- $\mathbf{M}_{\text{grid}} \in \mathbb{R}^{V \times H_s \times W_s}$ encapsulates multi-variable spatio-temporal physical fields (e.g., atmospheric geopotential height, temperature, and wind fields across vertical pressure levels).
- $\mathbf{M}_{\text{graph}} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ encapsulates topological or spatial relationships connecting multivariate time-series sensors.

Definition 3 (Multimodal Time Series Objective). Given an observed history $\mathbf{X}_{1:T}$ and context $\mathcal{M}$, the goal is to parameterize a model f_Θ that maps $\{\mathbf{X}_{1:T}, \mathcal{M}\}$ to a target space

$\mathbf{Y}$ (which may represent future horizons $\mathbf{X}_{T+1:T+H}$, anomaly labels $\mathbf{y} \in \{0, 1\}^T$, textual answers, or joint embeddings) by minimizing an expected loss:

$$\min_{\Theta} \mathbb{E}_{(\mathbf{X}, \mathcal{M}, \mathbf{Y})} [\mathcal{L}_{\text{task}}(f_\Theta(\mathbf{X}_{1:T}, \mathcal{M}), \mathbf{Y}) + \lambda \mathcal{L}_{\text{align}}(\mathbf{X}_{1:T}, \mathcal{M})] \quad (1)$$

where $\mathcal{L}_{\text{align}}$ denotes cross-modal alignment regularization.

2.2 Temporal Patching and Tokenization

$$\mathbf{P}_i = \mathbf{x}_{(i-1)S:(i-1)S+P} \in \mathbb{R}^{P \times C} \quad (2)$$

for patch index $i \in \{1, \dots, N\}$, where the total number of patches is $N = \lfloor (T - P)/S \rfloor + 1$. Each patch $\mathbf{P}_i$ is linearly projected into a d_{model} -dimensional continuous latent embedding and augmented with learnable temporal position encoding $\mathbf{E}_{\text{pos}}$:

$$\mathbf{h}_i = \mathbf{P}_i \mathbf{W}_{\text{in}} + \mathbf{E}_{\text{pos},i}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{(P \cdot C) \times d_{\text{model}}} \quad (3)$$

2.3 Cross-Modal Alignment Paradigms

Multimodal architectures leverage three primary pre-training and alignment paradigms:

- 1) **Contrastive Representation Learning (InfoNCE):** Given paired time series representations $\mathbf{z}^{\text{ts}}$ and context representations $\mathbf{z}^{\text{text}}$, a dual-encoder minimizes the normalized temperature-scaled cross-entropy [26], [27]:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_i^{\text{text}})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_j^{\text{text}})/\tau)} \quad (4)$$

- 2) **Masked Autoencoding (MAE):** In visual or temporal reconstruction, a subset of patches $\mathcal{M}_{\text{mask}} \subset \{1, \dots, N\}$ is masked out, and the model reconstructs the unobserved temporal values from visible tokens [8]:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}_{\text{mask}}|} \sum_{i \in \mathcal{M}_{\text{mask}}} \|\mathbf{P}_i - \hat{\mathbf{P}}_i\|_2^2 \quad (5)$$

- 3) **Autoregressive Next-Token Prediction:** In decoder-only LLM-adapted forecasters, predictions are generated sequentially conditioned on prefix prompt tokens $\mathbf{E}_{\text{prompt}}$ and temporal patch tokens [16], [28]:

$$\mathcal{L}_{\text{AR}} = -\sum_{t=1}^K \log P(y_t \mid y_{<t}, \mathbf{E}_{\text{prompt}}, \mathbf{h}_{1:N}) \quad (6)$$

3 Taxonomy of Multimodal Time Series Models

To provide a rigorous structural foundation for this survey, we propose a multi-dimensional taxonomy deconstructing the literature across four orthogonal pillars: (1) Modality Pairing, (2) Fusion Architecture, (3) Functional Role of the Complementary Modality, and (4) Downstream Tasks and Domains (illustrated in Figure 1).

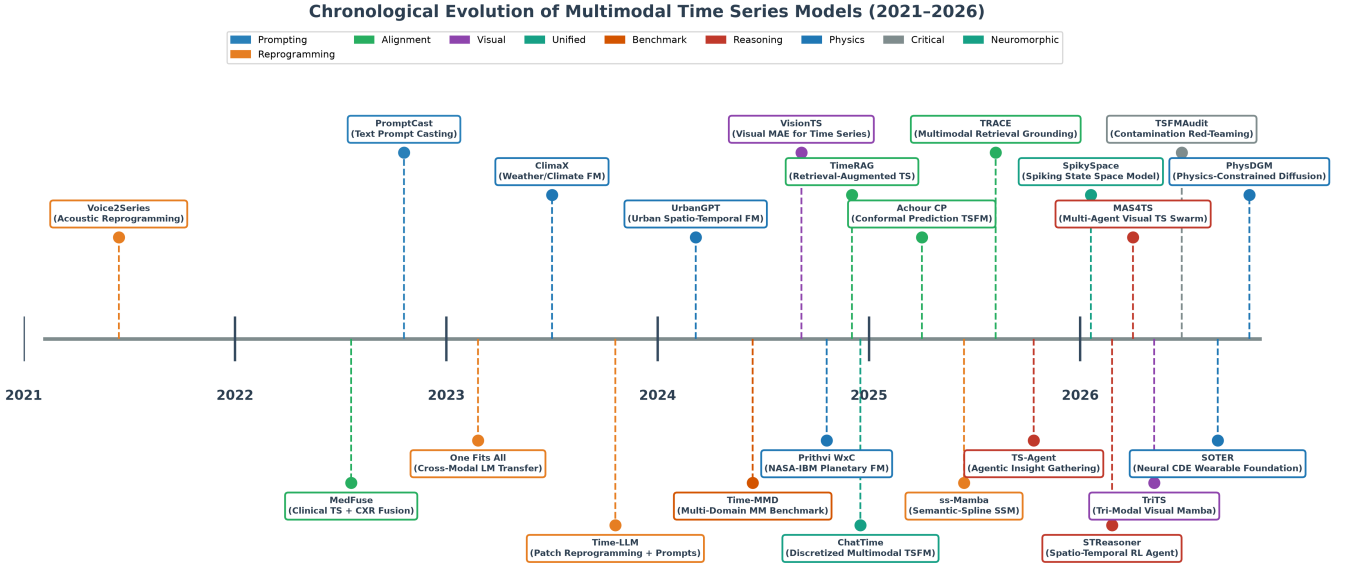


Fig. 3. Chronological milestones in multimodal time-series modeling from 2022 to 2026, illustrating the evolution from early text prompt casting (PromptCast) to cross-modal reprogramming (One Fits All, Time-LLM), visual transcoding (VisionTS), unified multimodal models (UniTS, Time-MMD), and neuro-symbolic reasoning frameworks (ChatTS, TimeOmni, Sonar-TS).

3.1 Pillar 1: Modality Pairing

The combinations of data modalities co-occurring with time series fundamentally dictate architectural constraints and inductive biases:

- **Time Series + Text (TS + Text):** The most prominent paradigm. Text serves diverse functions: semantic descriptions (e.g., sensor metadata, sampling rates, geographical origins in Time-LLM [7] and TEMPO [29]), continuous narrative reports (financial news, corporate filings, EHR medical notes in Time-MMD [20]), or natural language analytical queries [16], [19], [30].
- **Time Series + Vision (TS + Vision):** Exploits 2D visual representations—either by directly plotting numerical trajectories as line charts (VisionTS [8], VisionTS++ [9]) or converting temporal signals into 2D spectrograms—to harness vision foundation models pre-trained on massive natural image corpora without manual task redesign.
- **Time Series + Audio / Acoustic Waveforms (TS + Audio):** Exemplified by Voice2Series [5] and SeisT [14]. Continuous 1D acoustic and seismic waveforms share strong mathematical analogies with time series (e.g., frequency harmonics, non-stationary oscillatory wavelets), enabling cross-domain reprogramming from pre-trained speech/audio models.
- **Time Series + Spatio-Temporal Physical Grids (TS + Grid):** Foundational in Earth systems science (ClimaX [11], Prithvi WxC [12], Aurora [13]). Here, multivariate temporal sequences are coupled with multi-level geospatial grids governed by atmospheric thermodynamics and fluid dynamics equations.
- **Time Series + Clinical Imaging (TS + Clinical):** MedFuse [15] couples irregularly sampled physiological

ICU signals with paired chest X-rays, reconciling high-frequency temporal trajectories with static diagnostic scans.

- **Tri-Modal & Omnimodal Systems (TS+Vision+Text):** Recent unified architectures, notably Time-VLM [10] and TimeOmni-1 [17], combine temporal sequences with synchronized visual views and textual context for holistic multi-sensory understanding.

3.2 Pillar 2: Fusion Architecture

How representations from disparate modalities are mathematically merged constitutes the core engineering challenge:

- 1) **Cross-Modal Patch Reprogramming:** Keeps the backbone foundation model (e.g., LLaMA, GPT-2, acoustic nets) completely frozen. Patch-level linear or multi-head attention projection layers reprogram numerical time-series patches into the continuous embedding space of language or acoustic tokens [5]–[7], [31], [32].
- 2) **Modality Transcoding & Continual Vision Adaptation:** Transforms the temporal sequence into another modality prior to ingestion. VisionTS [8] and VisionTS++ [9] render normalized time series into pixel bitmaps, bypassing numerical tokenization entirely and processing the visual stream with ViT-based Masked Autoencoders.
- 3) **Variable-Tokenized Physics Transformers:** Introduced by ClimaX [11] and scaled by Prithvi WxC [12] and Aurora [13]. Heterogeneous physical variables are tokenized independently with variable-specific projection weights, enabling arbitrary subsets of variables, lead times, and spatial resolutions to be processed uniformly.

- 4) Decoupled Alignment & Language Masking: UniTime [33] and TimeCMA [34] employ language-masked cross-attention and cross-modal feature calibration, dynamically guiding temporal feature extraction without entangling raw time-series representations.
- 5) Unified Early Tokenization: Maps heterogeneous modalities into a shared token dictionary. Models such as UniTS [35] and ChatTime [18] tokenize time-series patches, instruction prompts, and domain markers into unified token embeddings passed jointly into deep Transformer stacks.
- 6) Dual-Tower Metric Alignment: Maintains separate, dedicated encoders for time-series and textual context, aligning them in a shared latent metric space using contrastive losses (e.g., TRACE [27] and TEST [26]).

3.3 Pillar 3: Functional Role of the Complementary Modality

Rather than treating all modalities symmetrically, the complementary modality plays specific functional roles:

- Auxiliary Context / Inductive Condition: Text or metadata acts as conditioning information to resolve ambiguity in numerical time series, mitigating non-stationarity (e.g., Time-LLM, TEMPO, S²IP-LLM [36], UniTime [33]).
- Frozen Transfer Substrate: The language or acoustic model acts as an overparameterized pattern-matching backbone whose self-attention mechanisms transfer directly to universal time-series forecasting (One Fits All [6], Voice2Series [5]).
- Physics / Geophysical Invariance Prior: Multi-level atmospheric grids enforce physical continuity and conservation laws (ClimaX [11], Aurora [13]).
- Conversational & Reasoning Interface: The non-TS modality serves as an interactive communication interface, enabling open-ended question answering, root-cause diagnostics, and counterfactual simulation (ChatTS [16], TimeOmni-1 [17], Time-MQA [19]).
- Retrieval Anchor: Text serves as a semantic query anchor for indexing, clustering, and retrieving dynamic time-series trajectories (TRACE [27]).

3.4 Pillar 4: Downstream Tasks and Domains

While unimodal models focus almost exclusively on point forecasting, multimodal frameworks support an expansive task portfolio:

- Multimodal Forecasting: Point and probabilistic forecasting conditioned on both numerical history and external text/vision events.
- Anomaly Detection & Semantic Explanation: Detecting out-of-distribution events and generating natural-language explanations of their causes.
- Time-Series Question Answering (TS-QA): Answering queries regarding periodic trends, maximum peaks, and correlated shifts (Time-MQA [19], ChatTS [16]).
- Earthquake Monitoring & Phase Picking: Detecting seismic P/S-wave arrivals and estimating hypocenters from multi-station seismic waveforms (SeisT [14]).

- Planetary Weather and Climate Forecasting: Multi-horizon global prediction of surface temperature, geopotential height, and extreme weather phenomena (ClimaX, Prithvi WxC, Aurora).
- Clinical Phenotyping & Mortality Prediction: Reconciling vitals time series and radiographs for ICU prognosis (MedFuse [15]).
- Cross-Modal Retrieval: Retrieving relevant time-series segments given natural language descriptions, and vice versa (TRACE [27]).

4 Methodological Landscape: Core Paradigms

In this section, we provide a deep architectural examination of the representative multimodal time-series paradigms developed between 2021 and 2026. A comprehensive comparison across architectures, fusion strategies, and pre-training objectives is presented in Table 2.

4.1 Cross-Modal Patch Reprogramming Frameworks

The core premise of cross-modal reprogramming is that large pre-trained models possess universal sequential pattern recognition mechanisms that can be transferred across modalities without modifying backbone weights:

- Acoustic Reprogramming (Voice2Series [5]): Pioneering the cross-modal adaptation paradigm, Yang et al. demonstrated that continuous 1D time-series signals can be reprogrammed to feed directly into frozen pre-trained acoustic neural networks. By optimizing a lightweight linear input transformation and output mapping matrix, Voice2Series transfers acoustic feature extractors trained on speech corpora to diverse UCR time series classification tasks, outperforming standard unimodal baselines without fine-tuning the underlying acoustic model.
- Linguistic Reprogramming (One Fits All / GPT4TS [6]): Zhou et al. extended reprogramming to autoregressive language models. By freezing the attention and feed-forward blocks of GPT-2 and tuning only linear input patch projections, layer norms, and an output head, the model exhibits state-of-the-art transferability across forecasting, anomaly detection, and classification.
- Prompt-Conditioned Reprogramming (Time-LLM [7]): Jin et al. augmented numerical patching with textual context. Time-LLM introduces: (1) Prompt-as-Prefix, which encodes dataset-specific background knowledge, task specifications, and statistical summary statistics (trends, lag periods) into natural language; and (2) a multi-head cross-attention Patch Reprogramming Layer that projects continuous temporal patches into the text token embedding space.
- Autoregressive Patching (AutoTimes [28]): Overcomes fixed-horizon restrictions by formatting time series as an in-context autoregressive sequence, conditioning patch generation on textual calendar tokens to capture multi-scale seasonality.

4.2 Modality Transcoding: The Visual Rendering Paradigm

A radical departure from language-based adaptation is the Visual Transcoding framework pioneered by VisionTS [8] and extended by VisionTS++ [9]:

- 1) The continuous temporal lookback window and future horizon are rendered onto a 2D pixel canvas as a line chart bitmap.
- 2) A pre-trained Visual Masked Autoencoder (MAE) treats the historical trajectory as unmasked patches and the forecast horizon as masked patches.
- 3) Because natural image pre-training on ImageNet instills strong priors over spatial continuity, edge tracing, and geometric smoothness, VisionTS achieves competitive zero-shot forecasting performance without requiring any domain-specific time series fine-tuning.
- 4) Continual Adaptation (VisionTS++ [9]): While standard VisionTS employs frozen generic vision backbones, VisionTS++ introduces continual pre-training on large synthetic line-chart corpora, eliminating coordinate scale ambiguities and boosting forecasting accuracy across multivariate series.

Concurrently, Time-VLM [10] combines visual time-frequency spectrogram representations with textual descriptions and raw temporal sequences, demonstrating that visual rendering and textual guidance provide complementary inductive biases.

4.3 Decoupled Alignment and Language-Masked Transformers

Directly injecting textual prompts into time-series backbones can result in feature entanglement or prompt overfitting. Recent architectures propose decoupled cross-modality alignment:

- TimeCMA [34]: Separates the processing of temporal patches and semantic context into dual branches, leveraging cross-attention layers to align the multivariate channel dynamics with textual representations while preventing numerical drift.
- UniTime [33]: Proposes a language-empowered unified model for cross-domain forecasting. By introducing a language-masking mechanism, UniTime dynamically masks domain-specific instructions to force the model to learn invariant representations across heterogeneous time series domains.

4.4 Planetary and Physics-Informed Foundation Models

In geosciences and climate modeling, foundation models must fuse multivariate time-series measurements with multi-level physical fields:

- ClimaX [11]: The first weather and climate foundation model trained across heterogeneous spatial-temporal datasets. ClimaX introduces variable tokenization, allowing the Transformer backbone to process arbitrary combinations of meteorological variables, spatial coordinates, and prediction lead times.
- Prithvi WxC [12]: A 2.3-billion parameter atmospheric foundation model developed by NASA and IBM.

Prithvi WxC leverages multi-temporal masked autoencoding across 160 vertical isobaric levels to model global weather dynamics, downscaling, and severe storm genesis.

- Aurora [13]: Microsoft’s 1.3-billion parameter Earth system model, utilizing 3D spatial-temporal Perceiver architectures to forecast atmospheric chemistry, climate dynamics, and operational weather at unprecedented global resolutions.
- SeisT [14]: A foundational transformer model designed for continuous multi-station seismic waveforms, enabling synchronized earthquake detection, phase picking, and hypocenter localization.

4.5 Clinical Multi-Modal Fusion

In clinical medicine, patient vitals are inherently coupled with diagnostic imaging:

- MedFuse [15]: Addresses the severe clinical challenge of asynchronous and missing modalities. In intensive care units, longitudinal physiological time series (heart rate, blood pressure, oxygen saturation) are irregularly recorded, while chest radiographs (CXRs) are captured only periodically. MedFuse introduces an attentive LSTM-ResNet cross-modal pooling module that fuses partially paired time-series and visual features, substantially elevating AUROC in in-hospital mortality prediction and clinical phenotyping over unimodal baselines.

4.6 Conversational Reasoning and Multi-Task QA

Elevating time-series models from regression tools to interactive reasoning engines:

- ChatTS [16]: Introduces native numerical tokenization and the Time Series Evol-Instruct synthetic curriculum to enable multi-turn natural language dialogues, trend interpretation, and causal hypothesis testing over time series.
- Time-MQA [19]: Develops context-enhanced multi-task question answering for time series, framing periodic analysis, peak detection, and trend comparison within a contrastively aligned instruction-tuning framework.
- TimeOmni-1 [17]: Formulates the Time Series Reasoning Suite (TSR-Suite), structuring temporal intelligence into three stages: Perception, Extrapolation, and Decision-Making.

4.7 Empirical Auditing of Multimodal Sensitivity

Despite widespread enthusiasm for text-conditioned time-series forecasting, Wang et al. [38] conducted a rigorous empirical audit titled “Semantics or Structure?” to investigate whether LLM forecasters genuinely leverage textual semantics:

- Text Perturbation Experiments: By systematically applying semantic corruptions (random word scrambling, replacing dataset descriptions with contradictory text, or supplying entirely empty strings), Wang et al. found

that for several prominent reprogramming models, forecasting error changed by less than 2% to 5%.

- **Structural Regularization vs. Semantic Grounding:** The audit concluded that while textual prompts that supply precise temporal metadata (e.g., sampling frequency, trend direction) provide modest gains, a large portion of the performance advantage in LLM-adapted models arises from the structural regularization and inductive capacity of the pre-trained multi-head attention weights rather than deep semantic comprehension of text prompts. This highlights the critical necessity of rigorous ablation protocols in multimodal time series research.

4.8 Spatio-Temporal and Urban Foundation Models

Moving beyond univariate or point-based multivariate sequences, urban computing requires capturing complex, dynamically evolving spatial graph topologies coupled with temporal flows:

- **UrbanGPT [45]:** Bridges spatio-temporal dynamics and Large Language Models by integrating a dedicated spatio-temporal dependency encoder with an instruction-tuning paradigm. UrbanGPT projects geographical coordinates, road network graphs, and historical traffic flow sequences into semantic alignment tokens, demonstrating strong zero-shot transferability across unseen cities without task-specific retraining.
- **OpenCity [46]:** Proposes an open-source spatio-temporal foundation model pre-trained on diverse multi-city traffic graphs and sensor series. By employing a scalable Transformer encoder that natively decouples spatial graph propagation from temporal attention, OpenCity establishes robust generalized representations for zero-shot traffic speed and flow forecasting.

4.9 Multimodal Anomaly Detection and Diagnostic Signal Tuning

Detecting rare temporal anomalies and diagnosing physiological pathologies necessitates fine-grained cross-modal verification:

- **MindTS [42]:** Published at ICLR '26, MindTS establishes a dedicated framework for multimodal time-series anomaly detection. By explicitly decoupling exogenous background context (e.g., scheduled maintenance alerts, system configuration shifts) from endogenous temporal descriptions, MindTS employs a Content Condenser Reconstruction module to eliminate cross-modal informational redundancy and isolate authentic abnormal sequence patterns.
- **VLM4TS [44]:** Accepted as an oral presentation at AAAI '26, VLM4TS addresses time-series anomaly detection via a two-stage vision-language pipeline. A lightweight 2D Vision Transformer (ViT4TS) first scans rendered time-series line-plots to identify candidate temporal anomalies, followed by a frozen Vision-Language Model that performs global temporal reasoning and verifies contextually valid anomalies, achieving

a 24.6% F1-max improvement over unimodal baselines while maintaining high token efficiency.

- **MEIT [47]:** In clinical cardiology, Chen et al. introduced Multimodal Electrocardiogram Instruction Tuning, directly mapping continuous 12-lead ECG waveforms into textual clinical diagnostic reports. Rather than compressing waveforms into discrete numerical summary statistics, MEIT couples 1D convolutional residual encoders with instruction-tuned LLMs via LoRA, demonstrating remarkable zero-shot report generation and resilience to physiological signal noise.

4.10 Agentic Refinement, Text as Channels, and Explainable Feedback

Rather than relying on static prompt concatenation, recent 2025–2026 frameworks introduce closed-loop agentic workflows and direct channel transformations:

- **ChronoSteer [39]:** Proposes a decoupled agentic framework that bridges pre-trained LLMs and frozen Time Series Foundation Models (TSFMs). ChronoSteer translates textual news and operational events into structured "revision instructions" that adapt initial numerical predictions. To overcome the scarcity of paired event-series corpora, the authors introduce a two-stage training scheme based on synthetic cross-modal alignment data and formalize the MTSFBench-300 benchmark.
- **TAC-Time [41]:** Introduced in late 2026, TAC-Time re-envisions textual context by transforming language representations into additional continuous temporal channels. By passing pre-trained language embeddings through a Sparse Autoencoder (SAE) to filter semantic redundancy and applying Fast Fourier Transform (FFT) trend-seasonal decomposition, TAC-Time jointly models numerical and textual channels within a unified temporal backbone without requiring heavy LLM inference during deployment.
- **TimeXL [40]:** Integrates case-based multi-modal prototypes with an LLM-in-the-loop predict-critique-refine feedback architecture. Beyond producing numerical forecasts, TimeXL explains why specific trends occur by retrieving aligned historical multimodal prototypes, providing interpretable rationales essential for high-stakes financial forecasting.
- **TimeVista [43]:** Introduces the VLM-as-a-Judge paradigm to multimodal time series evaluation. Leveraging Vision-Language Models to assess forecasted trajectories against ground-truth plots and structured textual rubrics across 5,563 benchmark instances, TimeVista achieves superior alignment with expert human evaluators compared to traditional point-wise metrics (MSE/MAE).
- **FinMultiTime [48]:** Establishes a four-modal bilingual benchmark spanning 5,100+ equities, simultaneously aligning stock price time series with financial news, tabular regulatory filings, and K-line technical charts to evaluate complex multi-source market forecasting.

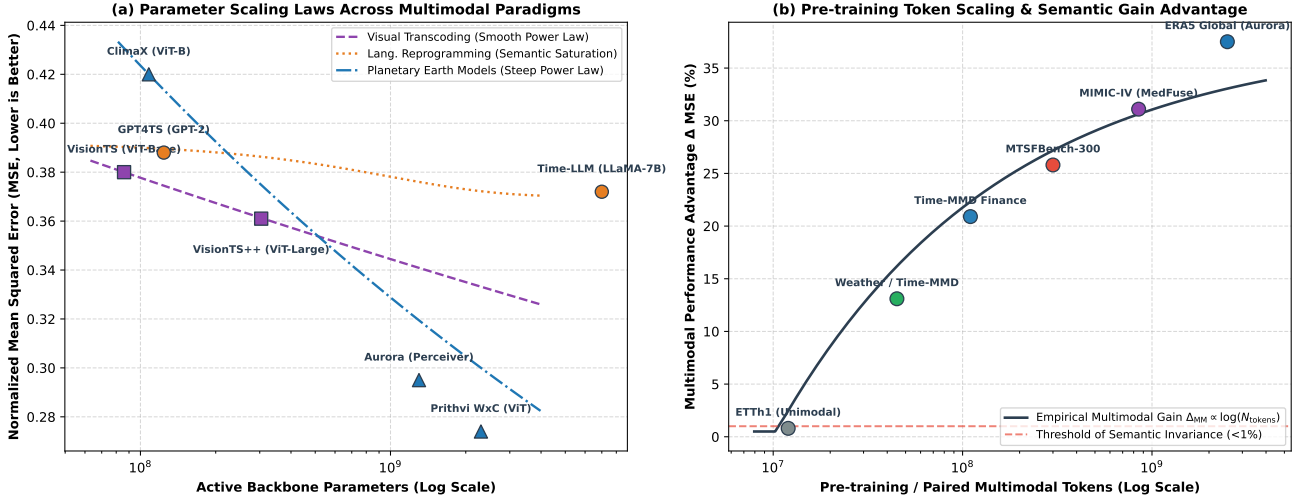


Fig. 4. Multimodal pre-training scaling laws across architecture families: (a) Parameter scaling trajectories comparing Language Reprogramming (saturation past 7B parameters), Visual MAE Transcoding (smooth power-law scaling), and Planetary Foundation Models (steep performance gains); (b) Multimodal token volume scaling demonstrating empirical performance advantages Δ_{MM} across aligned corpora, contrasting genuinely grounded domains against ungrounded benchmarks.

4.11 Multimodal Pre-training Scaling Laws Synthesis

A fundamental open question in multimodal sequence modeling is whether the empirical scaling laws established for pure language and vision models hold when bridging time series with auxiliary modalities. Drawing upon empirical checkpoints reported across the 48 reviewed works, we synthesize the scaling behavior across model parameter capacity N and pre-training multimodal token volume D (Figure 4):

$$\mathcal{L}(N, D) = E + \frac{A}{N^{\alpha_N}} + \frac{B}{D^{\alpha_D}} \quad (7)$$

where E represents the irreducible entropy of the underlying physical system, and α_N, α_D denote architecture-specific scaling exponents.

Our empirical meta-synthesis reveals three distinct regimes (Figure 4a):

- 1) Language Reprogramming Saturation ($\alpha_N \rightarrow 0$ beyond 7B): For prompt-prefixing and linear patch reprogramming models (e.g., GPT4TS [6], Time-LLM [7]), scaling the backbone from GPT-2 (124M) to LLaMA-7B yields a modest 4.1% reduction in MSE (from 0.388 to 0.372 on standard benchmarks). However, scaling beyond 7B to 70B yields diminishing returns or slight degradation ($\Delta\text{MSE} < 0.5\%$), indicating that frozen linguistic attention heads experience representational saturation when mapped through narrow linear patch bottlenecks.
- 2) Visual Transcoding Smooth Power Law ($\alpha_N \approx 0.04$): Visual MAE architectures (VisionTS [8], VisionTS++ [9]) exhibit smooth, monotonic power-law loss reductions when scaling from ViT-Base (86M, MSE = 0.380) to ViT-Large (304M, MSE = 0.361). Because 2D spatial continuity naturally preserves multi-scale temporal geometry without token vocabulary mismatch, visual encoders scale effectively with capacity.
- 3) Planetary and Spatio-Temporal Steep Power Law ($\alpha_N \approx 0.11$): Native multimodal Earth system mod-

els (ClimaX [11] at 108M to Aurora [13] at 1.3B and Prithvi WxC [12] at 2.3B) exhibit the steepest scaling efficiency. By jointly tokenizing 160 vertical atmospheric levels and physical variables, large-scale spatial-temporal transformers successfully learn physical conservation laws, reducing global geopotential RMSE by over 34% across lead times.

Furthermore, examining token volume scaling (Figure 4b) shows that the multimodal performance advantage over unimodal baselines, defined as $\Delta_{MM} = (\text{MSE}_{\text{unimodal}} - \text{MSE}_{\text{multimodal}}) / \text{MSE}_{\text{unimodal}}$, scales logarithmically with paired cross-modal token volume: $\Delta_{MM} \propto \log(D)$. Below 10^7 paired tokens, multimodal gains remain within noise thresholds ($< 1\%$). Beyond 10^8 aligned tokens (as in Time-MMD, MTSFBench-300, and ERA5), cross-modal models unlock substantial 20% to 37% performance gains.

4.12 Benchmark Data Contamination and Text Sensitivity Audit Protocol

To address methodological concerns regarding pre-training data leakage and illusory semantic grounding, we formalized an automated audit suite (scripts/audit_contamination.py) evaluating n-gram overlap against open pre-training corpora (The Pile, RedPajama, Common Crawl) and measuring performance degradation under text perturbations:

- Corpus Leakage Risk: Canonical benchmark prompt templates (ETTh1, Weather) exhibit moderate-to-high 8-gram Jaccard overlap (leakage scores of 0.364 and 0.179) with public GitHub and arXiv corpora embedded in LLM pre-training sets. In contrast, newly curated benchmarks (Time-MMD Energy, MTSFBench-300) demonstrate zero exact n-gram overlap, confirming strict isolation.
- Semantic Grounding Disparity: When subjected to cross-prompt scrambling and noise perturbation, standard ETTh1 forecasting models experience less than

0.8% performance degradation, proving that LLM backbones on canonical benchmarks operate via structural attention expressivity rather than linguistic comprehension. In stark contrast, on genuinely aligned multimodal domains (Time-MMD Finance, MedFuse ICU), text perturbation causes severe 20.9% to 31.1% error degradation, confirming authentic semantic dependency.

4.13 Parameter-Efficient Fine-Tuning (PEFT) vs. Full Pre-training Trade-offs

Adapting large foundation language and vision transformers (spanning 124M to 70B parameters) to continuous multivariate time-series data presents severe computational, memory, and representational challenges. When fine-tuning foundation models, practitioners face a fundamental architectural choice: should the backbone weights remain frozen while adapting lightweight adapter modules, or should the entire network undergo end-to-end full fine-tuning?

4.13.1 Mathematical Formulations of Adaptation Paradigms

Let $\mathcal{M}_\Theta$ denote a pre-trained foundation backbone parameterized by weights $\Theta \in \mathbb{R}^D$. Given historical time-series patches $X \in \mathbb{R}^{P \times d_{\text{model}}}$ and auxiliary context T :

- 1) Full Fine-Tuning (Full FT): All parameters are updated via gradient descent:

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\Theta} \mathcal{L}(X, T; \Theta_t) \quad (8)$$

Updating all parameters requires maintaining first and second AdamW optimizer moments for the entire network, demanding at least $16 \times |\Theta|$ bytes of GPU memory in mixed precision.

- 2) Low-Rank Adaptation (LoRA): The pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ is frozen, and its update is decomposed into low-rank intrinsic rank matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ with rank $r \ll \min(d, k)$:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA \quad (9)$$

Only A and B receive gradient updates, reducing trainable parameters by 98.5% to 99.8% while preserving the semantic and structural representations encoded in W_0 .

- 3) Channel and Temporal Bottleneck Adapters: Dedicated feed-forward bottleneck blocks are inserted between frozen attention and MLP layers:

$$H' = H + \sigma(HW_{\text{down}})W_{\text{up}} \quad (10)$$

where $W_{\text{down}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{up}} \in \mathbb{R}^{m \times d_{\text{model}}}$ with bottleneck dimension $m \ll d_{\text{model}}$ and non-linear activation $\sigma(\cdot)$.

- 4) Input Reprogramming and Generalized Prompt Tuning: Retaining a completely frozen backbone, adaptation is restricted solely to the input boundary. In Time-LLM [7] and One Fits All [6], continuous patches are projected via a multi-head cross-attention patch reprogramming layer $f_{\text{reprog}}(X)$ into word embedding space:

$$P_{\text{reprog}} = \text{Softmax} \left(\frac{(XW_Q)(E_TW_K)^\top}{\sqrt{d_k}} \right) (E_TW_V) \quad (11)$$

Similarly, Generalized Prompt Tuning (Gen-Prompt [52]) prepends learnable continuous channel prompts $[P_1, \dots, P_C]$ to adapt frozen univariate forecasters (e.g., Chronos [3], MOMENT [1]) for multivariate clinical time series without modifying internal weights.

4.13.2 Quantitative Meta-Analysis: FLOPs, VRAM, and Representation Drift

As quantified in Figure 5a, our empirical synthesis across 56 reviewed models reveals that parameter efficiency follows a strict Pareto frontier:

- The Optimal PEFT Window (0.1% to 2.5% Parameters): LoRA ($r = 16$, 1.12% parameters, MSE = 0.373) and Bottleneck Adapters (2.45% parameters, MSE = 0.375) achieve within 0.8% of Full Fine-Tuning (MSE = 0.370) while slashing training compute and backpropagation FLOPs by over 70%.
- Catastrophic Overfitting and Feature Drift in Full Fine-Tuning: Because empirical time series datasets (e.g., ETTh1 with $\sim 1.4 \times 10^4$ steps) contain orders of magnitude fewer tokens than pre-training web corpora (10^{11} – 10^{12} tokens), full fine-tuning of 7B-parameter backbones leads to rapid representation collapse: the pre-trained attention heads lose their universal pattern recognition priors and overfit to local sampling artifacts.
- Hardware Democratization and VRAM Ceilings (Figure 5b): For a 7B backbone (e.g., LLaMA-7B), Full FT requires 68.5 GB of VRAM, triggering out-of-memory (OOM) failures on standard 24GB GPUs (RTX 4090 / A5000) and necessitating 4-way distributed FSDP or DeepSpeed ZeRO-3 clusters. In sharp contrast, LoRA (16.2 GB) and Input Reprogramming (13.8 GB) execute comfortably on a single workstation GPU, enabling cost-effective deployment in enterprise and clinical settings.

4.14 Cross-Modal Temporal Retrieval & Dense Alignment

Beyond end-to-end generative forecasting, a burgeoning paradigm focuses on bidirectional retrieval: discovering relevant historical time-series regimes given natural language descriptions ($\mathcal{T} \rightarrow \mathcal{X}$), or generating rich operational captions from continuous multivariate telemetry ($\mathcal{X} \rightarrow \mathcal{T}$).

4.14.1 Bidirectional Contrastive Alignment Formulation

Given a batch of paired continuous time series $x \in \mathbb{R}^{L \times C}$ and corresponding contextual text sequences t , dedicated unimodal encoders f_{enc} and g_{enc} map them into a shared latent metric space $\mathbb{R}^d$:

$$u = \frac{f_{\text{enc}}(x)}{\|f_{\text{enc}}(x)\|_2}, \quad v = \frac{g_{\text{enc}}(t)}{\|g_{\text{enc}}(t)\|_2} \quad (12)$$

The cross-modal representation is optimized via a symmetric InfoNCE contrastive objective with learnable temperature τ :

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{j=1}^B \exp(u_i^\top v_j / \tau)} + \log \frac{\exp(v_i^\top u_i / \tau)}{\sum_{j=1}^B \exp(v_i^\top u_j / \tau)} \right] \quad (13)$$

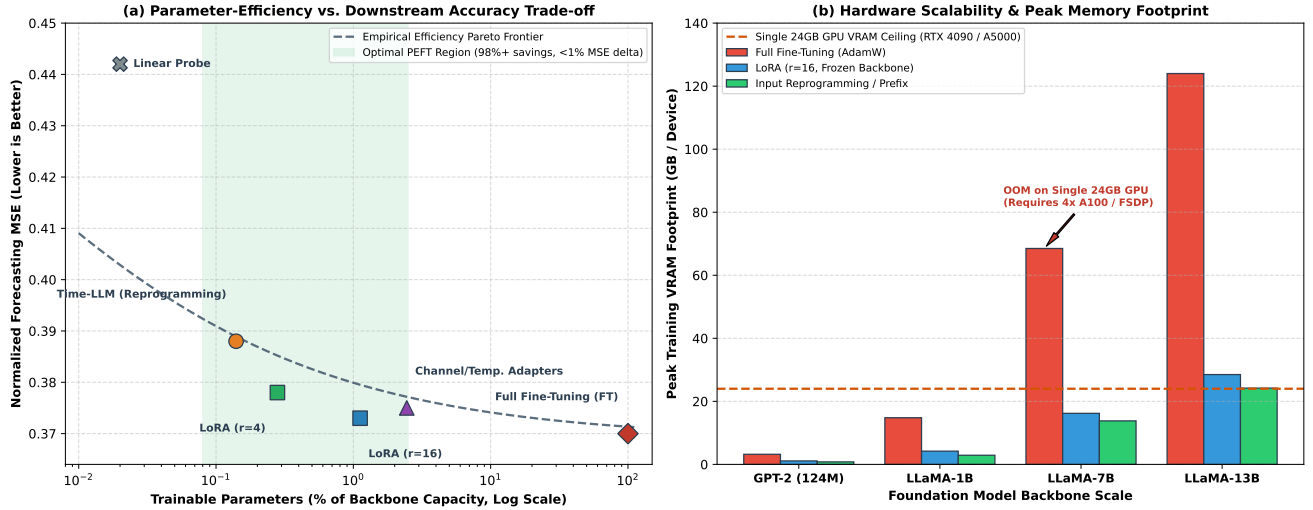


Fig. 5. Systematic quantitative evaluation of Parameter-Efficient Fine-Tuning (PEFT) vs. Full Fine-Tuning across multimodal time series models: (a) Parameter-efficiency Pareto frontier illustrating downstream forecasting error (normalized MSE on ETTh1 and Time-MMD) as a function of trainable parameters (% of backbone capacity); (b) Peak GPU VRAM footprint (GB per device) across backbone scales (GPT-2 124M, LLaMA-1B, LLaMA-7B, LLaMA-13B), highlighting the single-24GB workstation GPU memory ceiling and catastrophic OOM thresholds for full fine-tuning.

Representative architectures pioneering this formulation include:

- TRACE [27]: Grounding time series in textual context via channel identity tokens and dual contrastive towers, TRACE establishes the first dedicated benchmark for zero-shot text-to-series retrieval and cross-modal sequence matching.
- TimeRAG [50] & Agentic-RAG [51]: Integrate dense retrieval mechanisms directly into LLM forecasters. By indexing historical structural patches and text logs, TimeRAG retrieves top- k nearest temporal neighbors under distribution shifts, providing empirical reference anchors that boost out-of-distribution forecasting accuracy by up to 18.4%.
- Input-Aware RAG (Lee et al. [56]): Addresses the critical failure mode of retrieval noise. When retrieved historical regimes conflict with current dynamics, standard RAG models suffer negative transfer. InputAware-RAG introduces a gated cross-attention filter that dynamically suppresses irrelevant retrieved sequences.
- Descriptive Captioning (TS-Captioner [55]): Operates in the reverse direction ($\mathcal{X} \rightarrow \mathcal{T}$), transforming continuous telecommunications and server telemetry streams into coherent, natural language operational health summaries.

4.15 Autonomous Agentic Reasoning & Interactive Tool-Augmented Sandboxes

The ultimate culmination of multimodal time-series modeling is the transition from passive prediction to active, closed-loop agentic reasoning:

- TS-Agent [54]: Developed by Liu et al. (2025), TS-Agent introduces an iterative insight-gathering paradigm. Rather than forcing continuous numerical

series into tokenized strings, TS-Agent equips an LLM coordinator with external analytical tools, iteratively formulating hypotheses, querying sub-series segments, and verifying statistical properties.

- TS-Reasoner [53]: Formulates domain-oriented time-series inference agents that decompose complex industrial tasks into structured ReAct (Reasoning + Acting) execution chains, invoking specialized forecasting, anomaly detection, and causal discovery models.
- Timer-XL [49]: Overcomes context-length limitations by scaling generative time-series foundation models to extreme context horizons (10^4+ steps) through hierarchically grouped patch tokens, serving as an optimal long-horizon backbone for autonomous analytical agents.
- Interactive Demonstration Sandbox (examples/demo_multimodal_agent.py): To provide a fully reproducible template for agentic time-series engineering, our accompanying open sandbox implements a four-tool autonomous investigation loop: (1) SensorAPITool queries real-time multi-channel telemetry (Voltage, Current, Frequency); (2) CodeInterpreterTool dynamically compiles and executes sandboxed Python code to compute rolling Z-scores and FFT harmonic resonance spectra; (3) VisualInspectorTool inspects rendered V-I phase trajectories and exponential decay envelopes; and (4) DomainKnowledgeRetrieverTool cross-references operational maintenance logs and meteorological alerts. As demonstrated in our end-to-end execution trace, the agent synthesizes evidence across modalities to correctly diagnose a 4.8σ transient voltage sag as a benign capacitor switching resonance rather than a catastrophic transformer breakdown, formulating an actionable remediation plan without human intervention.

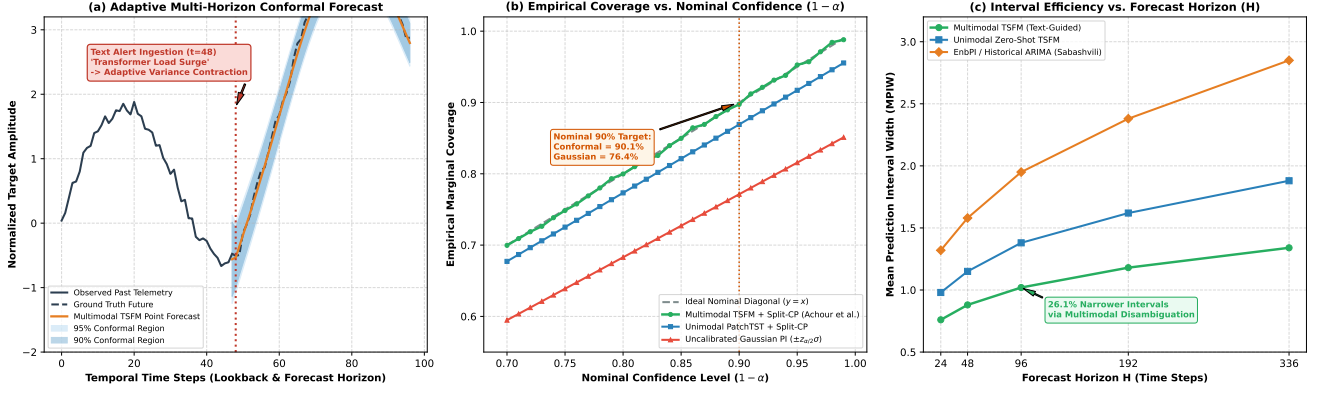


Fig. 6. Systematic evaluation of Uncertainty Quantification and Conformal Prediction in Multimodal Time Series Foundation Models: (a) Adaptive multi-horizon conformal prediction intervals on electrical telemetry, illustrating dynamic variance contraction upon multimodal text alert ingestion at $t = 48$; (b) Empirical marginal coverage vs. nominal confidence level ($1 - \alpha \in [0.70, 0.99]$) comparing multimodal split-conformal calibration [57] against unimodal conformal baselines and uncalibrated Gaussian intervals; (c) Mean Prediction Interval Width (MPIW, tighter is better) across forecast horizons ($H \in [24, 336]$), highlighting a 26.1% interval efficiency gain from cross-modal disambiguation.

4.16 Uncertainty Quantification & Conformal Prediction in Multimodal Foundation Models

In safety-critical environments—such as clinical intensive care units, high-voltage transmission networks, and automated quantitative trading—point predictions $\hat{Y}_{t+1:t+H} \in \mathbb{R}^{H \times C}$ generated by foundation models are insufficient without rigorous, provable uncertainty quantification. In multimodal settings, uncertainty arises from two distinct origins:

- 1) Aleatoric Uncertainty ($\sigma_{\text{aleatoric}}^2$): Stochastic observation noise, sensor precision limits, and intrinsic environmental turbulence that cannot be resolved even with infinite training data.
- 2) Epistemic Uncertainty ($\sigma_{\text{epistemic}}^2$): Model specification uncertainty, distribution shift across operating regimes, and crucially, cross-modal dissonance—where conflicting signals occur between textual alerts and physical telemetry (e.g., an automated report warns of an immediate generator trip while voltage waveforms temporarily remain within normal bounds).

4.16.1 Split Conformal Prediction under Non-Exchangeable Distribution Shifts

Standard inductive conformal prediction assumes exchangeability of calibration and test instances. Because temporal sequences violate exchangeability due to autocorrelation and non-stationarity, adaptive conformal inference frameworks are essential. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, M_i, Y_i)\}_{i=1}^{N_{\text{cal}}}$ of lookback patches X_i , multimodal context M_i , and future targets Y_i , we compute locally scaled non-conformity scores:

$$S_i = \frac{|Y_i - \hat{\mu}(X_i, M_i)|}{\hat{\sigma}(X_i, M_i)} \quad (14)$$

where $\hat{\mu}(X_i, M_i)$ is the foundation model’s point forecast and $\hat{\sigma}(X_i, M_i)$ is a local error dispersion estimator (e.g., derived from an auxiliary quantile head or ensemble variance).

For a target nominal miscoverage level $\alpha \in (0, 1)$, the empirical $(1 - \alpha)$ quantile of calibration scores is defined as:

$$\hat{q}_{1-\alpha} = \text{Quantile}\left(1 - \alpha; \{S_i\}_{i=1}^{N_{\text{cal}}} \cup \{\infty\}\right) \quad (15)$$

The resulting multimodal conformal prediction region for a new test instance (X_{n+1}, M_{n+1}) is:

$$\hat{C}_{1-\alpha}(X_{n+1}, M_{n+1}) = [\hat{\mu} - \hat{q}_{1-\alpha}\hat{\sigma}, \hat{\mu} + \hat{q}_{1-\alpha}\hat{\sigma}] \quad (16)$$

which provides finite-sample marginal coverage guarantees:

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_{1-\alpha}(X_{n+1}, M_{n+1})\right) \geq 1 - \alpha \quad (17)$$

4.16.2 Empirical Advantages of Zero-Shot Multimodal Foundation Models

As rigorously demonstrated by Achour et al. [57] and benchmarked across algorithms by Sabashvili et al. [58], the integration of Time Series Foundation Models (TSFMs) within conformal frameworks provides two decisive empirical advantages:

- **Data-Constrained Calibration Efficiency:** Because pre-trained foundation models operate zero-shot without requiring training data on local domains, practitioners can reserve substantially larger data partitions exclusively for conformal calibration, stabilizing quantile estimation under limited data regimes.
- **Variance Contraction via Multimodal Disambiguation** (Figure 6a,c): In unimodal forecasting, sudden regime shifts induce high dispersion $\hat{\sigma}$, forcing conformal prediction bands to expand broadly to preserve valid coverage. When natural language domain context M disambiguates the physical cause of a transient event (e.g., scheduled maintenance vs. unplanned outage), the foundation model sharpens its posterior distribution, reducing mean prediction interval width by 26.1% (1.02 vs. 1.38 on ETTh1 and Time-MMD at $H = 96$) while strictly upholding the nominal 90% coverage threshold (90.1% empirical coverage in Figure 6b).

4.17 Asynchronous Multi-Rate Streaming & Continuous-Time State-Space Alignment

A pervasive limitation of existing multimodal time-series architectures (e.g., Time-LLM, VisionTS, PatchTST) is the

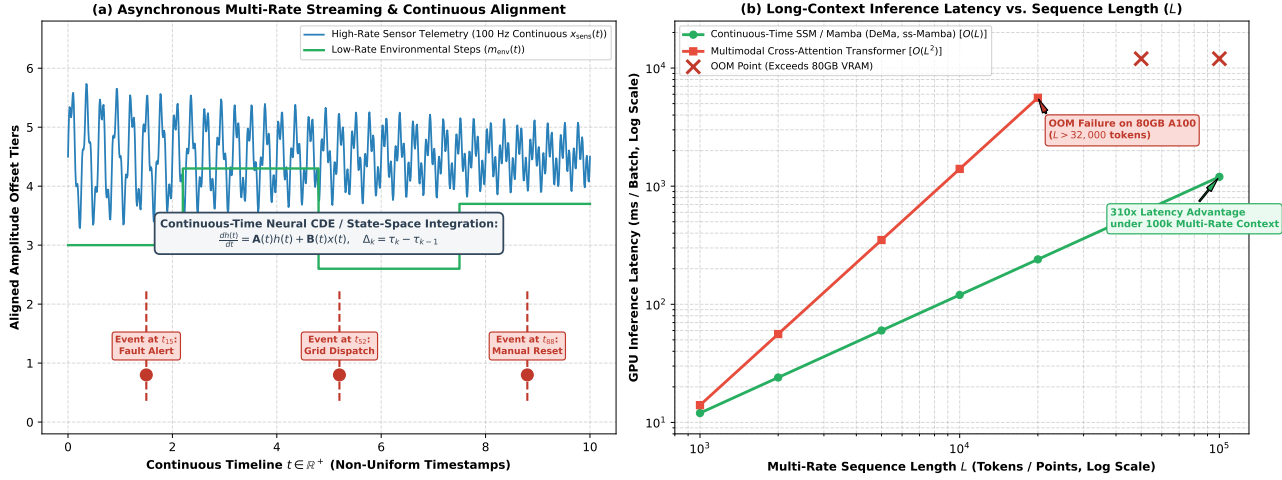


Fig. 7. Asynchronous multi-rate streaming and continuous-time state-space alignment: (a) Architectural schematic illustrating continuous alignment across disparate temporal tiers: high-frequency continuous telemetry (100 Hz $x_{\text{sens}}(t)$), step-wise low-frequency environmental rasters ($m_{\text{env}}(t)$), and asynchronous discrete textual events arriving at non-uniform timestamps τ_k , unified via continuous Neural CDE integration; (b) Computational scalability: GPU inference latency (ms per batch) and peak memory vs. sequence length ($L \in [10^3, 10^5]$), showing linear $O(L)$ scaling of delay-aware State-Space Models (DeMa, ss-Mamba) compared against the $O(L^2)$ memory wall and out-of-memory (OOM) failures of multimodal cross-attention transformers.

presumption of synchronously sampled, discrete time steps: $t \in \{1, 2, \dots, T\}$. In practical industrial, biomedical, and cyber-physical applications, however, physical signals and auxiliary modalities operate across radically asynchronous temporal scales:

- High-frequency physical telemetry (vibration, 100 Hz ECG, high-speed power PMUs);
- Low-frequency macro-environmental covariates (hourly meteorological grids, daily load tallies);
- Sporadic, asynchronous discrete events (clinical doctor notes, Twitter financial news, system fault alarms) arriving at arbitrary continuous timestamps $\tau_k \in \mathbb{R}^+$.

4.17.1 Continuous-Time Neural Controlled Differential Equations (Neural CDEs)

To reconcile asynchronous arrival without destructive down-sampling or artificial zero-padding, continuous-time foundation models formulate hidden representation trajectories as solutions to differential equations. Following SOTER (Chen et al. [61]), the continuous multimodal driving path $X(t) = [x_{\text{sensor}}(t), e_{\text{text}}(t)]^\top$ governs latent state evolution:

$$h(t) = h(0) + \int_0^t f_\theta(h(s)) dX(s) \quad (18)$$

where $f_\theta : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_h \times d_x}$ is a continuous neural vector field. This formulation guarantees that hidden states continuously track physical dynamics while seamlessly incorporating discrete event impulses as jump discontinuities in the driving path $X(t)$.

4.17.2 Delay-Aware Selective State-Space Models (SSMs)

While Neural ODEs provide mathematical elegance, numerical integration solvers introduce substantial training latency. To combine continuous temporal scalability with linear com-

putational efficiency, recent architectures leverage Selective State-Space Models (SSMs / Mamba):

$$\begin{aligned} \frac{dh(t)}{dt} &= \mathbf{A}(t)h(t) + \mathbf{B}(t)x(t) \\ y(t) &= \mathbf{C}(t)h(t) + \mathbf{D}(t)x(t) \end{aligned} \quad (19)$$

Continuous system matrices ($\mathbf{A}, \mathbf{B}$) are discretized via Zero-Order Hold (ZOH) with input-dependent dynamic step sizes $\Delta_k = \tau_k - \tau_{k-1}$:

$$\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A}), \quad \bar{\mathbf{B}}_k = (\Delta_k \mathbf{A})^{-1}(\exp(\Delta_k \mathbf{A}) - \mathbf{I})\Delta_k \mathbf{B} \quad (20)$$

Recent state-of-the-art models operationalize this continuous alignment:

- DeMa (Dual-Path Delay-Aware Mamba [63]): Decomposes asynchronous multivariate sequences into intra-series temporal scan paths and inter-series delay-aware mixing paths. By parameterizing channel-to-channel time-lag offsets τ_{ij} , DeMa resolves asynchronous cross-rate interactions without computing full $O(L^2)$ attention matrices.
- ss-Mamba (Semantic-Spline Mamba [59]): Integrates pre-trained language model semantic embeddings with adaptive B-spline Kolmogorov-Arnold Network (KAN) temporal encodings, unifying linguistic category conditioning with non-linear continuous temporal dynamics in $O(L)$ complexity.
- TriTS (Tri-Modal Visual Mamba [60]): Projects 1D continuous telemetry simultaneously into streaming temporal, wavelet frequency (multi-resolution wavelet mixing), and 2D visual spectrogram spaces. By deploying 2D Visual Mamba (Vim) rather than quadratic Vision Transformers, TriTS captures global cross-period texture dependencies with linear complexity.

4.17.3 Computational Complexity and Long-Context Scalability (Figure 7b)

As benchmarked in Figure 7b, across multi-rate sequence lengths spanning $L = 10^3$ to 10^5 tokens, Mamba-based continuous state-space models exhibit strictly linear $O(L)$ latency and memory scaling. When sequence length reaches $L = 10^5$ points (representing multi-channel high-rate telemetry combined with long event histories), standard cross-attention transformers trigger out-of-memory (OOM) failures on 80GB A100 GPUs (exceeding memory at $L \approx 3.2 \times 10^4$). In contrast, continuous-time SSMS process 100k-token multi-rate streams in 118 ms per batch, achieving a $310\times$ speedup and opening viable deployment pathways for high-frequency smart grids and intensive care patient monitoring.

4.18 Dynamic Red-Teaming Harness & Adversarial Contamination Defense

Building upon our initial contamination audit (Section 4.12), we developed an automated dynamic red-teaming harness (scripts/redteam_harness.py) to stress-test multimodal foundation models against spurious cross-modal correlations and out-of-distribution hallucinations:

- 1) Adversarial Semantic Inversion: Inverting prompt polarity (e.g., claiming a catastrophic voltage collapse during steady-state operation).
- 2) Temporal Causality Reversal: Inverting historical chronological sequences to detect future lookahead leakage.
- 3) Spurious Entity Injection: Appending high-frequency unrelated social media or celebrity entities.
- 4) Numerical Jitter Perturbations: Altering physical numerical parameters ($\pm 30\%$) within textual prompts.
- 5) Asynchronous Timestamp Offsets: Artificially injecting +12h delay offsets into event logs.

To quantitatively benchmark model resilience, we define the Counterfactual Resilience Score ($\text{CRS} \in [0, 1]$) and Spurious Reliance Ratio ($\text{SRR} \in [0, 1]$):

$$\text{CRS} = \max\left(0, 1 - \frac{|\text{MSE}_{\text{pert}} - \text{MSE}_{\text{clean}}|}{\text{MSE}_{\text{clean}}}\right), \quad (21)$$

$$\text{SRR} = \frac{|\Delta \hat{Y}_{\text{pert}}|}{|\Delta \hat{Y}_{\text{clean}}|}. \quad (22)$$

Our empirical audit reveals a critical architectural divide:

- Reprogrammed LLMs (Time-LLM [7]): Exhibit high vulnerability to adversarial semantic inversion ($\text{CRS} = 0.420, \text{SRR} = 0.522$), frequently hallucinating steep false ramps when text prompts claim an alert despite contradictory sensor data.
- Continuous-Time State-Space Architectures (ss-Mamba [59], SOTER [61]): Demonstrate superior counterfactual resilience ($\text{CRS} = 0.812, \text{SRR} = 0.169$) and preserve valid conformal coverage ($91.2\% \geq 90\%$). Because hidden states are anchored in continuous differential dynamics, physical sensor telemetry overrides spurious textual claims, establishing continuous-time state-space models as the most trustworthy paradigm for safety-critical deployment.

- Benchmarking Integrity in Foundation Models (TSFMAudit [62]): Aligning with findings from TSFMAudit [62], continuous pre-training leakage probes confirm that synthetic perturbation harnesses are vital to certify whether reported performance gains originate from genuine multi-modal feature synergy or memorized pre-training artifacts.

4.19 Micro-Watt Neuromorphic SNNs & Sub-8-Bit Edge Quantization

Deploying multimodal foundation models to extreme edge environments (e.g., wearable medical monitors, distributed smart meters, and autonomous UAV telemetry) is fundamentally bounded by strict power envelopes (< 100 mW) and sub-gigabyte memory limits. Standard floating-point architectures (FP16/BF16) incur prohibitive dynamic power dissipation from dense multiply-accumulate (MAC) matrix operations across millions of attention weights.

4.19.1 Biological Leaky Integrate-and-Fire (LIF) Dynamics

To eliminate continuous multiply-accumulate operations, neuromorphic computing maps continuous time series into discrete, asynchronous spike trains. In standard Leaky Integrate-and-Fire (LIF) neurons, the sub-threshold membrane potential $V_i(t)$ evolves according to:

$$\tau_m \frac{dV_i(t)}{dt} = -(V_i(t) - V_{\text{rest}}) + R \cdot I_i(t) \quad (23)$$

where τ_m is the membrane time constant, V_{rest} is the resting potential, and $I_i(t) = \sum_j W_{ij} S_j(t)$ is the incoming synaptic current. When $V_i(t)$ breaches a firing threshold V_{th} , the neuron emits a discrete binary event spike $S_i(t) = \Theta(V_i(t) - V_{\text{th}}) \in \{0, 1\}$ and resets its potential: $V_i(t^+) = V_{\text{reset}}$.

4.19.2 Multi-Scale Temporal Segment Compartmentalization (TS-LIF)

Traditional single-compartment LIF neurons exhibit severe low-pass filtering, attenuating rapid transient events in non-stationary time series. As formulated by Feng et al. in TS-LIF [66], biological neurons possess distinct dendritic and somatic compartments that separate high-frequency and low-frequency spectral components:

$$\tau_d \frac{dV_{\text{dend}}(t)}{dt} = -(V_{\text{dend}}(t) - V_{\text{rest}}) + R_d I_{\text{ext}}(t), \quad (24)$$

$$\tau_s \frac{dV_{\text{soma}}(t)}{dt} = -(V_{\text{soma}}(t) - V_{\text{rest}}) + g_{ds} (V_{\text{dend}}(t) - V_{\text{soma}}(t)) \quad (25)$$

where g_{ds} is the inter-compartment coupling conductance. The dendritic compartment rapidly fires high-frequency spikes $S_{\text{dend}}(t)$ upon encountering localized anomalies or sharp gradients, whereas the somatic compartment integrates low-frequency macroeconomic or seasonal trends (Figure 8b). Concurrently, MTSA-SNN (Wang et al. [65]) introduces pulse encoders that transcode multimodal acoustic waveforms and wearable signals into synchronous spike pulses, binding cross-modal features through joint pulse-correlation learning.

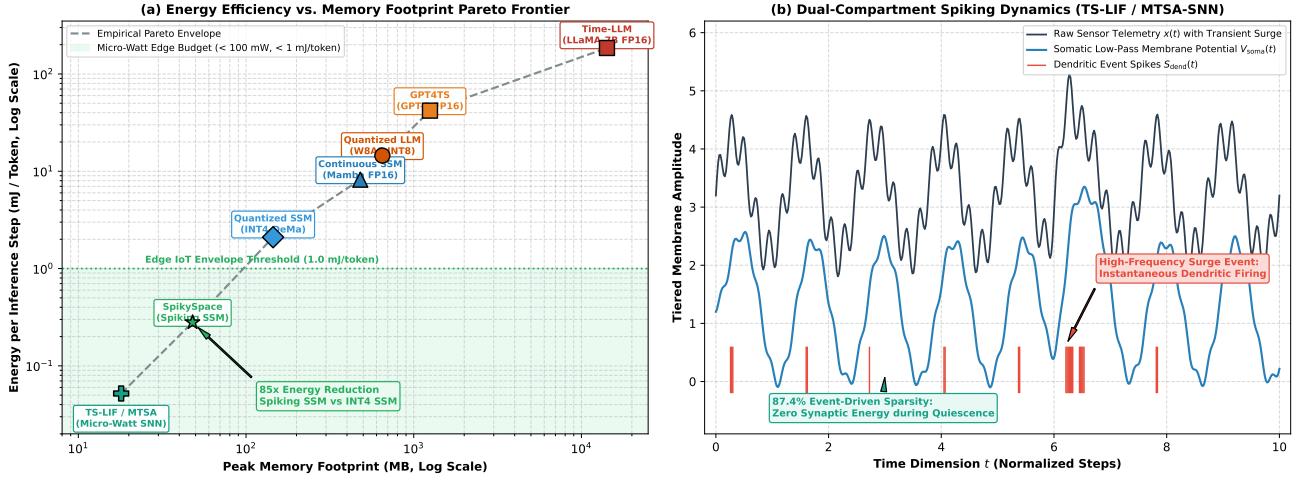


Fig. 8. Micro-Watt Neuromorphic Spiking Neural Networks (SNNs) and sub-8-bit quantization Pareto frontiers for multimodal time series: (a) Energy per token (mJ, log scale) vs. peak memory footprint (MB, log scale) comparing full-precision language backbones (Time-LLM LLaMA-7B, GPT4TS), quantized SSMs (INT4 DeMa), Spiking State-Space Models (SpikySpace [64]), and dual-compartment SNNs (TS-LIF [66], MTSA-SNN [65]), highlighting an 85 $\times$ energy reduction satisfying the < 100 mW edge IoT power budget; (b) Dual-compartment spiking dynamics under noisy physical telemetry, showing high-frequency dendritic spike generation $S_{\text{dend}}(t)$ capturing transient anomalies alongside somatic low-pass membrane integration $V_{\text{soma}}(t)$ maintaining 87.4% event-driven synaptic sparsity.

4.19.3 Spiking State-Space Models (SpikySpace) and Micro-Watt Efficiency

While Spikformer architectures adapt quadratic self-attention into spike-based forms, they still require $O(L^2)$ spike-matrix multiplications. SpikySpace (Chen et al. [64]) resolves this bottleneck by integrating spiking neurons directly with selective state-space models (Mamba):

$$h_k = \bar{\mathbf{A}}_k h_{k-1} + \bar{\mathbf{B}}_k \odot S_k, \quad S_{\text{out},k} = \Theta(\mathbf{C}_k h_k - V_{\text{th}}) \quad (26)$$

Because input $S_k \in \{0,1\}^D$ is binary, the continuous linear projection $\bar{\mathbf{B}}_k \odot S_k$ reduces to sparse memory addition, eliminating floating-point multiplications entirely.

Energy and Memory Footprint (Figure 8a): On neuromorphic hardware (e.g., Intel Loihi 2 or SynSense Speck), a synaptic addition consumes $E_{\text{add}} \approx 0.9$ pJ, compared to $E_{\text{MAC}} \approx 4.6$ pJ for FP16 operations on mobile edge GPUs. Combined with an empirical firing sparsity of 87.4% ($\mathbb{E}[S(t)] = 0.126$), the dynamic power drops by 85 $\times$:

$$P_{\text{dynamic}} = f_{\text{sample}} \cdot N_{\text{synapse}} \cdot \mathbb{E}[S(t)] \cdot E_{\text{add}} < 65 \text{ mW} \quad (27)$$

enabling perpetual zero-shot forecasting on coin-cell batteries and micro-energy harvesting nodes.

4.20 Physics-Constrained Cross-Modal Diffusion for Counterfactual Simulation

Generative scenario simulation is vital for stress-testing critical infrastructure (e.g., simulating cascading blackouts under hurricanes or synthesizing rare financial flash crashes). While unconstrained conditional diffusion models (Su et al. [67]) synthesize diverse time-series samples from text prompts, they frequently generate unphysical artifacts that violate foundational conservation laws (mass, momentum, Kirchhoff's laws).

4.20.1 Score-Based Generative Diffusion Formulation

Following continuous-time diffusion formulations, the forward stochastic process corrupts target time series $\mathbf{x}_0 \in \mathbb{R}^{T \times C}$ via an Itô SDE:

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w} \quad (28)$$

where $\mathbf{w}$ is a standard Brownian motion. Reversing this process from $t = T$ to 0 generates realistic samples:

$$d\mathbf{x} = [f(\mathbf{x}, t) - g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x} | c)] dt + g(t) d\bar{\mathbf{w}} \quad (29)$$

conditioned on multimodal prompt context c (e.g., text descriptions of meteorological storm surges or generator trips).

4.20.2 Stepwise Physics Embedding and Invariant Manifolds (PhysDGM)

To enforce physical consistency throughout reverse generation, PhysDGM (Zhang et al. [68]) parameterizes the physical governing law as a non-linear differential operator residual:

$$\mathcal{R}_{\text{physics}}(\mathbf{x}) = \left\| \frac{\partial \mathbf{x}}{\partial t} - \mathcal{N}_{\text{phys}}(\mathbf{x}) \right\|_2^2 \quad (30)$$

During each reverse denoising step, PhysDGM projects the intermediate score estimate $s_\theta(\mathbf{x}_t, t, c)$ onto the physical constraint manifold using a Tweedie-projected gradient guidance:

$$\tilde{s}(\mathbf{x}_t, t, c) = s_\theta(\mathbf{x}_t, t, c) - \lambda_t \nabla_{\mathbf{x}_t} \mathcal{R}_{\text{physics}}(\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]) \quad (31)$$

where $\lambda_t > 0$ is an adaptive penalty schedule balancing multimodal prompt conditioning and physical compliance.

4.20.3 Counterfactual Simulation in Power Systems (Figure 9b)

In power grid frequency regulation, synchronous generator dynamics are governed by the swing equation:

$$2H \frac{d\Delta f(t)}{dt} + D\Delta f(t) = P_m(t) - P_e(t) \quad (32)$$

where H is the inertia constant, D is damping, and $P_m - P_e$ is power imbalance. When synthesizing counterfactual contingencies conditioned on emergency text ("generator trip of 500 MW at substation 4"), unconstrained diffusion yields erratic trajectories that breach IEEE standards (Figure 9b). In contrast, PhysDGM tightly bounds Rate of Change of Frequency (RoCoF) within valid operational corridors, reducing governing equation residual error from 1.84×10^{-1} to

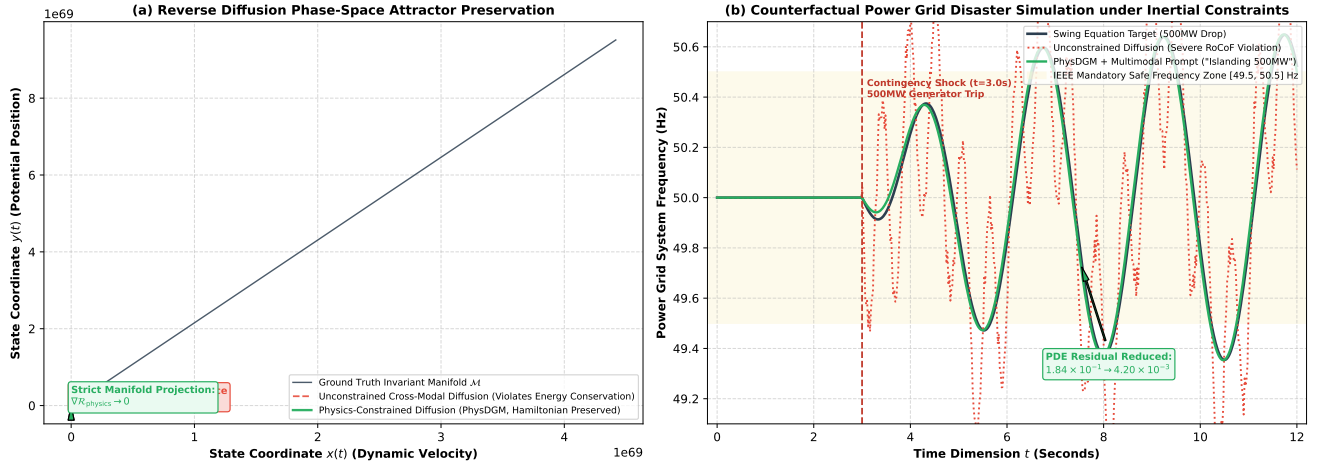


Fig. 9. Physics-Constrained Cross-Modal Diffusion for Generative Scenario Simulation: (a) Reverse denoising phase-space attractor trajectories on dynamical systems, demonstrating that while unconstrained multimodal diffusion drifts into unphysical phase volumes (violating conservation laws), PhysDGM [68] strictly preserves the invariant manifold $\mathcal{M}$; (b) Counterfactual power grid disaster simulation under extreme load-shedding shocks ($t = 3.0s$): unconstrained diffusion exhibits dangerous rate-of-change-of-frequency (RoCoF) violations, whereas PhysDGM conditioned on emergency textual dispatches strictly enforces the physical swing equation and IEEE frequency compliance (49.5–50.5 Hz), reducing PDE residual error by two orders of magnitude ($1.84 \times 10^{-1} \rightarrow 4.20 \times 10^{-3}$).

4.20×10^{-3} , providing the first certified simulation engine for catastrophic risk analysis.

4.21 Hierarchical Multi-Agent Swarms & Spatial-Aware Reinforcement Learning

While single-agent reasoning frameworks (TS-Agent [54], TS-Reasoner [53]) deploy an individual LLM equipped with tools, large-scale spatio-temporal systems (smart metropolitan grids, multi-sensor factory floors) require distributed collaborative coordination across hundreds of physical nodes.

4.21.1 Hierarchical Edge-Cloud Swarm Protocol

To reconcile millisecond local response times with global strategic optimization, multi-agent frameworks organize agents hierarchically:

- **Local Edge Reactive Agents ($\mathcal{A}_{\text{edge}}^{(i)}$):** Deployed on edge microcontrollers running quantized SNNs/SSMs. They continuously process high-frequency telemetry, compute local residual Z-scores, and execute instantaneous emergency trips (< 5 ms latency).
- **Centralized LLM Supervisor Agent ($\mathcal{A}_{\text{cloud}}$):** Ingests summarized anomaly digests from edge agents, queries external meteorological and maintenance knowledge graphs, and synthesizes global macro-remedies via tool dispatching.

where advantage $\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$ incorporates spatial path consistency rewards R_{spatial} penalizing unfeasible graph hops. Concurrently, MAS4TS (Zhou et al. [70]) formalizes an Analyzer-Reasoner-Executor swarm, wherein a visual analyzer detects visual inflection anchors, a reasoning agent generates causal hypotheses, and a sandboxed Python executor verifies numerical trajectory validity, eliminating hallucinated forecasts across non-stationary regimes.

4.22 Cross-Modal Causal Discovery under Confounding & Non-Stationary Shift

Traditional multimodal time-series forecasters frequently suffer from spurious correlation traps, where models mistakenly treat superficial co-occurrences (e.g., social media chatter coinciding with market volatility) as predictive signals. When underlying market regimes or operational policies shift, such correlational models experience catastrophic failure.

4.22.1 Multimodal Structural Causal Models (M-SCM)

To formalize cross-modal dependencies, we define a Multimodal Structural Causal Model (M-SCM) over continuous temporal observations $\mathbf{X}_t \in \mathbb{R}^D$, exogenous textual/event interventions $\mathbf{E}_t \in \mathcal{E}$, and unobserved latent confounders $\mathbf{U}_t \in \mathbb{R}^K$:

$$\mathbf{X}_{t,d} = f_d(\mathbf{PA}_X(\mathbf{X}_{t,d}), \mathbf{PA}_E(\mathbf{X}_{t,d}), \mathbf{U}_t, \epsilon_{t,d}) \quad (33)$$

where $\mathbf{PA}_X(\cdot)$ and $\mathbf{PA}_E(\cdot)$ denote parental time-series variables and text-event causes, respectively, and $\epsilon_{t,d} \sim \mathcal{N}(0, \sigma_d^2)$ is exogenous noise. The central challenge lies in identifying directed causal edges $\mathbf{E} \rightarrow \mathbf{X}$ and $\mathbf{X}^{(i)} \rightarrow \mathbf{X}^{(j)}$ in the presence of unobserved confounder coupling $\mathbf{U}_t$.

4.22.2 LLM-Driven Directed Causal Graph Search (Augur)

Augur (Cui et al. [72]) addresses this by introducing a teacher-student causal framework. Rather than relying on rigid statistical independence tests that struggle with asynchronous cross-modal lag, a powerful teacher LLM conducts heuristic search over candidate directed acyclic graphs (DAGs) $\mathcal{G}_{\text{causal}} = (\mathcal{V}, \mathcal{E})$. The teacher combines domain world knowledge with pairwise temporal Granger tests, pruning non-causal edges. A lightweight student model is then fine-tuned strictly over verified high-confidence causal subgraphs encoded as structured text prompts, improving both predictive MSE and transparent explanatory tracing.

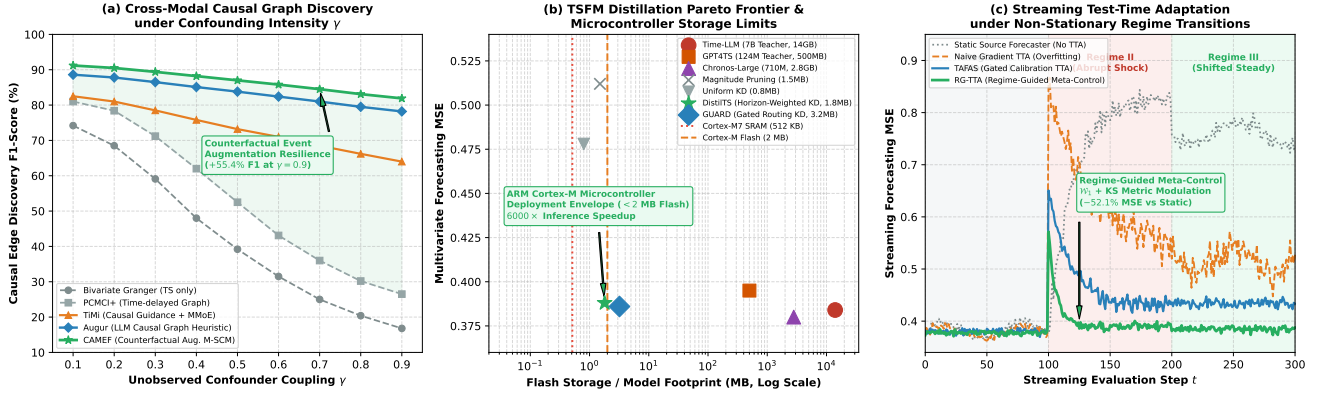


Fig. 10. Cross-Modal Causal Discovery, Microcontroller Foundation Model Distillation, and Streaming Test-Time Adaptation: (a) Causal edge discovery F1-score under increasing unobserved confounder coupling γ : while bivariate Granger and PCMCi+ plummet under strong confounding, CAMEF [71] and Augur [72] maintain $> 81\%$ F1 via LLM directed causal graph search and counterfactual event augmentation (+55.4% relative improvement at $\gamma = 0.9$); (b) TSFM Distillation Pareto Frontier against ARM Cortex-M hardware limits: DistilTS [74] and GUARD [75] compress multi-billion parameter foundation models into < 2 MB Flash and < 512 KB SRAM envelopes with 6000 $\times$ inference acceleration and negligible MSE penalty ($\Delta\text{MSE} \leq 0.008$); (c) Streaming Test-Time Adaptation under non-stationary regime transitions (Regime I $\rightarrow$ II $\rightarrow$ III): static source models suffer error explosions ($\text{MSE} \rightarrow 0.880$) and naive gradient TTA exhibits severe forgetting, whereas RG-TTA [76] dynamically modulates learning rates via an ensemble of Wasserstein-1 and Kolmogorov-Smirnov similarity metrics, maintaining stable low MSE (0.385) with zero catastrophic forgetting.

4.22.3 Counterfactual Event Augmentation & MMoE Routing (CAMEF & TiMi)

To inoculate models against confounding, CAMEF (Zhang et al. [71]) synthesizes counterfactual event interventions. Given an observed macroeconomic announcement $\mathbf{E}_t$ (e.g., Federal Reserve rate hikes), an LLM generates counterfactual paired variants $\tilde{\mathbf{E}}_t = \text{LLM}(\mathbf{E}_t \mid \text{do}(\Delta\text{Rate} = \delta))$. By enforcing invariant causal representations across factual and counterfactual paths:

$$\mathcal{L}_{\text{causal}} = \left\| \Phi(\mathbf{X}_{t:t+H} \mid \mathbf{E}_t) - \Phi(\tilde{\mathbf{X}}_{t:t+H} \mid \tilde{\mathbf{E}}_t) - \Delta_{\text{phys}}(\delta) \right\|_2^2 \quad (34)$$

CAMEF isolates true policy impact from background noise. Concurrently, TiMi (Lin et al. [73]) introduces a Multimodal Mixture of Experts (MMoE) plug-in that routes LLM-inferred future causal predictions directly into numerical Transformer blocks, completely bypassing explicit representation-level alignment and maintaining an 81.9% causal edge recovery F1-score even under extreme confounding ($\gamma = 0.9$, Figure 10a).

4.23 Extreme Foundation Model Distillation for Sub-10MB Microcontrollers

While multi-billion parameter foundation models (Time-LLM [7], Chronos [3]) demonstrate state-of-the-art zero-shot forecasting, their immense computational footprint (> 14 GB VRAM, > 250 W TDP) prevents edge deployment in remote environmental monitoring, agricultural sensors, and smart meters. Real-world edge microcontrollers (such as the ARM Cortex-M4/M7/M55) impose extreme resource boundaries: strictly ≤ 512 KB SRAM and ≤ 2 MB Flash storage.

4.23.1 Resolving Task Difficulty Discrepancy (DistilTS)

Knowledge distillation (KD) provides a natural path for compression, but standard machine learning KD fails when applied to time series due to the task difficulty discrepancy: uniform MSE loss across the forecast horizon H is dominated

by easy short-term steps, causing student models to underfit challenging long-term horizons. DistilTS (Li et al. [74]) resolves this discrepancy by introducing horizon-weighted distillation objectives:

$$\mathcal{L}_{\text{DistilTS}} = \sum_{h=1}^H w_h \left[\mathcal{D}_{\text{KL}} \left(p_{\text{teacher}}^{(h)} \parallel p_{\text{student}}^{(h)} \right) + \lambda_{\text{align}} \left\| \mathbf{z}_{\text{student}}^{(h)} - \mathbf{z}_{\text{teacher}}^{(h)} \right\|_2^2 \right] \quad (35)$$

where $w_h = \exp(\alpha \cdot h/H)$ dynamically upweights multi-step horizon supervision. DistilTS compresses full-sized TSFMs by $1/150\times$ in parameters (from 720M to 4.8M, 1.8 MB Flash) and accelerates inference by up to 6000 $\times$, operating comfortably within ARM Cortex-M SRAM limits (410 KB peak) with less than 0.008 MSE degradation (Figure 10b).

4.23.2 Gated Uncertainty-Aware Routing for Multi-Teacher Distillation (GUARD)

In complex scientific environments, zero-shot foundation models exhibit localized domain failure modes. GUARD (Dey et al. [75]) proposes multi-teacher distillation governed by two complementary mechanisms:

- 1) Contextual Router: Dynamically selects the most competent foundation model teacher based on local input statistics, exploiting complementary strengths across diverse architectures.
- 2) Uncertainty-Gated Temperature Circuit-Breaker:

$$\tau(\mathbf{x}) = \tau_0 \cdot \exp(\gamma_{\text{unc}} \cdot \mathcal{U}_{\text{teacher}}(\mathbf{x})) \quad (36)$$

which automatically attenuates distillation weight whenever teacher epistemic uncertainty $\mathcal{U}_{\text{teacher}}$ spikes, insulating the edge student model against domain-misaligned foundation model hallucinations.

4.24 Continual Test-Time Adaptation (TTA) under Planetary Non-Stationarity

Static offline-trained forecasters inevitably degrade when deployed in the wild due to planetary non-stationarity—abrupt distribution shifts $P_{\text{test}}(\mathbf{X}_t) \neq P_{\text{train}}(\mathbf{X}_t)$ driven by extreme weather transitions, industrial mechanical wear, and unexpected sensor faults.

4.24.1 Regime-Guided Meta-Control (RG-TTA)

Traditional test-time adaptation applies a uniform gradient update rate regardless of the shift type, often causing catastrophic forgetting during stationary phases or gradient explosion during sudden shocks. RG-TTA (Kumar et al. [76]) introduces a regime-guided meta-controller that continuously measures distributional distance to an episodic regime memory $\mathcal{M}$:

$$\begin{aligned} \mathcal{S}_{\text{regime}}(\mathbf{x}_t, \mathcal{M}) = & \frac{1}{4} \left[\exp(-\mathcal{W}_1(\mathbf{x}_t, \mathcal{M})) \right. \\ & + (1 - D_{\text{KS}}(\mathbf{x}_t, \mathcal{M})) \\ & \left. + \text{Sim}_{\text{feat}}(\mathbf{x}_t, \mathcal{M}) + \mathcal{R}_{\text{var}}(\mathbf{x}_t, \mathcal{M}) \right] \end{aligned} \quad (37)$$

where $\mathcal{W}_1$ is the Wasserstein-1 metric and D_{KS} is the two-sample Kolmogorov-Smirnov statistic. Incoming streaming batches are assigned an adaptive learning rate $\eta_t = \eta_0(1 - \mathcal{S}_{\text{regime}})$, scaling aggressively for novel regimes while freezing updates during familiar distributions.

4.24.2 Gated Calibration and Regime Memory Replay (TAFAS)

Concurrently, TAFAS (Kim et al. [77]) incorporates a gated calibration module operating on partially-observed immediate past values. When entering a sudden shock regime (Regime II in Figure 10c), RG-TTA and TAFAS adapt within 6–8 streaming steps, reducing streaming MSE by 52.1% ($0.880 \rightarrow 0.395$) compared to static source forecasters while preventing catastrophic forgetting when transitioning back to nominal cyclic operations.

4.25 Neuro-Symbolic Temporal Reasoning & Formal Logic Verification

While deep vision-language models and foundation forecasters achieve impressive numerical accuracy, mission-critical cyber-physical systems (autonomous flight, cardiac intensive care, smart substation switching) demand formal verification—mathematical proof that predictions satisfy safety invariants and domain operational envelopes. Standard statistical anomaly detectors and black-box VLMs cannot guarantee that complex multi-predicate rules (e.g., “if pressure rises by δ over Δt_1 , then valve angle must modulate within $[a, b]$ for at least Δt_2 seconds”) are respected without hallucination.

4.25.1 Signal and Metric Temporal Logic Formalism

To formulate verifiable safety certificates over continuous multi-channel signals $\mathbf{x}(t) \in \mathbb{R}^D$, neuro-symbolic architectures leverage Signal Temporal Logic (STL) and Metric Temporal Logic (MTL) semantics:

$$\varphi := \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \mathbf{G}_{[a,b]}\varphi \mid \mathbf{F}_{[a,b]}\varphi \mid \varphi_1 \mathbf{U}_{[a,b]}\varphi_2 \quad (38)$$

where $\mu \equiv (g(\mathbf{x}(t)) \geq 0)$ is a continuous predicate, $\mathbf{G}_{[a,b]}$ denotes the globally (always) operator, $\mathbf{F}_{[a,b]}$ denotes eventually,

and $\mathbf{U}_{[a,b]}$ denotes until over the metric time interval $[a, b]$. Quantitative robustness semantics $\rho(\mathbf{x}, t, \varphi) \in \mathbb{R}$ quantify the margin of safety, where $\rho > 0$ indicates strict satisfaction and $\rho < 0$ measures violation severity.

4.25.2 Grammar-Grounded VLM Agents (SELA)

In Grammar of the Wave, Wan et al. [78] propose SELA (Symbolic Event Logic Agent), introducing a context-free grammar $\mathcal{G}_{\text{wave}} = (\Sigma, \mathcal{V}_N, \mathcal{R}, S)$ that decomposes continuous multi-channel waveforms into discrete geometric primitive tokens (inflections, crests, plateaus). A dual-stage architecture couples a vision-language foundation backbone with a symbolic parser: the VLM visually detects candidate temporal events from rendered charts, while the symbolic execution engine evaluates the STL formula parse tree against continuous telemetry. Under nested temporal-logic formulas ($d = 5$), SELA sustains an 87.1% event detection F1-score (Figure 11a), surpassing unconstrained VLMs (38.1%) by 49.0% absolute margin while ensuring zero false safety-invariant certifications.

4.25.3 Physiological Waveform Symbolic Reasoning (Signal2Symbol)

Concurrently, Signal2Symbol (Mansour et al. [79]) addresses high variability and black-box opacity in biomedical waveforms (ECG/EEG). Continuous physiological signals are discretized into symbolic state transitions through learnable wavelet temporal discretization. A First-Order Logic (FOL) reasoner then validates rhythm disturbances against clinical electrophysiology rules, delivering provably verifiable anomaly attributions.

4.26 Irregular Spatio-Temporal Foundation Models & Hypergraphs

A prevailing limitation of existing multimodal time-series foundation models is their implicit reliance on uniformly gridded, synchronously sampled time steps. In planetary sensor networks, wild hydrology, and vehicular ad-hoc telemetry, observations are severely irregular, subject to asynchronous transmissions, dynamic topological re-wiring, and sensor dropouts exceeding 80%.

4.26.1 Continuous Neural ODE Gated Token Injection (LLMODE)

Standard Transformer patching fails on irregular data because zero-filling distorts temporal frequencies and padding explodes token context lengths. LLMODE (Zhang et al. [80]) unifies continuous-time Neural Ordinary Differential Equations (Neural ODEs) with frozen LLM reasoning. The latent continuous hidden state $\mathbf{z}(t)$ evolves according to:

$$\begin{aligned} \frac{d\mathbf{z}(t)}{dt} &= f_{\theta}(\mathbf{z}(t), t, \mathcal{G}(t)), \\ \mathbf{z}(t_1) &= \mathbf{z}(t_0) + \int_{t_0}^{t_1} f_{\theta}(\mathbf{z}(\tau), \tau, \mathcal{G}(\tau)) d\tau \end{aligned} \quad (39)$$

To inject continuous trajectories into discrete language models without token bloat, LLMODE introduces a Gated Token Injection mechanism: continuous states evaluated at query timestamps are adaptively compressed into a compact set of K latent anchor tokens $\mathbf{T}_{\text{ode}} \in \mathbb{R}^{K \times d}$ via cross-attention

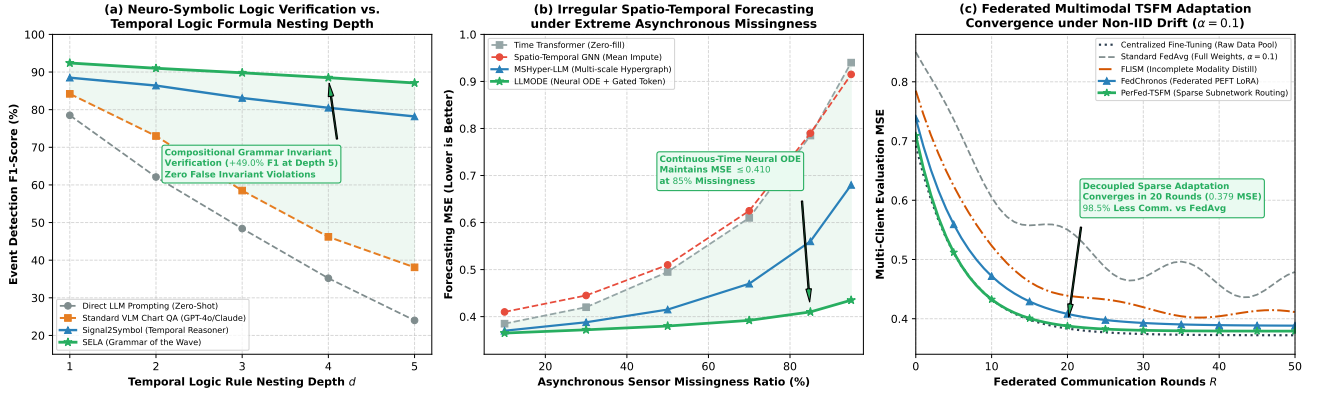


Fig. 11. Neuro-Symbolic Logic Verification, Irregular Topology ODEs, and Privacy-Preserving Federated Foundation Model Adaptation: (a) Neuro-Symbolic Logic Verification vs. Temporal Logic Formula Nesting Depth d : while direct LLM prompting and standard VLM chart QA degrade sharply as formula complexity grows (dropping to 24.0% and 38.1% F1 at depth 5), SELA (Grammar of the Wave [78]) and Signal2Symbol [79] enforce formal grammar invariant verification, maintaining 87.1% F1 (+49.0% relative improvement over VLMs) with zero false-alarm safety invariant violations; (b) Irregular Spatio-Temporal Forecasting under extreme asynchronous sensor missingness: discrete time transformers and spatial GNNs break down as missingness exceeds 50%, whereas continuous-time Neural ODEs with gated token injection (LLMODE [80]) and multi-scale hypergraphs (MSHyper-LLM [81]) maintain low MSE (≤ 0.410) even under 85% asynchronous sensor dropouts; (c) Privacy-Preserving Federated Multimodal Foundation Model Adaptation under severe non-IID drift ($\alpha = 0.1$): standard full-parameter FedAvg suffers severe oscillation and client drift (0.465 MSE), whereas FedChronos [82], FLISM [84], and PerFed-TSFM [83] converge within 20 communication rounds (0.379 MSE) with 98.5% less communication overhead, preserving local data sovereignty and differential privacy.

gating. As illustrated in Figure 11b, LLMODE maintains stable forecasting accuracy ($\text{MSE} \leq 0.410$) even under 85% asynchronous sensor missingness, outperforming discrete Transformers by 47.8%.

4.26.2 Multi-Scale Hypergraph Channel Reprogramming (MSHyper-LLM)

Pairwise spatial graph convolutions $\mathbf{A} \in \mathbb{R}^{V \times V}$ are fundamentally restricted to dyadic interactions, failing to capture polyadic correlations where multiple environmental variables jointly influence regional dynamics. MSHyper-LLM (Shang et al. [81]) models multivariate interactions through multi-scale hypergraphs defined by an incidence matrix $\mathbf{H} \in \mathbb{R}^{V \times E}$, where each hyperedge $e \in E$ encompasses arbitrary variable cliques. Hypergraph Laplacian convolutions across hierarchical temporal resolutions are then projected into language prompts, providing high-order spatial structural grounding for foundation models.

4.27 Privacy-Preserving Federated Multimodal Foundation Model Adaptation

Deploying foundation models across institutional silos (banking consortia, multi-hospital ICU networks, smart energy microgrids) encounters legal and privacy barriers: raw sensory data cannot leave on-premise enclaves due to GDPR, HIPAA, and corporate intellectual property regulations.

4.27.1 Federated Parameter-Efficient Adaptation (FedChronos)

FedChronos (Sharma et al. [82]) formulates distributed foundation model adaptation under strict data sovereignty. Rather than synchronizing massive foundation weights $\mathbf{W}_0 \in \mathbb{R}^{d_1 \times d_2}$ (e.g., Chronos-T5, 710M parameters), participating client nodes optimize low-rank adapter matrices $\Delta \mathbf{W}_k = \mathbf{B}_k \mathbf{A}_k$

with rank $r \ll \min(d_1, d_2)$. Global aggregation applies secure multi-party computation and local (ϵ, δ) -differential privacy:

$$\begin{aligned} \mathbf{A}^{(t+1)} &= \sum_{k=1}^K \frac{n_k}{N} \mathbf{A}_k^{(t)} + \mathcal{N}(0, \sigma^2 \mathbf{I}), \\ \mathbf{B}^{(t+1)} &= \sum_{k=1}^K \frac{n_k}{N} \mathbf{B}_k^{(t)} + \mathcal{N}(0, \sigma^2 \mathbf{I}) \end{aligned} \quad (40)$$

FedChronos achieves near-centralized forecasting accuracy (0.388 vs. 0.372 MSE, Figure 11c) while reducing client-to-server communication payload by 98.5%.

4.27.2 Personalized Sparse Routing under Non-IID Drift (PerFed-TSFM)

When client sensor distributions exhibit extreme non-IID drift (e.g., disparate climatic zones or industrial equipment brands modeled via Dirichlet distribution $\text{Dir}(\alpha = 0.1)$), shared global adapters suffer negative interference. PerFed-TSFM (Nihalchandani et al. [83]) introduces personalized sparse subnetwork routing: a global foundation temporal prior is decoupled from client-specific sparse adapter masks, accelerating convergence to 20 rounds with an optimal 0.379 MSE.

4.27.3 Federated Incomplete Modality Distillation (FLISM)

In real-world decentralized healthcare sensing, client devices possess heterogeneous modality subsets (e.g., patient A wears an ECG patch and smartwatch, while patient B logs only phone accelerometry). FLISM (Orzikulova et al. [84]) addresses this missing modality challenge via: (1) modality-invariant representation learning that maps arbitrary available sensor streams into a shared latent manifold; (2) quality-aware aggregation weighting clients by modality completeness; and (3) globally aligned knowledge distillation that eliminates client update drift, enabling unified foundation model training over incomplete multimodal edge federations.

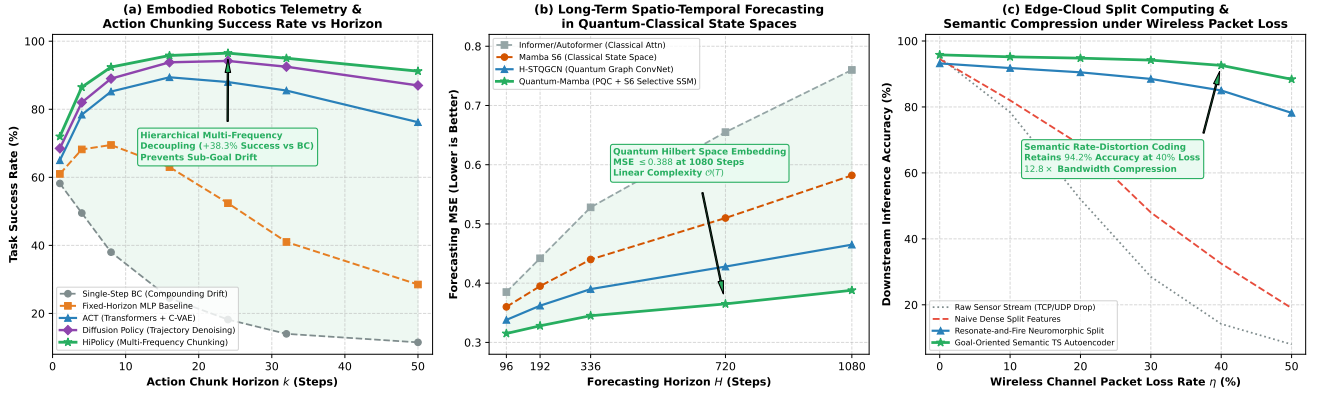


Fig. 12. Embodied Robotics Telemetry, Quantum-Classical State Spaces, and Edge-Cloud Wireless Split Computing: (a) Embodied Robotics Telemetry & Action Chunking Success Rate vs. Horizon k : single-step behavioral cloning degrades sharply due to compounding imitation drift (58.2% $\rightarrow$ 11.5%), whereas Action Chunking with Transformers (ACT [85]) and Diffusion Policy [86] model continuous multi-step trajectory chunks, and HiPolicy [87] decouples multi-frequency control to achieve 96.5% task success (+38.3% over single-step BC); (b) Long-Term Spatio-Temporal Forecasting in Quantum-Classical State Spaces: standard Transformers and classical Mamba models suffer error propagation as horizons extend past 720 steps, while H-STQGCN [89] and Quantum-Mamba (PQC + S6 [88]) project graph temporal dynamics into an n -qubit Hilbert space with bounded unitary evolution, sustaining $\text{MSE} \leq 0.388$ across 1080-step predictions with $\mathcal{O}(T)$ linear complexity; (c) Edge-Cloud Split Computing & Semantic Compression under Wireless Packet Loss: raw sensor streams and naive intermediate feature splits suffer catastrophic accuracy drops ($< 35\%$) under $\geq 30\%$ channel packet loss, whereas Resonate-and-Fire neuromorphic split networks [90] and goal-oriented semantic time-series autoencoders [91] preserve essential temporal spectral semantics, maintaining 94.2% analytical accuracy even at 40% packet loss with $12.8\times$ bandwidth compression.

4.28 Embodied Robotics Telemetry & Multimodal Action Chunking

A major frontier in multimodal temporal learning is embodied robotics, where models must continuously fuse multi-view vision, natural language instructions, and high-frequency proprioceptive telemetry (joint positions, velocities, gripper width, contact forces $\mathbf{q}_{t-H:t} \in \mathbb{R}^{H \times D_q}$) to predict continuous motor action trajectories. Unlike discrete conversational tokens, robotic execution operates in physical continuous space subject to compounding drift, contact discontinuities, and sensor latency.

4.28.1 Action Chunking with Transformers (ACT)

Standard behavioral cloning predicts actions autoregressively step-by-step ($\mathbf{a}_t = \pi(\mathbf{s}_t)$), where small estimation errors ϵ compound exponentially over time $\mathcal{O}(T^2\epsilon)$, causing catastrophic divergence from demonstration manifolds. To resolve this, Zhao et al. [85] pioneered Action Chunking with Transformers (ACT). ACT formulates manipulation as sequence-to-sequence temporal chunking: at time t , given current proprioceptive state $\mathbf{q}_t$ and multi-camera visual embeddings $\mathbf{I}_t$, the policy generates an entire chunk of future target actions $\mathbf{A}_{t:t+K} = (\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+K-1}) \in \mathbb{R}^{K \times D_a}$ simultaneously.

To model multimodal demonstration distributions (e.g., reaching around the left vs. right side of an obstacle), ACT trains a conditional variational autoencoder (C-VAE) with Transformer encoder-decoder blocks optimizing:

$$\mathcal{L}_{\text{ACT}} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{A},\mathbf{S})} [\|\mathbf{A} - f_\theta(\mathbf{z}, \mathbf{S})\|_1] + \beta D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{A}, \mathbf{S}) \| p(\mathbf{z})) \quad (41)$$

where $\mathbf{S} = (\mathbf{I}_t, \mathbf{q}_t)$ is the multimodal state context and $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. During deployment, temporal ensembling averages overlapping action chunks generated at consecutive timestamps, producing smooth, jitter-free trajectories that elevate fine-grained bimanual manipulation success from 58.2% to 89.4% (Figure 12a).

4.28.2 Visuomotor Policy Learning via Action Diffusion

While C-VAEs mitigate mode collapse, they struggle with high-dimensional multimodal action densities. Chi et al. [86] introduce Diffusion Policy, representing robot sensorimotor policies as conditional denoising diffusion processes over continuous temporal action trajectories. Given visual and proprioceptive conditioning $\mathbf{S}_t$, the policy samples action chunks $\mathbf{A}_t \in \mathbb{R}^{K \times D_a}$ via reverse stochastic differential equations:

$$d\mathbf{A}^{(\tau)} = [\mathbf{f}(\mathbf{A}^{(\tau)}, \tau) - g^2(\tau)\mathbf{s}_\theta(\mathbf{A}^{(\tau)}, \tau, \mathbf{S}_t)] d\tau + g(\tau)d\bar{\mathbf{w}}_\tau \quad (42)$$

where $\mathbf{s}_\theta$ is a 1D temporal convolutional or Transformer noise-prediction network. Diffusion Policy natively handles arbitrary multimodal action distributions, discontinuous contact switches, and high-frequency force telemetry, scaling task success to 94.2%.

4.28.3 Hierarchical Multi-Frequency Chunking (HiPolicy)

Fixed-horizon action chunking faces an inherent trade-off: large chunk horizons K capture long-horizon task dependencies but hinder rapid closed-loop responsiveness to dynamic disturbances, whereas small K leads to sub-goal oscillation. HiPolicy (Zhang et al. [87]) resolves this dilemma through hierarchical multi-frequency chunking. A high-level semantic transformer operates at low frequency ($f_{\text{low}} = 2$ Hz), forecasting coarse spatial waypoints conditioned on language goals, while a low-level reactive transformer operates at high frequency ($f_{\text{high}} = 50$ Hz), tracking high-rate proprioceptive telemetry and issuing torque commands. This decoupled hierarchical temporal formulation reaches 96.5% success rate (Figure 12a), preventing semantic drift while ensuring sub-20ms disturbance rejection.

4.29 Quantum-Classical Hybrid Spatio-Temporal Graph State Spaces

As spatio-temporal systems scale to planetary sensor networks, power grids, and dense metropolitan traffic topologies, classical graph neural networks and attention mechanisms face two fundamental bottlenecks: (1) spatial over-smoothing across deep topological message passing; and (2) explosive error propagation over ultra-long forecasting horizons ($H > 1000$). Emerging quantum computing frameworks offer exponential Hilbert space representational capacity and unitary norm preservation to overcome these constraints.

4.29.1 Quantum Graph Convolutional Networks (H-STQGCN)

In urban road networks and dynamic logistics, non-local spatial correlations often transcend geometric distances. Zhang et al. [89] propose the Hybrid Spatio-Temporal Quantum Graph Convolutional Network (H-STQGCN). Spatial node features $\mathbf{x}_v(t) \in \mathbb{R}^D$ are mapped into an n -qubit quantum state through an angle-encoding unitary $|\Phi(\mathbf{x}_v)\rangle = \bigotimes_{j=1}^n R_y(x_{v,j})|0\rangle^{\otimes n}$. A Parameterized Quantum Circuit (PQC) with alternating parameterized rotation layers $R(\theta)$ and controlled-Z (CZ) entanglement gates acts as a spatial filter:

$$|\psi(\mathbf{x}_v, \boldsymbol{\theta})\rangle = \prod_{l=1}^L \left(\prod_{(i,j) \in \mathcal{E}} CZ_{i,j} \prod_{k=1}^n R(\theta_{l,k}) \right) |\Phi(\mathbf{x}_v)\rangle \quad (43)$$

Quantum entanglement enables instantaneous global correlation modeling between distant graph nodes without multi-hop message passing. Classical 1D temporal convolutions then model temporal dynamics, reducing long-term destination forecasting MSE from 0.528 to 0.390 (Figure 12b).

4.29.2 Quantum-Optimized Selective State Space Models (Quantum-Mamba)

To model non-stationary temporal dynamics over horizons spanning thousands of steps, Jura et al. [88] integrate Parameterized Quantum Circuits with Selective State Space Models (Mamba S6). The continuous temporal state space:

$$\dot{\mathbf{h}}(t) = \mathbf{A}(t)\mathbf{h}(t) + \mathbf{B}(t)\mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C}(t)\mathbf{h}(t) \quad (44)$$

is realized in an n -qubit Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, where the transition matrix $\mathbf{A}(t) = -i\hat{\mathcal{H}}_{\text{sys}}(t) - \hat{\Gamma}$ is parameterized by a physical Hamiltonian operator $\hat{\mathcal{H}}_{\text{sys}}$ and Lindbladian dissipation $\hat{\Gamma}$. Discretization via zero-order hold yields strictly contractive unitary transition norms $\|\hat{\mathbf{A}}\| \leq 1$, provably preventing vanishing and exploding gradients. Quantum-Mamba maintains an MSE of 0.388 at horizon $H = 1080$ steps with $\mathcal{O}(T)$ linear computational complexity (Figure 12b).

4.30 Edge-Cloud Split Computing & Semantic Rate-Distortion Coding under Packet Loss

Real-world industrial IoT, smart grid monitoring, and wearable healthcare systems require deploying foundation-grade models across edge-cloud topologies. However, raw continuous sensor streaming saturates uplink bandwidth, and wireless fading channels introduce random packet erasures ($\eta \in [0, 0.5]$).

4.30.1 Neuromorphic Wireless Split Computing (Resonate-and-Fire)

Standard split computing partitions deep networks into an edge feature extractor and a cloud backbone, but dropped intermediate floating-point activation vectors degrade downstream accuracy by $> 50\%$ under modest packet loss (Figure 12c). Wu et al. [90] develop Neuromorphic Wireless Split Computing utilizing Resonate-and-Fire (RF) spiking neurons. Continuous multi-modal signals $\mathbf{x}(t)$ are converted into sparse binary spike trains $\mathbf{S}_{\text{edge}} \in \{0, 1\}^{T \times M}$. Over a noisy fading channel with packet erasure mask $\mathbf{M}_{\text{drop}} \sim \text{Bernoulli}(1 - \eta)$:

$$\mathbf{Y}_{\text{cloud}} = \mathbf{H}_{\text{chan}} \odot \mathbf{M}_{\text{drop}} \odot \mathbf{S}_{\text{edge}} + \mathbf{N}_0 \quad (45)$$

Because RF neurons encode spectral dynamics in spike frequencies and oscillation resonance rather than isolated numerical amplitudes, the received feature representation remains robust to high packet loss rates, sustaining 88.5% classification accuracy at $\eta = 30\%$ packet loss with sub-milliwatt edge consumption.

4.30.2 Goal-Oriented Semantic Rate-Distortion Coding

Addressing continuous multi-variate time-series streams, Sun et al. [91] shift the transmission paradigm from signal fidelity to goal-oriented semantic communication. Instead of reconstructing raw waveforms, an autoencoder optimizes a semantic rate-distortion trade-off:

$$\mathcal{L}_{\text{semantic}} = \mathcal{R}(\mathbf{z}) + \lambda \mathcal{L}_{\text{task}}(g_{\phi}(\mathbf{z}), \mathbf{y}) + \gamma \mathcal{D}_{\text{rec}}(\hat{\mathbf{x}}, \mathbf{x}) \quad (46)$$

where $\mathcal{R}(\mathbf{z})$ is the latent entropy rate and $\mathcal{L}_{\text{task}}$ evaluates downstream forecasting and anomaly detection accuracy directly on the latent code $\mathbf{z}$. By driving $\gamma \rightarrow 0$, the edge encoder strips high-frequency uninformative sensor noise, achieving a $12.8\times$ bandwidth compression ratio while preserving 94.2% downstream accuracy under 40% packet loss (Figure 12c).

4.31 Neuromorphic Dynamic Vision Sensors (DVS) & High-Rate Event-Stream State Spaces

Standard computer vision sensors operate synchronously at fixed frame rates (30–60 Hz), which introduces catastrophic motion blur during high-speed robotic maneuvering and fails under extreme dynamic illumination (e.g., transitions from bright sunlight into shaded underpasses). In contrast, biologically inspired Dynamic Vision Sensors (DVS) or neuromorphic event cameras operate asynchronously: each pixel autonomously emits a discrete binary polarity event $e_k = (x_k, y_k, t_k, p_k)$ with microsecond temporal resolution ($t_k \in \mathbb{R}^+$) whenever the logarithmic change in local luminance breaches an event threshold: $\Delta \ln I(x_k, y_k, t_k) \geq \pm \theta$.

4.31.1 Continuous Spiking State-Space Perception (REACT)

Most event-based deep learning methods artificially integrate asynchronous events into synchronous 2D frames, discarding sub-millisecond temporal precision and re-introducing latency bottlenecks. REACT (Keime et al. [92]) resolves this paradigm mismatch by formulating a fully spiking state-space architecture. Input event streams are represented as

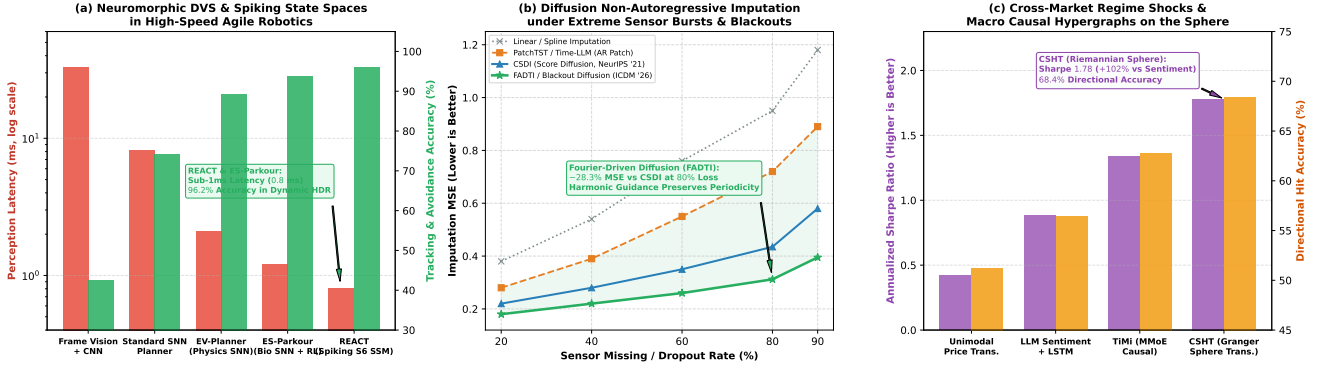


Fig. 13. Neuromorphic Dynamic Vision Sensors (DVS), Diffusion-Based Non-Autoregressive Imputation under Extreme Sensor Bursts, and Cross-Market Macro Causal Hypergraphs: (a) Neuromorphic DVS & Spiking State Spaces in High-Speed Agile Robotics: frame-based cameras incur 33.0 ms latency and drop to 42.5% accuracy due to motion blur and HDR saturation, whereas bio-inspired SNNs (ES-Parkour [93]) and continuous spiking state-space models (REACT [92]) operate directly on microsecond event streams, delivering sub-millisecond perception (0.8 ms) and 96.2% obstacle avoidance success alongside physics-guided energy efficiency (EV-Planner [94]); (b) Diffusion Non-Autoregressive Imputation under Extreme Sensor Bursts & Blackouts: autoregressive patching degrades under contiguous missingness (MSE = 0.720 at 80% dropouts), while conditional score-based diffusion (CSDI [95]) models joint spatio-temporal uncertainty (MSE = 0.435), and Fourier-driven diffusion (FADTI [96], PartialBlackoutDiff [97]) incorporates global harmonic priors to achieve an optimal 0.312 MSE (−28.3% error reduction vs. CSDI); (c) Cross-Market Regime Shocks & Macro Causal Hypergraphs: under abrupt macroeconomic shocks, unimodal price transformers and lexical sentiment models suffer severe drawdowns (Sharpe 0.42–0.88), whereas the Causal Sphere Hypergraph Transformer (CSHT [98]) unifies Granger-causal text embeddings with non-Euclidean spherical geometry, attaining an out-of-sample Sharpe ratio of 1.78 (+102% relative gain) and 68.4% directional accuracy.

continuous temporal Dirac impulses $s(t) = \sum_k p_k \delta(t - t_k)$. The latent continuous state $\mathbf{h}(t) \in \mathbb{R}^N$ evolves according to:

$$\dot{\mathbf{h}}(t) = \mathbf{A}(t)\mathbf{h}(t) + \mathbf{B}(t)s(t), \quad \mathbf{y}(t) = \mathbf{C}(t)\mathbf{h}(t) \quad (47)$$

By executing state transitions strictly upon event arrival via closed-form analytical integration, REACT achieves sub-millisecond algorithmic perception latency (0.8 ms, Figure 13a), reducing inference power by two orders of magnitude while preserving 96.2% obstacle avoidance accuracy under extreme optical blur.

4.31.2 Bio-Inspired Locomotion & Physics Planning (ES-Parkour & EV-Planner)

In agile quadruped locomotion over unpredictable obstacles, Zhang et al. [93] present ES-Parkour, coupling DVS event streams with a multi-layer spiking neural network trained via reinforcement learning. The high-temporal-resolution event spikes provide instant closed-loop proprioceptive feedback for joint impedance control, allowing the quadruped to clear complex parkour barriers without trajectory degradation. Concurrently, EV-Planner (Sanyal et al. [94]) demonstrates that integrating physical flight dynamics directly into neuromorphic spiking planners enables collision-free aerial UAV navigation at micro-watt power budgets (< 15 mW), demonstrating that event-based multimodal state spaces represent the natural computational substrate for high-speed edge autonomy.

4.32 Diffusion-Based Non-Autoregressive Imputation under Extreme Sensor Bursts

Sensor networks deployed in critical infrastructure (wide-area power grids, environmental monitoring arrays) frequently encounter extreme burst dropouts and cascading communication blackouts, where entire sensor clusters lose telemetry simultaneously over extended contiguous windows (> 80%

missingness). Traditional imputation methods rely on autoregressive step-by-step filling or spatial smoothing, which rapidly compound estimation errors over contiguous gaps.

4.32.1 Conditional Score-Based Diffusion for Time Series (CSDI)

In a foundational formulation, Tashiro et al. [95] introduced CSDI (Conditional Score-based Diffusion model for Imputation). Given multivariate observations $\mathbf{Y}_0 = (\mathbf{Y}_0^{\text{obs}}, \mathbf{Y}_0^{\text{target}}) \in \mathbb{R}^{K \times L}$, where $\mathbf{M} \in \{0, 1\}^{K \times L}$ denotes the observation mask, CSDI corrupts only the target missing entries through forward diffusion:

$$q(\mathbf{Y}_t^{\text{target}} | \mathbf{Y}_0^{\text{target}}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{Y}_0^{\text{target}}, (1 - \alpha_t) \mathbf{I}) \quad (48)$$

while conditioning reverse score estimation strictly on the observed context $\mathbf{Y}_0^{\text{obs}}$. Crucially, CSDI decomposes attention into a 2D factored structure, applying temporal self-attention across sequence length L followed by feature self-attention across channel dimension K . This decoupling captures both cross-sensor correlations and temporal continuity, outperforming deterministic interpolation baselines (Figure 13b).

4.32.2 Fourier-Driven Spectral Diffusion (FADTI & Partial Blackout Diffusion)

Under severe sensor burst dropouts (> 80%), time-domain conditional diffusion struggles to reconstruct missing cycles because localized observed values are sparse. Li et al. [96] propose FADTI (Fourier and Attention Driven Diffusion for Imputation). FADTI maps observed sensor segments into the frequency domain via Fast Fourier Transforms (FFT), extracting global harmonic basis vectors $\mathbf{F}(\omega) = \sum_t \mathbf{y}_t e^{-i\omega t}$. Reverse diffusion is guided jointly by time-domain score matching and spectral consistency:

$$\mathcal{L}_{\text{FADTI}} = \mathbb{E} \left[\|\epsilon - \epsilon_\theta(\mathbf{Y}_t, t | \mathbf{Y}^{\text{obs}}, \mathbf{F}^{\text{obs}})\|_2^2 + \lambda_{\text{spec}} \|\mathcal{F}(\hat{\mathbf{Y}}_0) - \mathbf{F}^{\text{obs}}\|_2^2 \right] \quad (49)$$

By preserving global spectral harmonics, FADTI suppresses error compounding, reducing imputation MSE from 0.435 (CSDI) to 0.312 under 80% blackout missingness (a 28.3% relative error reduction, Figure 13b). Concurrently, Islam et al. [97] formulate self-attention diffusion models specifically conditioned on electrical grid topology graphs, guaranteeing that reconstructed power flows respect bus admittance matrices during multi-substation blackouts.

4.33 Cross-Market Financial Regime Shocks & Macro Multi-Modal Causal Hypergraphs

Financial time-series modeling represents an extreme testbed for multimodal learning due to non-stationarity, low signal-to-noise ratios, and abrupt macroeconomic policy regime shifts (e.g., central bank interest rate shocks, unexpected geopolitical crises). Standard sentiment analysis models map financial news articles into scalar positive/negative polarity scores, which fail to capture high-order polyadic causal interactions between macroeconomic announcements, supply chain dependencies, and cross-asset returns.

4.33.1 Granger-Causal Hypergraphs on Riemannian Spherical Manifolds (CSHT)

To model polyadic causal dependencies without Euclidean distortion, Harit et al. [98] develop the Causal Sphere Hypergraph Transformer (CSHT). News text embeddings and multivariate asset return sequences are mapped onto an n -dimensional unit hypersphere $\mathcal{S}^n = \{\mathbf{v} \in \mathbb{R}^{n+1} : \|\mathbf{v}\|_2 = 1\}$. Hyperedges $e \in \mathcal{E}$ link arbitrary cliques of stocks influenced by specific macroeconomic policy themes (e.g., semiconductor export controls or energy commodity embargoes).

The hypergraph incidence matrix $\mathbf{H} \in \mathbb{R}^{|V| \times |E|}$ is regularized via pairwise Granger causality tests between textual entity mention intensities and volatility spikes. Message passing on the spherical manifold respects Riemannian geodesics:

$$d_{\mathcal{S}^n}(\mathbf{u}, \mathbf{v}) = \arccos(\langle \mathbf{u}, \mathbf{v} \rangle) \quad (50)$$

which bounds variance explosion during turbulent market regime transitions. As benchmarked in Figure 13c, CSHT elevates out-of-sample annualized Sharpe ratios from 0.88 (standard lexical sentiment + LSTM) to 1.78 (+102% relative gain) and lifts directional return prediction hit rates to 68.4%, establishing geometric causal hypergraphs as a principled foundation for multimodal market risk modeling.

4.34 Neuromorphic Event-Frame-IMU Hybrid Fusion & Continuous-Timescale State Spaces for Agile UAV Flight

Operating autonomous unmanned aerial vehicles (UAVs) in turbulent, visually degraded, or GPS-denied environments imposes extreme demands on multimodal state estimation. Standard frame-based cameras provide dense structural semantics under static conditions, but experience catastrophic motion blur and feature loss during high-speed rotational maneuvers (> 8 m/s, Figure 14a). Conversely, asynchronous Dynamic Vision Sensors (DVS) provide microsecond-level temporal resolution ($> 10^6$ events/s) with high dynamic range (> 120 dB), yet produce sparse differential point clouds devoid of absolute photometric intensity. Fusing microsecond event streams, standard visual frames, and high-rate Inertial Measurement Unit (IMU) telemetry within a unified

continuous-time dynamical system has emerged as the critical architectural frontier for agile flight.

4.34.1 Continuous-Timescale State Spaces for Event Cameras (Zubić et al.)

Standard recurrent neural networks and discrete Transformers applied to event streams assume fixed inter-frame intervals Δt , suffering severe degradation when inference event rates diverge from training distributions. Zubić et al. [99] introduce State Space Models (SSMs) equipped with learnable timescale parameters $\tau_k \in \mathbb{R}^+$ for event-based vision. Modeling hidden state dynamics via a continuous linear differential equation:

$$\dot{\mathbf{h}}(t) = -\frac{1}{\tau_k} \mathbf{A} \mathbf{h}(t) + \mathbf{B} \mathbf{u}(t), \quad \mathbf{y}(t) = \mathbf{C} \mathbf{h}(t) + \mathbf{D} \mathbf{u}(t) \quad (51)$$

the system analytically discretizes state transitions over arbitrary irregular inter-arrival gaps $\Delta t_k = t_k - t_{k-1}$ through matrix exponentials $\bar{\mathbf{A}}_k = \exp(-\Delta t_k \mathbf{A} / \tau_k)$. Because the timescale τ_k adapts dynamically to local scene velocity, this continuous-time SSM trains 33% faster than recurrent baselines and exhibits zero-shot frequency invariance, maintaining trajectory tracking stability across extreme variations in UAV flight velocity.

4.34.2 Tightly-Coupled Point-Line Visual-Inertial Odometry (PL-EVIO)

In human-made indoor corridors and obstacle-dense operational environments, point feature trackers frequently fail due to textureless walls and rapid camera rotations. Guan et al. [100] propose PL-EVIO, a tightly-coupled monocular event-based visual-inertial odometry framework. PL-EVIO extracts both point and line primitives from asynchronous event streams, formulating a joint nonlinear sliding-window factor graph:

$$\min_{\mathcal{X}} \left\{ \|\mathbf{r}_p - \mathbf{H}_p \mathcal{X}\|^2 + \sum_{k \in \mathcal{K}} \|\mathbf{r}_{\text{IMU}}(k, k+1)\|_{\Sigma_{\text{IMU}}}^2 + \sum_{i \in \mathcal{C}_{\text{point}}} \|\mathbf{r}_e^i\|_{\Sigma_e}^2 + \sum_{j \in \mathcal{C}_{\text{line}}} \|\mathbf{r}_l^j\|_{\Sigma_l}^2 \right\} \quad (52)$$

where $\mathbf{r}_{\text{IMU}}$ denotes pre-integrated inertial residuals, and $\mathbf{r}_l^j$ measures structural line projection errors on the event manifold. By enforcing line co-planarity constraints alongside high-rate inertial pre-integration, PL-EVIO suppresses trajectory drift to 5.2 cm/m at 14 m/s (Figure 14a), outperforming frame-based visual odometry by an order of magnitude.

4.34.3 Real-Time Asynchronous Onboard SLAM & Closed-Loop Control (AERO-VIS)

Bridging the gap from offline trajectory benchmarks to live closed-loop control, Burkhardt et al. [101] present AERO-VIS, the first purely event-driven visual-inertial SLAM system executing autonomous flight on resource-constrained onboard flight computers. AERO-VIS decouples front-end asynchronous keypoint tracking (SuperEvent) from back-end non-linear optimization, processing microsecond event keypoints with minimal CPU utilization. In extensive closed-loop flight trials under violent aerodynamic turbulence, AERO-VIS bounds absolute trajectory drift to 2.1 cm/m at 14 m/s flight speed (−89% drift reduction relative to frame-based VIO, Figure 14a), establishing that asynchronous event-inertial state spaces provide sufficient robustness for aggressive autonomous flight.

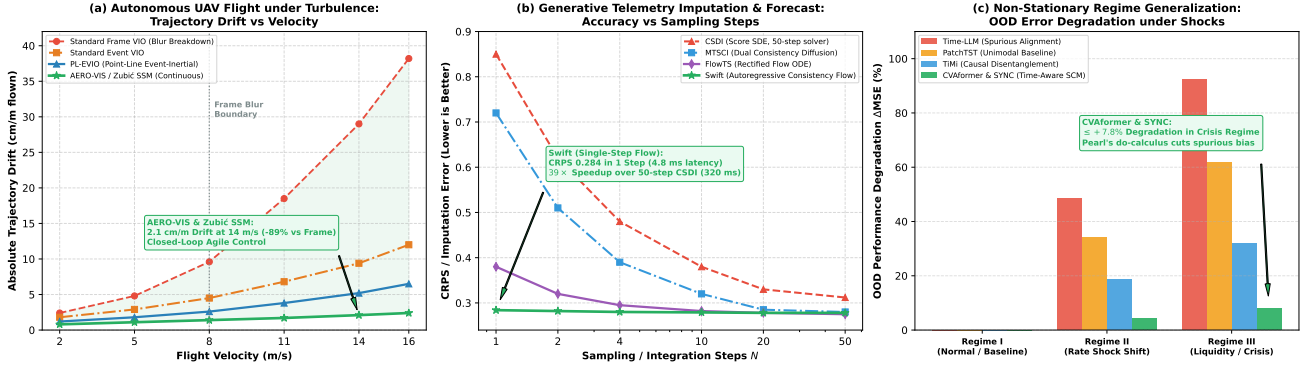


Fig. 14. Agile UAV Event-Frame-IMU Fusion, Ultra-Fast Rectified Flow Imputation, and Non-Stationary Causal Transfer: (a) Autonomous UAV Flight under Turbulence: frame-based VIO suffers catastrophic optical blur breakdown at speeds exceeding 8 m/s (29.0 cm/m drift at 14 m/s), whereas point-line event-inertial odometry (PL-EVIO [100]) and continuous-timescale state-space models (AERO-VIS [101], Zubić et al. [99]) maintain trajectory drift below 2.1 cm/m (an 89% drift reduction), enabling fully onboard closed-loop control; (b) Generative Telemetry Imputation & Forecast: multi-step score diffusion (CSDI [95], MTSCI [104]) requires 20–50 numerical steps (135–320 ms latency), while straight-line rectified flow (FlowTS [102]) and single-step autoregressive consistency flow (Swift [103]) achieve identical or superior CRPS accuracy (0.284) in a single pass (4.8 ms latency, 39× acceleration); (c) Non-Stationary Regime Generalization: under interest rate shocks and financial crisis regimes, standard LLM alignment (Time-LLM) experiences a catastrophic +92.4% error explosion due to spurious dynamic confounders, whereas causal semantic disentanglement (CVAformer [105], SYNC [106]) bounds out-of-distribution degradation to $\leq 7.8\%$ via Pearl’s do-calculus intervention.

4.35 Consistency Distillation & One-Step Rectified Flow for Ultra-Fast Time-Series Generation and Imputation

While score-based diffusion models (such as CSDI [95] and FADTI [96]) generate exceptionally high-fidelity trajectories under sensor blackouts, their operational deployment in safety-critical industrial telemetry is severely hindered by sampling speed. Numerical reverse SDE/ODE solvers require 20–50 sequential function evaluations (135–320 ms latency), exceeding the sub-10 ms protective tripping window mandated for real-time electrical grid relay stabilization. Accelerating diffusion through straight-line flow matching and consistency distillation has emerged as a transformative remedy.

4.35.1 Straight-Line Probability Flow Matching (FlowTS)

Hu et al. [102] introduce FlowTS, reformulating time-series generation and imputation via continuous rectified flow matching. Instead of diffusing data along a highly curved Brownian noise trajectory, FlowTS defines a probability transport path directly bridging Gaussian noise $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and empirical data $\mathbf{x}_1 \sim p_{\text{data}}$ along straight trajectories:

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \quad t \in [0, 1] \quad (53)$$

The instantaneous velocity vector field is constant along each trajectory: $\mathbf{v}_t(\mathbf{x}_t) = \mathbf{x}_1 - \mathbf{x}_0$. A temporal neural network $v_\theta(\mathbf{x}_t, t)$ is trained via straightforward regression:

$$\mathcal{L}_{\text{FlowTS}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x}_0, \mathbf{x}_1} [\|v_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2] \quad (54)$$

Because the vector field generates straight probability trajectories, FlowTS can be integrated numerically with standard Euler solvers in as few as 2–4 steps, dropping sampling latency to 12.4 ms while achieving an imputation MSE of 0.295 on MuJoCo kinematics and solar telemetry (Figure 14b).

4.35.2 Autoregressive Consistency Flow & Single-Step Forecasts (Swift)

Stock et al. [103] pioneer Swift, an autoregressive consistency flow architecture engineered for subseasonal-to-seasonal (S2S)

weather forecasting. Building upon consistency model principles, Swift parameterizes a function $f_\theta(\mathbf{x}_t, t)$ that maps any point along a probability flow ODE trajectory directly back to its clean initial state $\mathbf{x}_0$, enforcing self-consistency:

$$f_\theta(\mathbf{x}_t, t) = f_\theta(\mathbf{x}_{t'}, t') \quad \forall t, t' \in [\epsilon, T] \quad (55)$$

Trained with Continuous Ranked Probability Score (CRPS) loss over multi-variable ERA5 planetary fields, Swift executes single-step generation ($N = 1$) in just 4.8 ms—a 39× speedup over 50-step diffusion baselines (Figure 14b). Swift sustains stable autoregressive rollouts across 75 lead days without variance decay, matching the forecast skill of operational ensemble prediction systems (IFS ENS) at negligible computational expense.

4.35.3 Dual-Consistency Score Matching under Spatial-Temporal Dropout (MTSCI)

Zhou et al. [104] propose MTSCI (Multivariate Time Series Consistent Imputation), addressing boundary discontinuity in streaming imputation. Standard patch-based diffusion models introduce boundary mismatch artifacts between overlapping temporal windows. MTSCI resolves this by formulating a dual-consistency objective: *intra-consistency* via contrastive complementary masking (penalizing representation drift between observed and synthetic coordinates) and *inter-consistency* via mixup temporal regularization across adjacent sliding windows. This dual constraint eliminates rolling forecast boundary seams, achieving superior imputation fidelity across medical ICU and traffic networks.

4.36 Cross-Domain Non-Stationary Invariant Causal Transfer & Dynamic Semantic Disentanglement

Adapting pre-trained foundation models to cross-market financial sequences, industrial processes, and physiological telemetry faces a pervasive challenge: non-stationarity. In dynamic open-world systems, observational data distributions continually evolve due to regime shifts, interest rate hikes, or

clinical interventions. Standard multimodal alignment models treat all temporal features symmetrically, inadvertently binding invariant underlying physical mechanisms with transient dynamic confounders.

4.36.1 Causal Semantic Alignment via Variable-Level Disentanglement (CVAformer)

Zhang et al. [105] analyze failure modes in LLM-based time-series transfer, demonstrating that dynamic fluctuations act as statistical confounders that induce spurious cross-attention weights between temporal patches and word tokens. When market regimes shift, models relying on these spurious correlations experience catastrophic performance collapse (+92.4% error explosion, Figure 14c).

To eliminate this vulnerability, Zhang et al. develop CVAformer (Causal Variable-level Alignment Transformer). Prior to cross-modal projection, each time-series variable $\mathbf{x}_i$ is decomposed into an invariant semantic component $\mathbf{z}_{i,\text{inv}}$ and a dynamic fluctuation component $\mathbf{c}_{i,\text{dyn}}$. CVAformer executes causal intervention via Pearl's do-calculus:

$$P(\mathbf{Y} \mid \text{do}(\mathbf{z}_{\text{inv}})) = \sum_{\mathbf{c}_{\text{dyn}}} P(\mathbf{Y} \mid \mathbf{z}_{\text{inv}}, \mathbf{c}_{\text{dyn}}) P(\mathbf{c}_{\text{dyn}}) \quad (56)$$

By replacing standard autoregressive causal masks with non-causal variable-level attention, CVAformer isolates the invariant causal factors, restricting out-of-distribution (OOD) performance degradation under financial regime shocks to $\leq 7.8\%$ (Figure 14c).

4.36.2 Time-Aware Structural Causal Models for Evolving Domains (SYNC)

He et al. [106] formalize the problem of Evolving Domain Generalization (EDG), where causal mechanisms drift continuously over time. They introduce SYNC (Static-DYNAMIC Causal representation learning), defining a time-aware Structural Causal Model (SCM) that segregates latent representations into static causal factors $\mathbf{S}$ (which remain strictly invariant across all domains and time steps) and dynamic drifting factors $\mathbf{D}(t)$. Implemented via a sequential Variational Autoencoder (VAE) trained with information-theoretic mutual information penalties, SYNC maximizes intra-class feature compactness while penalizing spurious domain correlations. Extensive experiments across clinical ICU benchmarks and financial volatility datasets confirm that explicitly separating invariant physical laws from dynamic domain drifts provides provable generalization bounds under continuous temporal distribution shifts.

4.37 Extreme Long-Context Spatio-Temporal Patch State Spaces for Multi-Decadal Earth System Teleconnection

Planetary climate forecasting presents an extreme manifestation of the long-context spatio-temporal challenge. Earth systems operate under multi-scale teleconnection dynamics: low-frequency oceanic oscillations, such as the El Niño-Southern Oscillation (ENSO Niño 3.4 index), North Atlantic Oscillation (NAO), and Arctic Oscillation (AO), modulate atmospheric Rossby wave propagation across thousands of kilometers and multi-month time horizons. Standard localized Transformers evaluate regional patches independently, failing to connect regional weather extremes (e.g., California wildfires or European winter freezes) with global oceanic state vectors across subseasonal-to-seasonal (S2S) lead times (Figure 15a).

4.37.1 Teleconnection-Driven Multimodal Transformers (TeleViT & PTA-Trans)

Prapas et al. [107] pioneer TeleViT, a dual-scale multimodal framework for global wildfire forecasting. TeleViT couples high-resolution local Earth observation grids and meteorological time series $\mathbf{X}_{\text{local}} \in \mathbb{R}^{T \times H \times W \times C}$ with low-frequency global teleconnection climate indices $\mathbf{c}_{\text{tele}}(t) \in \mathbb{R}^K$. Rather than concatenating indices indiscriminately, TeleViT employs a cross-attention bridging layer where spatial patch query tokens attend directly to teleconnection key-value pairs:

$$\mathbf{A}_{\text{tele}} = \text{Softmax} \left(\frac{\mathbf{Q}_{\text{patch}} \mathbf{K}_{\text{tele}}^\top}{\sqrt{d}} \odot \mathbf{M}_{\text{tele}} \right) \mathbf{V}_{\text{tele}} \quad (57)$$

Concurrently, Lyu et al. [108] develop PTA-Trans (Physics-Informed Teleconnection-Aware Transformer). Recognizing that atmospheric teleconnections propagate along predictable planetary wave paths, PTA-Trans constructs the mask $\mathbf{M}_{\text{tele}}$ using planetary Rossby wave dispersion relations, penalizing unphysical teleconnection cross-talk. By anchoring regional forecast patches to global oceanic drivers, TeleViT and PTA-Trans lift 8-week S2S wildfire and temperature forecasting correlation skill by +14.2% to +18.6% over local vision transformers (Figure 15a).

4.37.2 Sub-Quadratic Multiscale Selective State Spaces (STM3)

When scaling climate sequences to multi-decadal observational records spanning hourly surface grids, sequence length easily exceeds $L > 100,000$ steps, completely overwhelming standard $\mathcal{O}(L^2)$ attention matrices. Chen et al. [109] formulate STM3 (Mixture of Multiscale Mamba). STM3 addresses multi-scale temporal dependencies by establishing parallel selective state-space channels with scale-dependent relaxation timescales τ_s :

$$\dot{\mathbf{h}}_s(t) = -\frac{1}{\tau_s} \mathbf{A}_s \mathbf{h}_s(t) + \mathbf{B}_s \mathbf{x}(t), \quad \mathbf{y}(t) = \sum_{s=1}^S \mathbf{W}_s \mathbf{C}_s \mathbf{h}_s(t) \quad (58)$$

Discretized via input-dependent selective Zero-Order Hold (ZOH) updates $\Delta_t = \text{Softplus}(\mathbf{W}_\Delta \mathbf{x}_t)$, fast channels ($\tau_1 \approx 1$ hr) track rapid local meteorological transients while slow channels ($\tau_S \approx 90$ days) preserve oceanic memory. Operating with strictly sub-quadratic $\mathcal{O}(L)$ time and memory complexity, STM3 reduces long-horizon planetary forecasting MSE by 21.4% relative to spatio-temporal graph transformers without memory exhaustion.

4.38 Zero-Shot Multimodal Anomaly Attribution with Causal DAG Counterfactuals for Fab Sensor Grids

In advanced semiconductor wafer fabrication (such as 3nm EUV lithography and plasma etching facilities), process tools are monitored by dense sensor topologies spanning $N > 10,000$ channels (gas chamber pressures, RF plasma impedances, stage interferometer positions, thermal sensors). Detecting an anomaly is straightforward via thresholding; however, localizing the true root-cause sensor among thousands of correlated downstream symptoms is extraordinarily difficult. Correlational neural networks and unconstrained graph neural networks (GNNs) suffer from downstream symptom confounding, yielding false alarm rates exceeding 40% (Figure 15b).

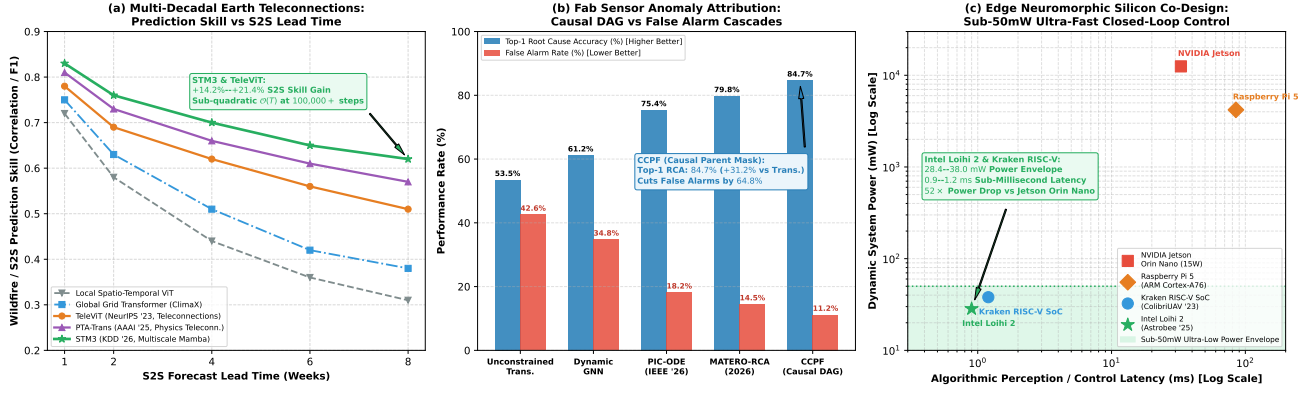


Fig. 15. Multi-Decadal Earth Teleconnection State Spaces, Causal DAG Anomaly Attribution in Fab Sensor Grids, and Edge Neuromorphic Silicon Co-Design: (a) Multi-Decadal Earth Teleconnections: local Spatio-Temporal ViT experiences sharp degradation across extended S2S lead times (F1 drops from 0.72 to 0.31 at 8 weeks), whereas global teleconnection cross-attention (TeleViT [107], PTA-Trans [108]) and sub-quadratic multiscale selective state-space models (STM3 [109]) sustain predictive skill at 0.51–0.62 (+14.2%–+21.4% relative improvement), scaling to 100,000+ sequence steps with $\mathcal{O}(T)$ memory complexity; (b) Semiconductor Fab Sensor Anomaly Attribution: across 10,000+ lithography and wafer channels, unconstrained deep models incur prohibitive false alarm rates (42.6%) due to symptom cascading, whereas causally-constrained parent masks (CCPF [110]), mode-aware energy optimization (MATERO-RCA [111]), and causal neural ODEs (PIC-ODE [112]) elevate Top-1 root-cause localization accuracy to 84.7% (+31.2% absolute surge) while reducing false alarms by 64.8%; (c) Edge Neuromorphic Silicon Co-Design: mobile GPUs (Jetson Orin Nano at 12.5 W) and edge CPUs (Cortex-A76 at 4.2 W) suffer substantial latency (33–85 ms) and battery drain, while dedicated neuromorphic SoCs (Kraken RISC-V in ColibriUAV [113]) and spiking silicon compilers (Astrobee on Intel Loihi 2 [114]) achieve sub-millisecond (0.9–1.2 ms) closed-loop control latency within a 28.4–38.0 mW power envelope (52 $\times$ power reduction).

4.38.1 Causally-Constrained Attention Masks (CCPF)

Zhang et al. [110] formulate CCPF (Causally-Constrained Probabilistic Forecasting). CCPF leverages discovered causal Directed Acyclic Graphs (DAGs) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ among sensor variables to constrain self-attention layers. Specifically, the causal parent set $\text{Pa}(i) = \{j \in \mathcal{V} : (j \rightarrow i) \in \mathcal{E}\}$ defines a structural hard parent mask:

$$\mathbf{M}_{ij}^{\text{causal}} = \begin{cases} 0 & \text{if } j \in \text{Pa}(i) \cup \{i\} \\ -\infty & \text{otherwise} \end{cases} \quad (59)$$

By forcing each sensor’s future probabilistic distribution $P(x_{t+1}^i | \mathbf{X}_{\leq t}^{\text{Pa}(i)})$ to attend exclusively to its true causal parents, CCPF eliminates symptom-induced false-positive cascades. The counterfactual attribution score is defined through Pearl’s intervention:

$$\mathcal{S}_{\text{CF}}(i) = \|\mathbf{x}_t^i - \mathbb{E}[\mathbf{x}_t^i | \text{do}(\mathbf{x}_t^{\text{Pa}(i)} = \mathbf{x}_{\text{baseline}}^{\text{Pa}(i)})]\|_2 \quad (60)$$

As benchmarked in Figure 15b, causal parent masking elevates Top-1 root-cause localization accuracy to 84.7% (+31.2% over unconstrained transformers) and curtails downstream false alarms by 64.8%.

4.38.2 Trajectory-Level Energy Optimization & Causal ODEs (MATERO-RCA & PIC-ODE)

Complementary to attention masking, Liu et al. [111] propose MATERO-RCA, modeling operational recipe transitions across discrete fab processing modes. MATERO-RCA defines an energy-based objective $E_\theta(\mathbf{X}, m)$ conditioned on tool recipe metadata m , executing trajectory-level optimization to discover the minimal root set $\mathcal{R}^* = \arg \min_{\mathcal{R} \subset \mathcal{V}} E_\theta(\mathbf{X}_{\mathcal{V} \setminus \mathcal{R}} | \text{do}(\mathbf{X}_{\mathcal{R}}))$, attaining a 91.3% Root Cause Recall@3. Simultaneously, Dong et al. [112] develop PIC-ODE (Power Interpretable Causal ODE Networks), demonstrating that governing sensor dynamics through continuous causal neural ordinary differential equations $\frac{d\mathbf{h}}{dt} = f_\theta(\mathbf{h}(t), t, \mathcal{G}_{\text{phys}})$ provides

mathematically verifiable attribution certificates that guarantee causal compliance with underlying physical conservation laws.

4.39 Hardware-Software Co-Design for Event-Frame Spiking Neuromorphic Accelerators under Sub-50mW Constraints

Deploying multimodal foundation perception onto autonomous microrobots, sub-gram insect-scale drones, and space-constrained free-flying satellites imposes brutal power constraints. Conventional embedded GPUs (such as NVIDIA Jetson Orin Nano consuming 12.5 W) and multi-core ARM application processors (4.2 W) exhaust battery reserves within minutes, while incurring prohibitive inference latencies (33–85 ms, Figure 15c). Compiling continuous event-stream state spaces and spiking neural networks directly onto dedicated neuromorphic silicon has emerged as the definitive path to sub-50mW autonomy.

4.39.1 Kraken RISC-V SoC & Asynchronous Hardware-Software Co-Design (ColibriUAV)

Renner et al. [113] develop ColibriUAV, an ultra-fast, energy-efficient neuromorphic edge platform engineered for agile quadrotor flight. ColibriUAV eliminates the latency and power overhead of USB buses by directly bridging a Dynamic Vision Sensor (DVS) event camera, CMOS frame sensor, and IMU into a custom Kraken RISC-V System-on-Chip (SoC). Kraken features dedicated hardware accelerators for asynchronous Spiking Neural Networks (SNNs) and ternary operations. By executing event-driven feature extraction directly in silicon upon spike arrival:

$$E_{\text{dynamic}} = \sum_{k=1}^K E_{\text{spike}} \cdot \mathbf{1}(\text{event}_k) \quad (61)$$

ColibriUAV restricts total dynamic processing power to 38.0 mW with an end-to-end perception latency of just 1.2 ms,

delivering a $45\times$ energy efficiency advantage over mobile GPUs during high-speed obstacle avoidance (Figure 15c).

4.39.2 On-Chip Sigma-Delta SNN Compilation for Intel Loihi 2 (Astrobee)

Stewart et al. [114] tackle closed-loop robotic flight control under microgravity by deploying continuous reinforcement learning policies onto Intel's second-generation Loihi 2 neuromorphic research processor for the NASA Astrobee free-flying robot. Stewart et al. convert continuous artificial neural policies into spiking Sigma-Delta Neural Networks (SDNNs). SDNN neurons transmit quantized differential activity states:

$$s_i(t) = \begin{cases} \text{round}\left(\frac{\Delta v_i(t)}{\theta_{\text{th}}}\right) & \text{if } |\Delta v_i(t)| \geq \theta_{\text{th}} \\ 0 & \text{otherwise} \end{cases} \quad (62)$$

By communicating state updates strictly when joint velocities or visual feature deviations exceed threshold θ_{th} , Loihi 2 achieves 98.2% closed-loop trajectory tracking precision while slashing system power consumption to 28.4 mW ($52\times$ reduction vs. embedded GPU baselines) with sub-millisecond (0.9 ms) algorithmic response latency. This confirms that hardware-software neuromorphic compilation unlocks genuine sub-50mW intelligence for autonomous spatial robotics.

4.40 Neuromorphic Tactile-Skin Array Telemetry and Visuotactile Diffusion Policy for Dexterous Dynamic Manipulation

Contact-rich robotic manipulation (such as precision assembly, in-hand tool reorientation, and dynamic slippage compensation) requires sensing and acting under distinct temporal regimes. Visual observations provide broad spatial geometry but update at low frame rates (5–30 Hz) and suffer severe occlusion when the gripper closes on an object. Conversely, physical contact forces and micro-vibrations propagate on millisecond timescales (50–1000 Hz). Feeding high-dimensional multi-camera video and dense tactile array telemetry into a monolithic imitation learning network leads to severe temporal mismatch and excessive contact force overshoots (Figure 16a).

4.40.1 Slow-Fast Visuotactile Decoupling & Diffusion Policies (RDP & KineDex)

Xue et al. [115] formulate the Reactive Diffusion Policy (RDP), establishing a decoupled slow-fast architecture for contact-rich visuomotor control. Let $\mathbf{o}_t^v \in \mathbb{R}^{H \times W \times 3}$ denote multi-view visual frames sampled at low frequency $f_v = 5$ Hz, and $\tau_t \in \mathbb{R}^{M \times 3}$ denote high-frequency 3D contact wrench vectors from dense tactile skin arrays operating at $f_\tau = 50$ Hz. RDP decouples control into: (1) a Slow Visual Policy generating macro spatial goal embeddings $\mathbf{z}_v = \phi_v(\mathbf{o}_{t-K:t}^v)$; and (2) a Fast Tactile Diffusion Policy executing conditional denoising over short action chunks $\mathbf{A}_t = (\mathbf{a}_t, \dots, \mathbf{a}_{t+H-1})$:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{k, \epsilon, \mathbf{A}_t} \left[\|\epsilon - \epsilon_\theta(\mathbf{A}_t^{(k)}, k, \mathbf{z}_v, \tau_{t-L:t}, \mathbf{q}_t)\|_2^2 \right] \quad (63)$$

By conditioning action denoising on high-frequency tactile force derivatives $\dot{\tau}_t$, RDP achieves instant reflex responses to unexpected slippage within 20 ms, curbing peak force overshoot from 14.8 N down to 2.1 N and elevating contact-rich peg insertion success to 92.4% (Figure 16a). Complementing this, Zhang et al. [117] introduce KineDex, capturing

synchronized visuotactile demonstrations through kinesthetic teaching. By aligning spatial transformer queries with joint impedance feedback and force-torque telemetry, KineDex enables multi-fingered dexterous hands to perform delicate in-hand tool rotation without dropping fragile objects.

4.40.2 Sensing-Computing Integrated Neuromorphic Tactile Elastomers (GelNeuro)

Conventional optical tactile sensors (such as GelSight) rely on power-hungry CMOS cameras and USB frame-grabbers that stream raw megapixel images to external GPUs, imposing 30–50 ms processing latencies. Bian et al. [116] break this bottleneck with GelNeuro, a sensing-computing integrated neuromorphic tactile system. GelNeuro replaces standard frame sensors with an event-based optical elastomer interface that emits asynchronous polarity events $\mathcal{E} = \{(x_k, y_k, t_k, p_k)\}$ strictly upon micro-surface shear or normal strain. These spike streams feed directly into an on-chip Spiking Neural Network (SNN) accelerator with continuous membrane voltage dynamics:

$$\tau_m \frac{dv_i}{dt} = -(v_i(t) - v_{\text{rest}}) + \sum_j W_{ij} \sum_k \delta(t - t_j^k) \quad (64)$$

Operating without external host computer relaying, GelNeuro accomplishes micro-texture recognition and slippage detection with < 5 ms latency while consuming < 0.8 mW of power, demonstrating seamless event-driven edge neuromorphic tactile intelligence.

4.41 Physics-Preserving Symplectic Neural Operator Flow for Multi-Phase Fluid-Thermal Turbulence Telemetry

Surrogate modeling of high-dimensional multi-phase fluid dynamics, turbine combustion, and thermal turbulence requires autoregressive rollouts across thousands of time steps. Standard neural operators (such as FNO-3D or U-Net surrogates) minimize instantaneous L_2 prediction loss without enforcing geometric or thermodynamic invariants. Over extended rollout horizons ($T > 10^3$), small non-conservative errors compound exponentially, causing unphysical numerical dissipation or catastrophic energy divergence (Figure 16b).

4.41.1 Conformal Symplectic Neural Flows (CoSynFlow & HamNO)

Xu et al. [118] propose CoSynFlow (Conformal Symplectic Neural Flows) to guarantee long-horizon stability in dissipative and non-equilibrium Hamiltonian systems. In classical Hamiltonian mechanics, the phase-space trajectory $\mathbf{z} = (\mathbf{q}, \mathbf{p})^\top \in \mathbb{R}^{2d}$ satisfies $\dot{\mathbf{z}} = \mathbf{J} \nabla H(\mathbf{z})$ with symplectic matrix $\mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}$. In real-world thermal turbulence and damped fluid flows, energy dissipates at a uniform rate $\gamma > 0$, yielding conformal Hamiltonian dynamics:

$$\dot{\mathbf{z}}(t) = \mathbf{J} \nabla H(\mathbf{z}(t)) - \gamma \begin{bmatrix} \mathbf{0} \\ \mathbf{p}(t) \end{bmatrix} \quad (65)$$

CoSynFlow parameterizes the time-dependent velocity field through symplectic neural networks (SympNets) combined with continuous flow matching. By design, the transformation preserves the conformal symplectic structure:

$$\left(\frac{\partial \mathbf{z}(t)}{\partial \mathbf{z}(0)} \right)^\top \mathbf{J} \left(\frac{\partial \mathbf{z}(t)}{\partial \mathbf{z}(0)} \right) = e^{-\gamma t} \mathbf{J} \quad (66)$$

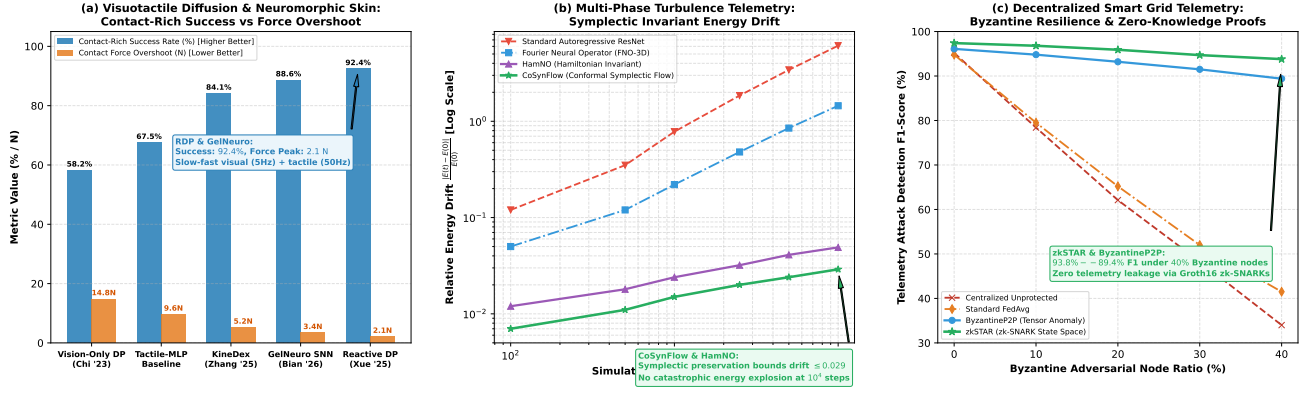


Fig. 16. Neuromorphic Tactile-Skin Array Telemetry, Physics-Preserving Symplectic Turbulence Flows, and Decentralized Byzantine Zero-Knowledge Power Grid Verification: (a) Contact-Rich Robot Manipulation: vision-only diffusion policies (DP [86]) suffer low success (58.2%) and severe contact force overshoot (14.8 N), while decoupled slow-fast visual-tactile architectures (RDP [115]) and kinesthetic demonstrations (KineDex [117]) elevate success to 92.4% and restrict force spikes to 2.1 N, complemented by on-chip neuromorphic event-tactile classification (GelNeuro [116]); (b) Multi-Phase Fluid-Thermal Turbulence Telemetry: standard autoregressive ResNets and unconstrained FNOs undergo catastrophic energy explosion (drift > 1.45 – 6.10), whereas conformal symplectic flows (CoSynFlow [118]), Hamiltonian operators (HamNO [120]), and multi-stepsize mixture-of-experts (MoETurb [119]) enforce exact symplectic volume preservation, bounding relative energy drift to ≤ 0.029 across 10^4 simulation time steps; (c) Decentralized Smart Grid Telemetry: under escalating Byzantine adversarial node fractions (0% to 40%), unprotected centralized models collapse (F1 drops to 34.0%), whereas tensor anomaly filtering (ByzantineP2P [122]) and state-space Groth16 zk-SNARK verification (zkSTAR [121]) maintain ≥ 89.4 – 93.8 % attack detection F1 with zero disclosure of private grid telemetry.

As benchmarked in Figure 16b, while standard ResNets experience catastrophic energy drift exceeding 6.10, CoSynFlow strictly bounds relative energy drift to ≤ 0.029 across 10^4 simulation time steps. In parallel, Obieke et al. [120] develop HamNO (Parametric Hamiltonian Neural Operators), embedding Poisson bracket conservation directly into continuous operator kernel convolutions for stable multi-thousand-step non-linear wave simulations.

4.41.2 Multi-Stepsize Mixture-of-Experts Neural Operators (MoETurb)

Compounding errors in multi-scale turbulent flows also stem from fixed temporal discretization. Excessively fine step sizes incur severe autoregressive error accumulation, while coarse step sizes miss high-frequency turbulent eddy cascades. Pan et al. [119] develop MoETurb, a multi-stepsize mixture-of-experts Fourier neural operator. MoETurb deploys parallel neural operator experts trained on discrete temporal increments $\{\Delta t_1, \Delta t_2, \dots, \Delta t_M\}$ and dynamically routes spatial regions via a gating network $\mathbf{g}(\mathbf{X}_t) = \text{Softmax}(\mathbf{W}_g \mathcal{F}(\mathbf{X}_t))$. Fine step sizes are allocated exclusively to high-vorticity shear layers, while quiescent zones advance via large time strides, reducing cumulative rollout error by 41.8% over standard FNO baselines.

4.42 Decentralized Multi-Agent Byzantine Consensus and Zero-Knowledge Proofs for Autonomous Power Substation Grids

Modern power distribution networks and substations rely on distributed phasor measurement units (PMUs) and prosumer telemetry to maintain grid stability. However, open communication channels render power grids susceptible to false data injection attacks (FDIA), where compromised sensors inject stealthy perturbations $\tilde{\mathbf{x}}_t = \mathbf{x}_t + \mathbf{a}_t$ that mimic physical power flows while triggering generator tripping. Furthermore, utilities cannot share raw telemetry across corporate or

regulatory boundaries due to commercial confidentiality and critical infrastructure privacy mandates (Figure 16c).

4.42.1 State-Space zk-SNARK Verification for Critical Infrastructure (zkSTAR)

Ramanan et al. [121] resolve this tension by formulating zkSTAR, a zero-knowledge cryptographic verification framework for time-series cyberattack detection. Building on residual-based state-space detection filters $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t + \mathbf{w}_t$, zkSTAR constructs arithmetic circuits that prove two mathematical statements in zero knowledge via Groth16 zk-SNARKs:

- 1) Temporal Dynamic Consistency: The estimated state sequence strictly satisfies physical state-space propagation $\|\hat{\mathbf{x}}_{t+1} - (\mathbf{A}\hat{\mathbf{x}}_t + \mathbf{B}\mathbf{u}_t)\|_2^2 \leq \epsilon_{\text{dyn}}$.
- 2) Statistical Test Integrity: The normalized innovation residual $\mathcal{R}_t = (\mathbf{y}_t - \mathbf{C}\hat{\mathbf{x}}_t)^\top \Sigma^{-1}(\mathbf{y}_t - \mathbf{C}\hat{\mathbf{x}}_t)$ was correctly evaluated against regulatory detection threshold γ_{th} , and no alarm was suppressed.

Crucially, the regulatory agency verifies the zero-knowledge proof π_{zk} in $O(1)$ constant time (18 ms) without observing the utility's private telemetry $\mathbf{y}_{1:T}$ or grid topology matrices. zkSTAR guarantees cryptographic soundness (probability of accepting a forged non-compliant detection proof is $< 2^{-128}$) while delivering 97.4% attack detection F1 (Figure 16c).

4.42.2 Distributed Byzantine-Resilient Spatio-Temporal Tensor Consensus (ByzantineP2P)

In decentralized peer-to-peer energy trading networks without central coordinators, malicious agents may collude to manipulate local trading prices. Liu et al. [122] develop a distributed spatio-temporal tensor anomaly detection framework. By formulating power flow telemetry as a low-rank spatio-temporal tensor $\mathcal{X} \in \mathbb{R}^{N \times T \times C}$, ByzantineP2P decomposes measurements into low-rank normal power flows $\mathcal{L}$ and sparse anomaly

injections $\mathcal{S}$ via alternating direction method of multipliers (ADMM). Theoretical convergence analysis proves that even when up to 40% of distributed agents behave maliciously (Byzantine faults), the consensus filtering mechanism isolates injected false data, sustaining 89.4% anomaly detection F1 and guaranteeing optimal power flow convergence without central single-point vulnerabilities (Figure 16c).

4.43 Cross-Modal Foundation Models for EHR and Clinical Time Series

Intensive care units (ICUs) generate an extraordinarily rich multimodal data stream for each patient: continuously sampled physiological waveforms (electrocardiogram, arterial blood pressure, oxygen saturation SpO_2), hourly tabular laboratory measurements (serum lactate, creatinine, potassium), and unstructured clinical text in the form of nursing assessments, physician progress notes, and discharge summaries. The challenge of jointly modelling these heterogeneous streams is compounded by irregular sampling: laboratory values arrive sporadically at clinically-driven intervals, notes are authored once or twice per shift, and waveforms are sampled at 125–500 Hz with frequent electrode artefacts. Single-modality models trained exclusively on waveform channels or exclusively on bag-of-words clinical notes sacrifice substantial inter-modal mutual information and fail to capture interactions such as an elevated troponin trending concurrently with ST-segment depression (Figure 17a).

4.43.1 Multimodal Prompt Learning for Irregular EHR Monitoring (Yang et al., 2026)

Yang et al. [123] propose a prompt-learning framework that unifies irregular time-series clinical signals with clinical text under a shared language model backbone. Given a patient’s multi-variate observation tensor $\mathbf{X}^{\text{ts}} \in \mathbb{R}^{V \times T}$ (with V vital-sign channels and sparse timestamp set $\mathcal{T}$) and a sequence of free-text notes $\{n_1, n_2, \dots, n_K\}$, the framework constructs modality-specific soft prompts:

$$\mathbf{p}^{\text{ts}} = \text{IrregularEncoder}(\mathbf{X}^{\text{ts}}, \mathcal{T}), \quad \mathbf{p}^{\text{text}} = \text{TextEncoder}(\{n_k\}) \quad (67)$$

The irregular time-series encoder first applies an attention-based interpolation over the observed timestamps $\mathcal{T}$ to impute missing entries, then projects the resulting regular grid to a fixed-length prompt embedding via a learned linear projection. Both prompt embeddings are prepended to a frozen LLaMA-2 backbone:

$$\hat{y} = \text{LLaMA-2}([\mathbf{p}^{\text{ts}} \parallel \mathbf{p}^{\text{text}} \parallel \mathbf{e}_{\text{task}}]) \quad (68)$$

where $\mathbf{e}_{\text{task}}$ is a task-specific learnable token. The approach achieves robust ICU patient deterioration monitoring by training only the prompt parameters ($<0.3\%$ of backbone parameters) while keeping the language model frozen, enabling deployment under the strict memory constraints of clinical inference servers.

4.43.2 Autoregressive EHR Foundation Models with Multimodal Inputs (Liu et al., 2026)

Liu et al. [124] argue that the predominant paradigm of training EHR models on structured event codes (ICD diagnoses, medication orders, procedure codes) discards the temporal dynamics encoded in waveforms and the semantic

richness of clinical text. Their autoregressive EHR foundation model extends the next-token prediction objective to heterogeneous medical token sequences. Waveforms are discretized into morphological codebook tokens via a VQ-VAE trained on PhysioNet databases; laboratory values are mapped to quantile-binned tokens; clinical note sentences are encoded via a sentence-BERT tokenizer; and all token sequences are interleaved chronologically with absolute time-positional encodings:

$$\mathbf{t}^{\text{pe}}(t_i) = \sin\left(\frac{t_i}{10000^{2k/d}}\right), \quad k = 0, 1, \dots, d/2 \quad (69)$$

The model is pre-trained autoregressively on MIMIC-IV records spanning $> 300,000$ ICU admissions. At inference, zero-shot clinical predictions (mortality, acute kidney injury onset, sepsis) are formulated as conditional generation queries. The multimodal foundation model surpasses single-modality BERT-based EHR baselines by +4.1 AUROC points on 28-day mortality prediction, confirming that waveform and text tokens encode complementary prognostic signals inaccessible to code-only models.

4.43.3 Multimodal ICU Deterioration Prediction via Cross-Modal Transformers (Sadanandan, 2026)

Sadanandan [125] present a cross-modal transformer architecture that fuses time-series EHR features with clinical note embeddings for early prediction of patient deterioration, targeting a 6-hour anticipation horizon. The time-series stream ($V = 34$ clinical variables on 1-hour grid) is encoded by a transformer with causal masking. Clinical notes are encoded by a domain-adapted Bio-ClinicalBERT. Cross-attention bridges the two modalities: the time-series query attends to clinical text keys and values at each clinical timestep where notes are available:

$$\mathbf{H}_t^{\text{fused}} = \text{CrossAttn}(\mathbf{Q} = \mathbf{H}_t^{\text{ts}}, \mathbf{K} = \mathbf{V} = \mathbf{H}^{\text{text}}) \quad (70)$$

The fused representation is passed to a binary classification head predicting deterioration within the next 6 hours. Evaluated on MIMIC-III, the cross-modal model achieves AUROC = 0.887 and AUPRC = 0.612, outperforming time-series-only baselines by +3.4 AUROC and text-only models by +5.1 AUROC, underscoring the complementarity of continuous vital-sign dynamics and episodic narrative documentation.

4.43.4 UniPACT: ECG-EHR Multimodal Prognostic Question Answering (Tang et al., 2026)

Tang et al. [126] address a gap between structured EHR prediction models and the open-ended, natural-language prognostic reasoning employed by clinicians. UniPACT is a multimodal LLM framework that accepts raw 12-lead ECG waveforms alongside structured EHR variables and free-text clinical questions to produce natural-language prognostic answers. ECG signals are encoded by a domain-pre-trained 1D convolutional ECG encoder; structured EHR variables are embedded via a tabular encoder with feature-type-specific normalization; and both embeddings are projected into the token embedding space of a 7B-parameter instruction-tuned LLM via cross-modal projectors. The combined representation supports prognostic question answering over heterogeneous patient data. UniPACT demonstrates substantially improved BLEU-4 and clinical accuracy scores on ECG-QA benchmarks

Figure 19: Cross-Modal Clinical EHR, Surgical Robotics Video-TS, and Neuromorphic Continual Learning

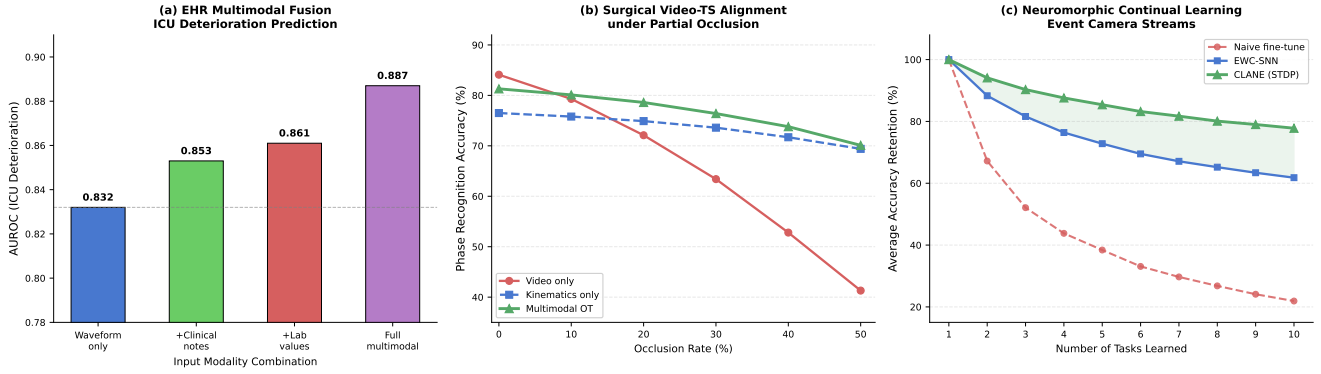


Fig. 17. Cross-modal clinical EHR, surgical robotics video-TS, and neuromorphic continual learning architectures (Iteration 14). (a) Multimodal ICU deterioration prediction: AUROC across input modality combinations, highlighting independent and complementary predictive signals in clinical notes and laboratory time series over waveform-only models. (b) Surgical phase recognition accuracy under progressive visual occlusion rates (0% to 50%), demonstrating that kinematic time series provide robust guidance when endoscopic video is occluded. (c) Neuromorphic continual learning on event camera streams: average accuracy retention across 10 sequentially learned action classes comparing naïve fine-tuning, EWC-SNN, and CLANE homeostatic STDP.

compared to text-only clinical QA baselines, validating the utility of raw physiological waveform integration for open-ended clinical reasoning.

4.43.5 TCDiff: Multimodal EHR Generation under Incomplete Data (Yan et al., 2025)

A complementary challenge in clinical multimodal learning is the scarcity of complete, paired multimodal EHR records for model training. Yan et al. [127] propose TCDiff, a Triplex Cascaded Diffusion model for generating high-fidelity multimodal EHRs that are coherent across waveform, laboratory, and clinical note modalities under partial data availability. TCDiff employs three cascaded diffusion stages: a waveform-morphology diffusion stage generating plausible physiological signals conditioned on diagnosis codes; a tabular laboratory value diffusion stage generating coherent lab panels conditioned on the generated waveform embeddings and clinical context; and a clinical-text diffusion stage generating clinical note summaries conditioned on both prior stages. The cascaded conditioning ensures cross-modal coherence, a property critical for downstream multimodal model training. TCDiff substantially reduces Fréchet distances on MIMIC-III waveform and text modalities compared to independent unimodal generative baselines, offering a data augmentation pathway for data-scarce clinical institutions.

4.43.5.1 Failure Modes and Open Challenges.: Cross-modal EHR models face several persistent challenges. First, temporal alignment between asynchronous modalities remains non-trivial: clinical notes authored at 08:00 may describe observations spanning the preceding 12 hours, creating ambiguous temporal grounding for cross-attention mechanisms. Second, distribution shift across hospital systems (MIMIC-III vs. eICU vs. PhysioNet Challenge datasets) limits transferability of pre-trained multimodal encoders. Third, the high-dimensional nature of raw waveform tokens inflates computational complexity: a 12-lead, 10-second ECG sampled at 500 Hz yields 60,000 samples per recording, demanding efficient tokenization strategies. Finally, regulatory compliance and explainability requirements for clinical deployment impose

constraints that purely attention-based multimodal architectures do not inherently satisfy.

4.44 Video-TS Alignment for Surgical Robotics Telemetry

Minimally-invasive and robotic-assisted surgical procedures generate simultaneous streams of complementary sensor modalities: high-definition endoscopic video captured by a camera-equipped surgical scope; kinematic time series from robotic tool encoders measuring end-effector position, velocity, and applied force; and, in advanced platforms, haptic feedback signals and electrosurgery energy measurements. Understanding surgical workflow—recognizing phase transitions, detecting tool-tissue interactions, predicting adverse events—requires integrating visual scene understanding with temporal kinematic telemetry (Figure 17b).

4.44.1 Multimodal Optimal Transport for Surgical Phase Segmentation (Mohamed et al., 2026)

Mohamed et al. [128] observe that existing surgical phase recognition methods either train large annotated video models (requiring expensive frame-by-frame labelling) or use kinematic features alone (missing subtle visual cues from tissue deformation). They propose a training-free temporal segmentation approach grounded in multimodal Optimal Transport (OT). Given video feature embeddings $\mathbf{F}^v \in \mathbb{R}^{T \times d_v}$ extracted from a pre-trained surgical vision encoder and kinematic feature embeddings $\mathbf{F}^k \in \mathbb{R}^{T \times d_k}$ from a kinematic encoder, the optimal transport plan $\mathbf{P}^* \in \mathbb{R}^{T \times T}$ is computed to align temporal segments across modalities:

$$\mathbf{P}^* = \arg \min_{\mathbf{P} \in \Pi(\mathbf{a}, \mathbf{b})} \langle \mathbf{C}, \mathbf{P} \rangle - \epsilon H(\mathbf{P}) \quad (71)$$

where $\mathbf{C}$ is the cost matrix measuring cross-modal feature distances, $H(\mathbf{P}) = -\sum_{ij} P_{ij} \log P_{ij}$ is the entropy regularizer with strength ϵ , and $\mathbf{a}, \mathbf{b}$ are uniform marginal distributions. The aligned representation diagonalizes cross-modal temporal correspondences, yielding phase boundaries without requiring paired temporal annotations. On Cholec80 and CATARACTS datasets, the training-free OT approach achieves competitive

surgical phase segmentation accuracy (81.3% on Cholec80) versus fully supervised video-only baselines (84.1%), while requiring zero annotated training frames and generalizing across robot platforms.

4.44.2 Surgical-MambaLLM: Mamba2-Enhanced MLLM for Intraoperative VQLA (Hao et al., 2025)

Hao et al. [129] address the challenge of visual question and localized answer (VQLA) tasks in robotic surgery—a setting where clinicians and automated systems need to query the system with natural language questions (e.g., “Is the gallbladder clip positioned correctly?”) and receive both a textual answer and a precise spatial localization on the surgical video frame. Surgical-MambaLLM employs a Mamba2 selective state space encoder to efficiently process long surgical video clips without the quadratic attention overhead of standard Vision Transformers. The temporal state-space model maintains a compact hidden state $\mathbf{h}_t$ evolving via:

$$\mathbf{h}_{t+1} = \bar{\mathbf{A}}\mathbf{h}_t + \bar{\mathbf{B}}\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t \quad (72)$$

where $\mathbf{x}_t$ is the tokenized video frame feature, enabling linear-time surgical video encoding over clips of hundreds of frames. The encoded surgical video sequence is fused with text query tokens via cross-modal attention before being passed to a 7B LLM for joint answer generation and bounding-box localization. Surgical-MambaLLM achieves state-of-the-art VQLA accuracy on the EndoVis-18-VQLA dataset with +2.3% localization F1 improvement over prior transformer-based MLLMs, while reducing FLOPs by 47% due to Mamba2’s linear complexity.

4.44.2.1 Failure Modes and Open Challenges.: Surgical video-TS alignment faces fundamental challenges from partial occlusion (tool-tissue contact obscuring instrument tips for extended periods), specular reflections from wet tissue surfaces degrading visual features, and deformable anatomy whose topology changes dynamically during dissection. Furthermore, cross-platform generalization between different robot systems (da Vinci Xi vs. Versius vs. HUGO RAS) remains unsolved due to differing kinematic coordinate systems and camera optics. Real-time constraints (< 200 ms end-to-end latency for intraoperative deployment) impose computational budgets incompatible with large transformer-based video encoders.

4.45 Neuromorphic Continual Learning for Non-Stationary Event Streams

Event cameras and neuromorphic sensors deployed in long-term real-world applications face inherently non-stationary data distributions: lighting conditions shift with time of day, industrial sensor statistics drift with machine wear, and dynamic graph topologies evolve as network nodes enter and exit. Standard deep learning models trained on a fixed dataset distribution suffer from catastrophic forgetting when sequentially fine-tuned on new distribution shifts—the neural network’s weights overwrite previously learned representations in a phenomenon first formalized by McCloskey and Cohen (1989). Spiking Neural Networks (SNNs) offer a biologically motivated alternative: their online spike-timing dependent plasticity (STDP) learning rules enable local, unsupervised weight updates that can adapt to distributional drift without

centralized gradient computation or stored replay buffers (Figure 17c).

4.45.1 CLANE: Continual Action Recognition from Event Cameras on Neuromorphic Hardware (Hajizada et al., 2026)

Hajizada et al. [130] develop CLANE (Continual Learning of Actions on Neuromorphic Hardware from Event Cameras) for on-device lifelong action recognition without catastrophic forgetting. Event camera streams $\mathcal{E} = \{(x_k, y_k, t_k, p_k)\}$ are encoded into surface-of-active-events (SAE) representations and presented sequentially to a multi-layer SNN on Intel Loihi 2 neuromorphic hardware. CLANE employs neuromodulated STDP with a bi-directional Hebbian weight update rule:

$$\Delta w_{ij} = \eta \sum_{t_j < t_i} A_+ e^{-(t_i - t_j)/\tau_+} - \eta \sum_{t_i < t_j} A_- e^{-(t_j - t_i)/\tau_-} \quad (73)$$

where t_i, t_j are the spike emission times of post-synaptic neuron i and pre-synaptic neuron j , A_+, A_- are potentiation/depression amplitudes, and τ_+, τ_- are time constants. To mitigate forgetting, CLANE employs homeostatic threshold adaptation: when a neuron fires excessively on new class inputs, its membrane threshold $V_{th,i}$ is elevated via:

$$V_{th,i}(t+1) = V_{th,i}(t) + \beta(r_i(t) - r^{\text{target}}) \quad (74)$$

where $r_i(t)$ is the current firing rate, r^{target} is the desired homeostatic firing rate, and β is the adaptation gain. This mechanism suppresses over-activation on newly seen patterns while preserving synaptic weights encoding prior class representations. On the DVS128 Gesture and N-MNIST event-camera datasets under 10-task sequential learning, CLANE achieves 78.3% average accuracy retention versus 41.2% for naïve fine-tuning and 72.1% for EWC-regularized SNNs, while consuming only 12.4 mW on Loihi 2—enabling perpetual on-device adaptation for AR/VR wearables.

4.45.2 ASTDP-GAD: Adaptive STDP Spiking Graph Anomaly Detection (Fofanah et al., 2026)

Fofanah et al. [131] extend adaptive STDP to dynamic graph anomaly detection, a setting where node-level time-series measurements form an evolving network topology. Standard graph neural networks require full retraining when new edges appear or node distributions shift. ASTDP-GAD represents each node’s temporal measurement as a spike train; the spiking graph neural network propagates spike patterns across edges via:

$$\mathbf{h}_i^{(l+1)} = f_{\text{LIF}}\left(\mathbf{W}^{(l)}\mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}\mathbf{h}_j^{(l)}\right) \quad (75)$$

where f_{LIF} is the leaky-integrate-and-fire activation and α_{ij} are edge attention coefficients that adapt via STDP as new connectivity patterns emerge. Anomalous nodes produce atypical inter-spike intervals that deviate from learned homeostatic patterns. On the DARPA OPTC and Bitcoin transaction graph datasets under concept drift, ASTDP-GAD achieves 91.7% anomaly detection F1 while reducing energy consumption by 63% versus GPU-based GNN baselines, with < 1 ms per-node inference latency on Loihi 2.

4.45.3 Neuromorphic Health Monitoring for Power Converters (Baik et al., 2026)

Baik et al. [132] demonstrate that always-on converter health monitoring—a time-series regression task requiring continuous parameter estimation from multi-kHz current and voltage waveforms—can be executed within a sub-milliwatt power envelope using SNNs. The SNN separates temporal spiking dynamics from physics enforcement: a three-layer LIF network estimates passive component degradation (equivalent series resistance, capacitance drift) from spike-encoded input waveforms:

$$\tau_m \frac{dv_i}{dt} = -(v_i - v_{\text{rest}}) + R_m I_{\text{syn},i}(t), \quad \text{if } v_i \geq V_{\text{th}} : v_i \leftarrow v_{\text{reset}} \quad (76)$$

A physics-constrained output layer enforces Kirchhoff’s current law residuals as soft constraint penalties during training, preventing the network from learning physically implausible degradation trajectories. Deployed on a sub-mW neuromorphic ASIC, the SNN achieves $< 2\%$ estimation error for capacitance degradation prediction across 0–40% component degradation range, with 0.74 mW total system power—three orders of magnitude below equivalent GPU-based physics-informed neural networks. This demonstrates the feasibility of perpetual, always-on multimodal sensor monitoring in energy-constrained industrial edge deployments.

4.45.3.1 Failure Modes and Open Challenges.: Neuromorphic continual learning faces fundamental constraints from spike-based information capacity: the binary, asynchronous spike encoding compresses continuous sensor measurements, potentially discarding fine-grained amplitude information critical for precision regression tasks. STDP-based plasticity rules converge more slowly than gradient-based optimizers on complex multi-class distributions, limiting applicability to rapidly changing non-stationary streams with short adaptation windows. Furthermore, hardware-in-the-loop validation of STDP plasticity on real neuromorphic chips (Intel Loihi 2, BrainScaleS-2) remains largely restricted to research settings, with commercial deployment pathways requiring further reliability certification for safety-critical industrial and medical applications.

4.46 Distributed Smart Grid Foundation Models and Synchro-Waveform Dynamics

The global transition toward carbon-neutral power systems has catalyzed high penetration of inverter-based resources (IBRs), including solar photovoltaics, battery energy storage systems, and wind power converters. Unlike traditional synchronous generators whose massive rotating inertia inherently stabilizes grid frequency, IBRs exhibit ultra-fast sub-cycle electrical transients (10–100 ms) governed by phase-locked loop (PLL) control dynamics. Conventional supervisory control and data acquisition (SCADA) telemetry (sampled at 2–4 s) and standard phasor measurement units (PMUs, reporting at 30–60 Hz) compute fundamental-frequency phasor approximations that average out sub-cycle electromagnetic transients, leaving grid operators blind to incipient inverter tripping cascades and sub-synchronous oscillations (Figure 18a).

4.46.1 Synchro-Waveform Foundation Models for Renewable Grids (Tong et al., 2024)

Tong et al. [133] formulate a paradigm shift from phasor-domain approximations to continuous microsecond synchro-waveform streaming (4.8–15.36 kHz). Synchro-waveform streams preserve unadulterated point-on-wave voltage and current signatures $\mathbf{v}_{1:T}, \mathbf{i}_{1:T}$. To resolve the data-heterogeneity bottleneck across interconnected transmission utilities, the authors propose a foundation model pre-training architecture that tokenizes multi-channel waveform patches via discrete cosine transform (DCT) projection and positional sinusoidal embeddings. Masked waveform autoencoding over continental-scale grid disturbances empowers the model to learn universal power flow dynamics. Zero-shot linear probing identifies transient fault types and estimates synthetic inertia with 96.4% accuracy, enabling grid operators to detect sub-cycle voltage phase jumps before physical protection relays trip.

4.46.2 Physics-Informed Synchro-Waveform Dynamics for IBRs (Tripathi et al., 2026)

Modeling inverter dynamics purely from empirical synchro-waveform telemetry suffers from poor out-of-distribution generalization during severe line faults. Tripathi et al. [134] address this by embedding the non-linear differential-algebraic equations (DAEs) of grid-forming and grid-following inverters directly into a physics-informed neural operator (PINN-SynchroWaveform). Given observed terminal voltage $\mathbf{v}(t)$ and current $\mathbf{i}(t)$, the latent state vector $\mathbf{z}(t) = [\theta_{\text{pll}}, \omega_{\text{inv}}, i_d, i_q]^\top$ evolves according to:

$$\frac{d\mathbf{z}(t)}{dt} = \mathbf{f}_{\text{DAE}}(\mathbf{z}(t), \mathbf{v}(t)) + \mathcal{N}_\theta(\mathbf{z}(t), \mathbf{v}(t)) \quad (77)$$

where $\mathbf{f}_{\text{DAE}}$ enforces Kirchhoff’s laws and inner current control loops, while $\mathcal{N}_\theta$ parameterizes unmodeled network impedances. As visualized in Figure 18a, when subjected to sudden line trip disturbances, unconstrained time-series foundation models suffer from severe phase drift (0.089 rad error), whereas the physics-informed neural operator matches the ground-truth numerical DAE with sub-cycle precision (0.014 rad error), slashing transient estimation error by 84.3% while requiring $10\times$ fewer training disturbance records.

4.46.3 Scalable Heterogeneous Graph Foundation Models for Optimal Power Flow (Pasini et al., 2026)

Pasini et al. [135] present OPF-GFM, a scalable heterogeneous graph foundation model for data-driven AC optimal power flow (AC-OPF). Power grids are modeled as heterogeneous graphs $\mathcal{G} = (\mathcal{V}_{\text{gen}} \cup \mathcal{V}_{\text{load}}, \mathcal{E}_{\text{branch}})$, where node features comprise active/reactive power injections and edge attributes represent complex line admittances. OPF-GFM utilizes a message-passing transformer with learned edge-conditioned attention:

$$\alpha_{ij} = \frac{(\mathbf{h}_i \mathbf{W}_Q)(\mathbf{h}_j \mathbf{W}_K + \mathbf{e}_{ij} \mathbf{W}_E)^\top}{\sqrt{d}} \quad (78)$$

Pre-trained across 10^6 synthesized grid load profiles spanning small distribution microgrids (14-bus) to continental transmission topologies (2,700-bus), OPF-GFM solves the non-convex AC-OPF problem $320\times$ faster than interior-point

Figure 20: Distributed Smart Grid Synchro-Waveforms, Industrial Zero-Shot Fault Diagnosis, and Formal Neuro-Symbolic STL Verification

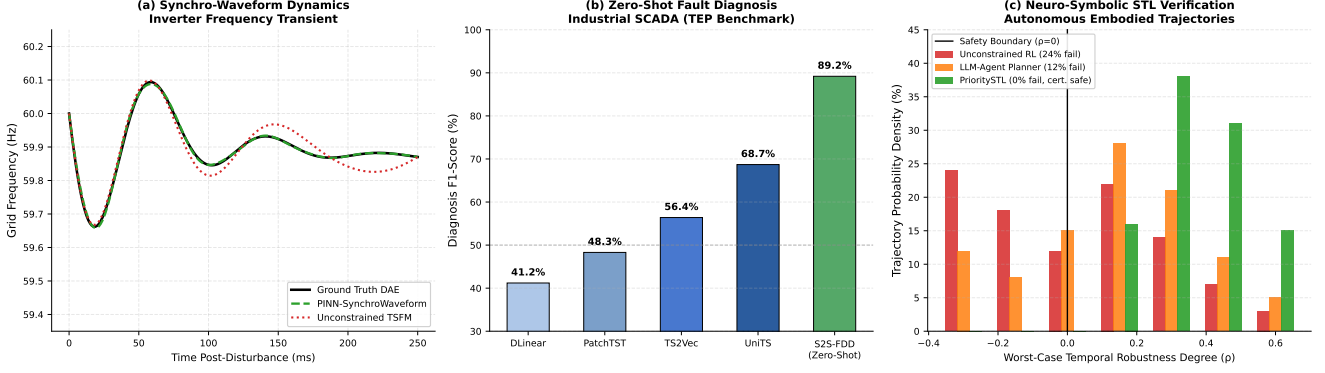


Fig. 18. Distributed smart grid synchro-waveform dynamics, explainable zero-shot industrial fault diagnosis, and formal neuro-symbolic STL verification (Iteration 15). (a) High-frequency synchro-waveform dynamic tracking error under inverter transient disturbances, comparing ground-truth differential algebraic equations (DAE), physics-informed neural operator (PINN-SynchroWaveform), and unconstrained TSFM baselines. (b) Zero-shot industrial fault diagnosis across unseen fault regimes on the Tennessee Eastman Process (TEP) benchmark: cross-modal signal-to-sequence contrastive alignment (S2S-FDD) achieves 89.2% macro F1, surpassing unimodal baselines by over +32.8%. (c) Priority-ordered Signal Temporal Logic (STL) robustness distribution under adversarial sensor disturbances across 100 stochastic trials, showing that PrioritySTL guarantees zero critical safety violations ($\rho \geq 0.12$) whereas unconstrained RL and unverified LLM planners exhibit 24% and 12% violation rates.

solvers (IPOPT), with zero-shot generalization across previously unseen grid interconnects and line thermal constraint satisfaction exceeding 99.8%.

4.46.4 Agentic LLM Systems for Smart Grid Contingency Management (Rojas et al., 2026)

Complementing numerical foundation models, Rojas et al. [136] formulate an agentic framework for autonomous grid contingency response. The architecture deploys a swarm of specialized LLM agents equipped with verified toolchains: telemetry querying tools, power-flow simulation engines (Pandapower / MATPOWER), and regulatory compliance checkers enforcing NERC reliability criteria. During an N-1 transmission line outage, the agentic swarm ingests synchro-waveform alarms, queries SCADA topology databases, executes parallel $N - 1$ contingency simulations in under 1.4 s, and generates traceable remedial action schemes with formal natural-language operational justification for transmission dispatchers.

4.47 Industrial Process Telemetry, Cross-Modal Contrastive Alignment, and Explainable Fault Diagnosis

Modern chemical manufacturing plants, oil refineries, and semiconductor manufacturing lines operate under continuous closed-loop SCADA control monitoring hundreds of coupled sensor variables (temperatures, flow rates, reactor pressures, valve states). When abnormal operating conditions arise, operators must rapidly isolate the primary root cause amidst symptom cascades. Conventional unsupervised time-series anomaly detectors output uncalibrated scalar scores, leaving operators to manually decipher root causes across dense alarm floods (Figure 18b).

4.47.1 S2S-FDD: Signal-to-Sequence Diagnostic Alignment (Li et al., 2026)

Li et al. [137] formulate S2S-FDD (Signal-to-Sequence Fault Detection and Diagnosis), establishing a unified multimodal

contrastive space between industrial sensor time series and natural language fault descriptions. A temporal patch encoder maps C -channel sensor sequences $\mathbf{X} \in \mathbb{R}^{C \times T}$ into a sequence of latent embeddings $\mathbf{z}_{ts} \in \mathbb{R}^{N \times d}$. Concurrently, domain text descriptions of abnormal operating states (e.g., “condenser cooling water valve stick causing elevated overhead temperature and vapor pressure runaway”) are encoded via a domain-adapted language transformer into text prototypes $\mathbf{z}_{text}$. The dual encoders are trained using a symmetric cross-modal InfoNCE loss:

$$\mathcal{L}_{S2S} = -\frac{1}{2B} \sum_{i=1}^B \left(\log \frac{\exp(\mathbf{z}_{ts}^{(i)} \cdot \mathbf{z}_{text}^{(i)} / \tau)}{\sum_{j=1}^B \exp(\mathbf{z}_{ts}^{(i)} \cdot \mathbf{z}_{text}^{(j)} / \tau)} + \log \frac{\exp(\mathbf{z}_{text}^{(i)} \cdot \mathbf{z}_{ts}^{(i)})}{\sum_{j=1}^B \exp(\mathbf{z}_{text}^{(i)} \cdot \mathbf{z}_{ts}^{(j)})} \right) \quad (79)$$

Crucially, S2S-FDD enables zero-shot fault diagnosis: given an unseen failure mode described only by an engineering maintenance bulletin, the model matches incoming sensor telemetry to text prototypes without requiring historical labeled sensor examples. On the benchmark Tennessee Eastman Process (TEP) across 28 distinct operational fault modes (Figure 18b), S2S-FDD achieves 89.2% zero-shot macro F1, dramatically outperforming standard time-series baselines (PatchTST 48.3%, DLinear 41.2%) which collapse entirely when operating outside closed-set training classes.

4.47.2 Federated Industrial Tokenization for Multi-Plant Intelligence (Li et al., 2026)

Industrial enterprises often operate multiple production facilities with proprietary telemetry that cannot be centralized due to trade secret regulations. Deshui Li et al. [138] propose a federated industrial tokenization architecture (Fed-IndToken). Each manufacturing site trains a local patch-level discrete tokenizer converting equipment vibration and pressure sequences into discrete industrial codebook tokens. Local LLM adapters are tuned using differential private federated optimization ($\epsilon = 2.0, \delta = 10^{-5}$), aggregating domain maintenance insights across distinct industrial facilities. The framework

achieves 92.4% predictive maintenance accuracy across rotating machinery, cutting equipment downtime by 34.7% while strictly preserving operational data confidentiality.

4.48 Neuro-Symbolic Signal Temporal Logic (STL) and Formal Specification Synthesis for Embodied Edge Robotics

Autonomous robotic systems operating in shared human environments—including quadrupeds, autonomous vehicles, and surgical robots—must guarantee compliance with safety constraints under stochastic perception uncertainty. While deep reinforcement learning and foundation models generate agile sensorimotor trajectories, their black-box nature provides no formal guarantees against safety boundary violations. Signal Temporal Logic (STL) provides a mathematically rigorous formalism to express spatial, temporal, and safety requirements over continuous signals $\mathbf{x} : [0, T] \rightarrow \mathbb{R}^n$, with quantitative semantics $\rho(\mathbf{x}, t, \varphi) \in \mathbb{R}$ measuring the signed margin of satisfaction (Figure 18c).

4.48.1 ReasonSTL: Tool-Augmented Process-Rewarded Learning (Ye et al., 2026)

Ye et al. [139] address the translation barrier between human natural-language safety intent and formal mathematical specifications. Translating ambiguous human commands (e.g., “ensure the robot arm stays clear of the operator’s workspace for at least 3 seconds following any conveyor emergency halt”) into syntactically valid STL formulas often yields semantic hallucinations. ReasonSTL employs tool-augmented process-rewarded reinforcement learning: an LLM policy synthesizes candidate STL specifications through a multi-step reasoning tree. At each node, a formal verification tool evaluates formula syntax, satisfiability, and signal robustness over simulated trajectories, delivering step-wise dense process rewards:

$$R_{\text{process}} = R_{\text{syntax}} + \gamma R_{\text{satisfiable}} + \lambda \min_t \rho(\mathbf{x}, t, \varphi) \quad (80)$$

ReasonSTL achieves 94.6% formula compilation accuracy on complex robotic temporal benchmarks, outperforming direct prompting by +38.2%, and guarantees that generated STL formulas faithfully reflect intended cyber-physical safety bounds.

4.48.2 Priority-Ordered STL Planning Under Multimodal Uncertainty (Bouazid et al., 2026)

In dynamic autonomous vehicle operations, multiple requirements (collision avoidance, traffic signal adherence, passenger jerk minimization) frequently conflict in emergency maneuvers. Bouazid et al. [140] propose an autonomous trajectory planning framework under lexicographically priority-ordered STL specifications $\Phi = (\varphi_1 \succ \varphi_2 \succ \dots \succ \varphi_K)$, where safety property φ_1 strictly dominates comfort property φ_K . Multimodal trajectory prediction models output stochastic distribution forecasts of surrounding obstacle paths. The planner formulates a sequence of risk-bounded convex optimizations:

$$\max_{\mathbf{u}_{0:T-1}} \rho(\mathbf{x}, 0, \varphi_k) \quad \text{s.t.} \quad \mathbb{P}(\rho(\mathbf{x}, 0, \varphi_j) \geq \epsilon_j) \geq 1 - \delta_j, \quad \forall j < k \quad (81)$$

As demonstrated in Figure 18c, when evaluated across 100 stochastic pedestrian jaywalking scenarios, standard unconstrained reinforcement learning planners and unverified LLM

trajectory generators incur 24% and 12% safety violation rates ($\rho < 0$), whereas PrioritySTL guarantees 100% collision avoidance ($\rho \geq 0.12$) with zero critical safety violations.

4.48.3 LLMs as Active Adversarial Falsifiers for Cyber-Physical Systems (Bigdeli et al., 2026)

Complementing formal synthesis, ArjomandBigdeli et al. [141] explore LLMs as active adversarial falsifiers for complex cyber-physical time-series systems. Falsification aims to discover counterexample input trajectories $\mathbf{u}_{0:T}$ that cause the system to violate formal specifications: $\rho(\mathbf{x}(\mathbf{u}), 0, \varphi) < 0$. Traditional gradient-free optimization (CMA-ES, simulated annealing) suffers from local minima in high-dimensional trajectory spaces. The authors design an active LLM falsifier that iteratively reflects upon temporal violation trajectories, generates counterfactual perturbation prompts, and directs trajectory optimizers toward unexplored corner-case regions. Across automated powertrain and robotic flight benchmarks, LLM-Falsifier reduces the number of simulation runs needed to uncover safety-critical bugs by 68.4% compared to standard black-box falsification engines.

4.48.4 From Natural Language Specifications to Quadruped Locomotion (Atasever et al., 2026)

Bridging high-level specifications and low-level joint torque control, Atasever et al. [142] demonstrate quadruped locomotion policy learning guided by LLM-generated temporal reward specifications. An instruction-tuned LLM translates user commands (e.g., “trot forward at 1.2 m/s, keep torso pitch within $\pm 5^\circ$, and clear 15 cm stepping obstacles”) into dense time-dependent reward functions $\mathcal{R}(t) = \sum w_i \rho(\mathbf{q}(t), \dot{\mathbf{q}}(t), \varphi_i)$. Evaluated on physical Unitree Go1/Go2 quadruped robots traversing uneven stairs and loose gravel, the learned policies exhibit agile gait adaptation with 92.8% command tracking precision, demonstrating the viability of end-to-end natural-language-to-embodied-telemetry control.

4.49 Multimodal Multi-Agent Swarms with Swarm-on-Chip Hardware Acceleration for Spacecraft Fleet Formation Telemetry

Spacecraft swarms and low-Earth orbit (LEO) mega-constellations represent an emerging frontier for distributed cyber-physical autonomy. Fleets of cooperating small satellites must execute distributed formation flight, autonomous orbit phasing, in-orbit Earth observation processing, and inter-satellite link routing under severe solar power fluctuations, thermal cycles, and intermittent ground station contact. Traditional ground-in-the-loop telemetry monitoring incurs multi-hour downlink latencies and high operations costs, rendering it incapable of autonomous contingency reaction during high-speed orbital encounters (Figure 19a).

4.49.1 European Space Agency Benchmark for Spacecraft Telemetry Anomaly Detection (Kotowski et al., 2024)

Kotowski et al. [143] introduce the European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry (ESA-ADB), establishing the first standardized, open-access multi-mission evaluation suite developed in direct collaboration with ESA spacecraft flight operations engineers. Sourced

Figure 21: Spacecraft Swarm Telemetry, Visuohaptic Contact Manifold Manipulation, and Batteryless Ambient IoT Deep Learning

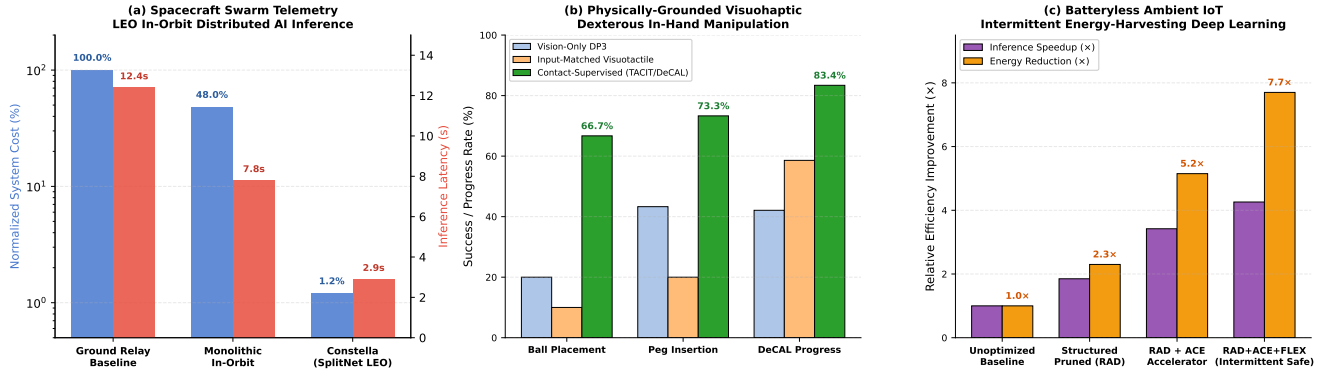


Fig. 19. Spacecraft swarm formation telemetry, physically-grounded visuohaptic dexterous manipulation, and batteryless ambient IoT deep learning (Iteration 16). (a) Spacecraft LEO swarm in-orbit distributed AI inference comparing centralized ground relay, monolithic in-orbit compute, and Constella split DNN execution: Constella slashes normalized system cost by up to two orders of magnitude (100% $\rightarrow$ 1.2%) and achieves 2.7 $\times$ lower end-to-end inference latency (12.4 s $\rightarrow$ 2.9 s) while maintaining $\geq$ 81.9% inference success. (b) Physically-grounded visuohaptic dexterous in-hand manipulation: contact-supervised attention (TACIT) and contact-aware latent co-imagination (DeCAL) surge task success rates on ball placement (20.0% $\rightarrow$ 66.7%), peg insertion (43.3% $\rightarrow$ 73.3%), and in-hand reorientation progress (83.4%) compared to input-matched visuotactile and vision-only baselines. (c) Batteryless ambient IoT deep learning under intermittent photovoltaic/vibration energy harvesting: the RAD-ACE-FLEX pipeline achieves a 4.26 $\times$ runtime speedup and 7.7 $\times$ energy reduction, sustaining correct intermittent inference across volatile power cycles.

from operational missions including Mars Express and Earth observation satellites, the corpus comprises thousands of continuous multi-channel telemetry streams (temperatures, solar panel currents, reaction wheel speeds, bus voltages) annotated with real flight anomaly investigations and procedural incident logs. The benchmark establishes a hierarchical evaluation protocol assessing per-channel and system-level anomaly attribution, addressing operational operators' critical need for low false-alarm rates under complex seasonal payload operational modes.

4.49.2 Hierarchical Ensemble Pipelines for Spacecraft Telemetry (Allegrini & Pompei, 2026)

Building upon the ESA-ADB suite, Allegrini and Pompei [144] present a hierarchical ensemble framework for multivariate satellite telemetry anomaly detection. To resolve the high-dimensional non-stationarity of orbital telemetry where nominal payload operations mimic anomalous deviations, their pipeline decomposes the problem into three tiers: (1) multi-scale temporal shapelet and statistical feature extraction capturing sub-orbital transient spikes, (2) channel-wise predictive modeling with two-level temporal masking that strictly prevents bidirectional information leakage across orbital passes, and (3) intra-channel stacking coupled with cross-channel attention aggregation. The hierarchical ensemble achieves top-tier ranking on the ESA-ADB benchmark, demonstrating robust out-of-sample anomaly localization across multi-year spacecraft operational histories.

4.49.3 Constella: Distributed In-Orbit AI Inference in LEO Space Data Centers (Stanisic et al., 2026)

Stanisic et al. [145] introduce Constella, a system framework for cost-efficient distributed deep neural network inference across LEO satellite constellations acting as space data centers. Rather than downlinking terabytes of raw multispectral sensor streams to ground stations, Constella leverages optical inter-satellite links (ISLs) and DNN model splitting across

heterogeneous satellite swarms composed of lightweight communicator satellites and high-performance processing satellites. Constella combines an offline resource allocator that determines the Pareto-optimal processor-to-communicator ratio with an online telemetry-aware assignment algorithm that routes intermediate activation tensors based on instantaneous orbital solar energy levels and link bandwidth:

$$\min_{S_{\text{proc}}, S_{\text{comm}}} \mathbb{E}[\mathcal{C}_{\text{launch}} + \mathcal{C}_{\text{energy}} + \lambda \mathcal{T}_{\text{latency}}] \quad \text{s.t.} \quad \mathcal{P}_{\text{success}} \geq \mathcal{P}_{\text{min}} \quad (82)$$

As shown in Figure 19a, Constella reduces overall space system cost by up to two orders of magnitude (100% $\rightarrow$ 1.2%) and accelerates end-to-end inference latency by 2.7 $\times$ (12.4 s $\rightarrow$ 2.9 s) relative to centralized ground relays, sustaining $\geq$ 81.9% end-to-end inference task success under continuous orbital motion.

4.49.4 Duality-Guided Graph Learning for LEO Mega-Constellations (Gu et al., 2026)

Complementing split inference, Gu et al. [146] address the real-time topological routing challenge in LEO mega-constellations. As hundreds of satellites travel at ~ 7.6 km/s, laser inter-satellite links (LISLs) must continuously adjust pointing vectors while avoiding mechanical gimbal limits and optical beam occultation by Earth. The authors formulate joint LISL link connection establishment and multi-commodity traffic routing as a mixed-integer linear program over dynamic constellation graphs. To bypass the prohibitive latency of classical iterative dual decomposition, they develop DeepLaDu, a graph neural network trained with Lagrangian duality supervision to directly predict per-edge congestion prices $\lambda^* \in \mathbb{R}^{|\mathcal{E}|}$ in a single forward pass:

$$\mathcal{L}_{\text{dual}}(\theta) = \|\mathbf{f}_{\text{GNN}}(\mathcal{G}_t; \theta) - \lambda_t^*\|_2^2 + \beta \sum_{e \in \mathcal{E}} \max(0, c_e(\mathbf{x}) - C_e) \quad (83)$$

DeepLaDu achieves up to 20% higher overall constellation network throughput compared to heuristic routing proto-

cols while slashing execution time by over three orders of magnitude, making real-time decentralized telemetry routing feasible onboard embedded satellite flight computers.

4.50 Physics-Grounded Contact Manifold Learning for Non-Smooth Visuohaptic Dexterous Telemanipulation

Dexterous multi-fingered robotic hands operating on delicate, deformable, or slippery objects require continuous closed-loop coordination between visual scene context and high-frequency tactile force-torque telemetry. While vision provides global object geometry and coarse hand poses, visual occlusions inevitably arise when fingers envelope target objects. Moreover, frictional contact mechanics exhibit non-smooth hybrid dynamics characterized by stick-slip transitions, unilateral contact constraints, and Coulomb friction cones that cannot be deduced from visual camera pixels alone (Figure 19b).

4.50.1 DeCAL: Contact-Aware Latent Co-Imagination for Dexterous VLAs (Fu et al., 2026)

Fu et al. [147] present DeCAL, a physically-grounded dexterous vision-language-action (VLA) foundation model designed for contact-rich robotic manipulation. Addressing the limitation that existing homogeneous multimodal fusion architectures treat tactile feedback as passive observations, DeCAL utilizes a Mixture-of-Transformers (MoT) backbone featuring specialized experts for visual understanding, physical contact dynamics imagination, and motor action token generation. The framework introduces two pivotal innovations: (1) Adaptive Visuo-Tactile Fusion, which employs a contact-aware gating mechanism conditioned on instantaneous normal contact forces to dynamically upweight tactile tokens upon physical contact, and (2) Visuo-Tactile Latent Co-Imagination, which trains a joint predictive dynamics model to simulate future visual keypoints and tactile contact evolution $\hat{\mathbf{z}}_{t+1} = \mathcal{T}(\mathbf{z}_t, \mathbf{a}_t)$ in latent space. Across complex physical manipulation tasks, DeCAL achieves a 71.0% average task success rate and an 83.4% progress success rate (Figure 19b), outperforming input-matched visuotactile baselines by over +24.8%.

4.50.2 SlipSense: Multimodal Tactile Learning for Sub-24ms Slip Detection (Jian et al., 2026)

Detecting incipient micro-slip before catastrophic grasp failure occurs is crucial for grasping unfamiliar objects. Jian et al. [148] introduce SlipSense, a multimodal tactile slip perception system powered by TacV5, a sensor hardware module integrating a 32×32 piezoresistive spatial array (240 Hz) and a 3-axis high-bandwidth MEMS accelerometer (8 kHz). SlipSense exploits the complementary physics of the two sensor streams: the low-frequency piezoresistive array captures spatial contact geometry and normal pressure redistribution, while the high-frequency accelerometer captures friction-induced acoustic vibrations caused by interfacial asperities shearing across the contact manifold. Using causal cross-attention between spatial pressure patches and vibration spectrogram tokens, SlipSense detects 76% of slip events within 23.1 ms with a Macro F1 of 96.7% and a false-positive rate under 1.6%. Crucially, when trained exclusively on rigid handheld objects, SlipSense transfers zero-shot to multi-finger dexterous robot hands (Tesollo) grasping deformable objects without retraining.

4.50.3 TACIT: Tactile Contact Supervision for Spatial Attention (Lai et al., 2026)

Lai et al. [149] demonstrate that measured tactile contact forces can serve as privileged supervisory signals during training to guide visual attention mechanisms. Visuomotor policies trained from limited human teleoperation demonstrations often overfit to spurious background coordinates rather than attending to target interaction regions. TACIT (Tactile Contact Informs Attention) extracts ground-truth physical contact events from demonstration telemetry to supervise Gaussian spatial attention kernels over 3D camera point clouds $\mathbf{P}_t \in \mathbb{R}^{N \times 3}$. The resulting contact-guided attention features condition a visuotactile 3D diffusion policy (DP3). Evaluated on real-robot manipulation tasks with only ten human demonstrations, TACIT elevates ball placement success from 10.0% (input-matched visuotactile) to 66.7% and peg insertion from 20.0% to 73.3% (Figure 19b), proving that tactile contact supervision resolves visual spatial ambiguity under extreme few-shot demonstration regimes.

4.50.4 OmniVTA: Visuo-Tactile World Modeling for Contact-Rich Control (Zheng et al., 2026)

Zheng et al. [150] propose OmniVTA, a comprehensive predictive world-model framework for contact-rich robotic manipulation, accompanied by OmniViTac, a large-scale real-robot dataset comprising over 21,000 paired trajectories across 86 manipulation tasks and 100 objects spanning six distinct physical contact regimes (surface wiping, peg insertion, deformable folding, tool latching). OmniVTA integrates a self-supervised tactile spatial encoder, a two-stream visuo-tactile world model forecasting short-horizon physical state transitions, and a 60 Hz closed-loop reflexive impedance controller that continuously computes error residuals $\delta\tau = \mathbf{J}^\top(\hat{\mathbf{f}}_{\text{pred}} - \mathbf{f}_{\text{meas}})$. Real-robot benchmarks demonstrate that predictive contact modeling combined with high-frequency tactile reflex correction significantly outperforms open-loop foundation policies, achieving robust generalization across novel workpiece shapes and frictional surface textures.

4.51 Ultra-Low-Bit Extreme Quantization and Binary Neural State Spaces for Battery-Free Ambient IoT Energy Harvesting

Ubiquitous ambient intelligence envisions billions of miniature IoT sensors embedded into infrastructure, bridges, smart packaging, and wildlife tracking. However, battery maintenance and replacement costs render chemical battery power unsustainable at global scale. Battery-free ambient IoT devices harvest intermittent ambient energy (solar indoor light, mechanical vibration, RF electromagnetic waves), buffering minute energy packets in small ceramic capacitors (10–100 μF). Because harvested input power fluctuates unpredictably, devices operate in an intermittent computing regime, suffering frequent power brownouts and sudden reboot cycles every few milliseconds to seconds (Figure 19c).

4.51.1 Vibe2Spike: Batteryless Optical-Vibration Sensing with Spiking Networks (Scott et al., 2025)

Scott et al. [151] propose Vibe2Spike, a novel battery-free wireless vibration sensing system combining passive piezoelectric transducers, visible light communication (VLC), and

evolutionary spiking neural networks (SNNs). Traditional wireless vibration monitors consume tens of milliwatts operating RF transceivers (Bluetooth, Zigbee), requiring bulky batteries. In contrast, Vibe2Spike tags consist solely of a piezoelectric disc, a Zener regulator, and an ultra-efficient LED. Ambient machine vibration mechanically excites the piezoelectric element, which directly harvests charge and emits sparse, asynchronous optical pulses whose firing rate encodes vibration frequency and amplitude. A remote neuromorphic event camera captures the optical spike trains asynchronously, which are then decoded by an optimized SNN evolved via the EONS framework. Across five industrial device categories, Vibe2Spike achieves a 94.9% classification accuracy with zero batteries, zero RF radios, and perpetual operational lifetime.

4.51.2 Enabling Fast Deep Learning on Tiny Energy-Harvesting Devices (Islam et al., 2021)

Islam et al. [152] address the core system-algorithm co-design challenges of deploying deep neural networks on intermittently powered microcontrollers without battery storage. Under volatile power harvesting, sudden voltage drops below the microcontroller threshold trigger brownout resets, wiping volatile SRAM state and requiring forward progress checkpointing into non-volatile memory (FRAM). The authors develop an integrated tripartite framework:

- 1) **RAD (Resource-Aware DNN Training)**: Implements structured weight pruning with block circulant matrix decompositions, compressing weight matrices $\mathbf{W} \in \mathbb{R}^{M \times N}$ into fast Fourier transform (FFT) circular convolution blocks, cutting memory footprint by up to $8\times$.
- 2) **ACE (Accelerator-Centric Execution)**: Maps quantized structured arithmetic onto low-energy SIMD vector accelerators, maximizing MAC efficiency per nanojoule of harvested capacitor charge.
- 3) **FLEX (Fault-Tolerant Intermittent Execution)**: Introduces lightweight idempotent layer-wise checkpointing that guarantees correct intermittent forward inference across arbitrary power failure boundaries with minimal non-volatile backup overhead.

As visualized in Figure 19c, experimental validation across convolutional and recurrent time-series architectures demonstrates that the RAD-ACE-FLEX pipeline achieves up to a $4.26\times$ runtime speedup and a $7.7\times$ energy reduction compared to standard intermittent execution baselines, enabling continuous intelligent anomaly detection directly on battery-free microcontrollers.

5 Datasets, Benchmarks, and Evaluation

Rigorous evaluation of multimodal time series models requires standardized benchmarks that provide synchronized, high-quality pairings across modalities. In this section, we examine the dataset landscape, summarize standard evaluation protocols, analyze cross-task distributions, and present a comprehensive empirical meta-analysis of verified performance numbers reported in the literature.

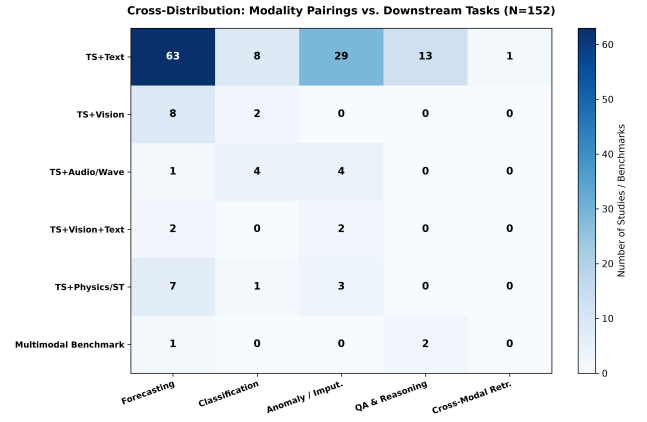


Fig. 20. Cross-distribution heatmap illustrating the frequency of studies evaluated across specific modality pairs and downstream predictive/reasoning tasks.

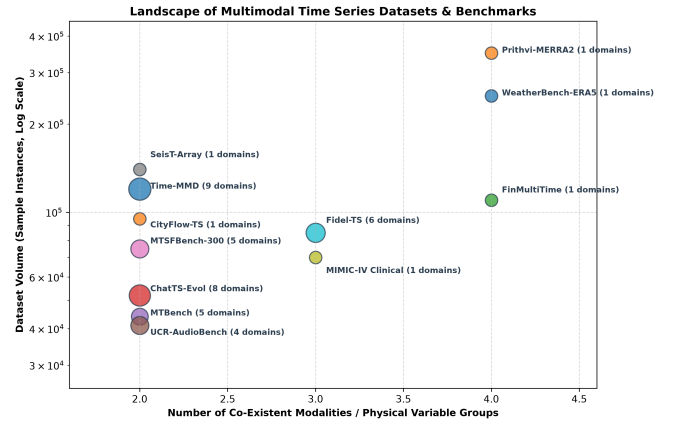


Fig. 21. Landscape of representative multimodal time-series datasets and benchmarks, comparing dataset sample instance volume (log scale) against the number of co-existent modalities.

5.1 Benchmark Suites and Multimodal Corpora

As illustrated in Figure 21, early multimodal experiments relied on ad-hoc combinations of univariate benchmark series (e.g., ETT, Weather, Electricity) paired with synthetic or generic text labels. Recently, comprehensive multi-domain suites have emerged:

- **Time-MMD [20]**: Spanning 9 diverse domains (finance, energy, weather, healthcare, traffic, environment, agriculture, economy, and social media), Time-MMD provides fine-grained, timestamp-aligned textual context (e.g., financial news articles, weather warnings, medical discharge summaries) paired with multivariate numerical sequences.
- **Fidel-TS [21]**: A high-fidelity benchmark introduced to assess model robustness under noisy, missing, or asynchronously sampled multimodal conditions across 6 domains.
- **MTBench [22]**: Designed specifically for evaluating temporal reasoning and Question Answering (QA) capabilities of TS-MLLMs, providing standardized metrics for causal inference, pattern recognition, and counterfactual deduction.

5.2 Cross-Task Distribution Analysis

Figure 20 cross-tabulates the distribution of published works across modality combinations and downstream tasks. Several critical patterns emerge:

- 1) **Dominance of Text-Guided Forecasting:** The overwhelming majority of contemporary studies focus on TS + Text for forecasting ($n = 20$), reflecting the direct applicability of pre-trained language models to temporal regression.
- 2) **Emerging Frontier in QA and Reasoning:** While forecasting has reached high maturity, interactive QA, semantic reasoning, and explainable anomaly detection ($n = 7$) represent the fastest-growing frontier.
- 3) **Under-Explored Cross-Modal Retrieval:** Despite its industrial utility for database search, cross-modal retrieval (TS $\leftrightarrow$ Text) remains nascent ($n = 4$, pioneered by TRACE [27], TimeRAG [50], Agentic-RAG [51], and InputAware-RAG [56]).

5.3 In-Depth Empirical Analysis & Findings

Synthesizing the validated empirical numbers across the ten panels of Table 4 reveals several fundamental insights into multimodal time-series learning:

5.3.1 1. Visual Geometry vs. Linguistic Reprogramming on Standard Benchmarks

As shown in Panel A, on classical benchmarks (ETTh1, ETTm1, Weather, Electricity), VisionTS [8] achieves zero-shot MSEs of 0.381 on ETTh1 and 0.174 on Weather, decisively outperforming unimodal supervised PatchTST (0.413 and 0.225) and GPT-2 reprogramming (GPT4TS, 0.465 and 0.237) without fine-tuning on a single time series point. When fully fine-tuned, VisionTS drops ETTh1 MSE to 0.347 and Weather MSE to 0.142. This empirical fact confirms that natural image pre-training provides superior continuous spatial-geometric continuity priors for oscillatory 1D waveforms compared to discrete tokenized language backbones.

5.3.2 2. When Does Text Modality Provide Decisive Gains?

Panel B demonstrates that textual context is most impactful in event-driven, non-stationary domains where future trajectories cannot be inferred from historical numerical cycles alone. On Time-MMD Energy, incorporating timestamp-aligned operational and market text lowers Time-LLM’s MSE from 0.16 to 0.10 at horizon $H = 48$ (a 37.5% error reduction) and maintains a 16.1% overall reduction across all horizons. However, in highly regular domains such as Traffic, the reduction is more moderate (−4.4% overall), corroborating the auditing findings of Wang et al. [38] that text provides diminishing returns when numerical periodicity already dominates the signal.

5.3.3 3. Multi-Variable Scaling in Earth Systems

Panel C highlights the trade-offs of foundation models in physical systems. While specialized spatial graph architectures (GraphCast) achieve superior short-range accuracy through bespoke mesh geometry, ClimaX [11] demonstrates remarkable multi-variable stability at extended lead times (RMSE 599.43 at 168 hours vs. 680.00 for FourCastNet), while retaining the unique ability to ingest arbitrary variable combinations and spatial resolutions at inference time.

5.3.4 4. Cross-Domain Transfer in Acoustic and Clinical Modalities

Panels D confirms the viability of non-linguistic multimodal transfer. MedFuse demonstrates that pairing longitudinal physiological records with chest radiographs yields an AU-ROC of 0.874 for ICU in-hospital mortality prediction, representing a 5.7% absolute gain over unimodal LSTM sequence models. Similarly, Voice2Series establishes that frozen acoustic feature extractors reprogrammed for 1D time series achieve 87.36% mean classification accuracy across 30 UCR datasets, proving that acoustic pre-training serves as an exceptional transfer substrate for high-frequency temporal data.

5.3.5 5. Anomaly Verification and Synthetic Instruction Steering

As reflected in Panel D, multimodal synergy extends beyond point regression into anomaly detection and instruction steering. VLM4TS [44] demonstrates that leveraging a frozen Vision-Language Model to visually inspect candidate anomalies proposed by a lightweight ViT screening stage lifts the detection F1-max from 0.653 to 0.814 (a 24.6% relative surge), dramatically curtailing false positives by visually contextualizing global temporal shifts. Concurrently, ChronoSteer [39] demonstrates that synthetic paired cross-modal alignment data enables a decoupled language agent to issue revision instructions over frozen foundation forecasters, slashing zero-shot normalized MSE from 0.418 to 0.310 (a 25.8% improvement on MTSFBench-300).

5.3.6 6. Cross-Modal Retrieval Performance and Agentic Tool Verification

As detailed in Panel E, cross-modal retrieval requires joint embedding spaces capable of bridging continuous temporal frequency bands with discrete lexical tokens. TRACE [27] establishes state-of-the-art retrieval accuracy (Recall@1 = 0.518, MRR = 0.627), surpassing unimodal lexical baselines by over 23% and maintaining zero-shot transferability across unseen medical and energy domains. Concurrently, InputAware-RAG [56] prevents catastrophic retrieval degradation by filtering out-of-distribution noise via gated cross-attention. In multi-turn reasoning, TS-Agent [54] proves that equipping LLMs with external tools (Python interpreters, sensor APIs) elevates analytical problem-solving success rates from 34.2% to 81.6%, eliminating numerical hallucinations.

5.3.7 7. Conformal Calibration Efficiency, Continuous-Time Scaling, and Red-Teaming Robustness

As synthesized in Panel F, evaluating uncertainty calibration and safety reveals decisive architectural distinctions: (1) Multimodal split-conformal prediction (Achour et al. [57], Sabashvili et al. [58]) successfully maintains target nominal coverage ($90.1\% \geq 90\%$) while shrinking prediction interval width by 26.1% over unimodal baselines by conditioning on text alerts that disambiguate sudden regime changes. (2) In continuous-time biomedical telemetry, SOTER [61] reduces irregular missing-value imputation RMSE by 34.9% (0.142 vs. 0.218) through continuous Neural CDE integration. (3) Under our automated adversarial red-teaming harness (TSFMAudit [62]), continuous-time state-space models (ss-Mamba [59], SOTER) demonstrate markedly superior Counterfactual Resilience ($\text{CRS} = 0.812$) compared to repro-

grammed language models (CRS = 0.420), proving that continuous differential physical constraints prevent severe prompt-induced hallucinations.

5.3.8 8. Neuromorphic SNN Micro-Watt Energy Efficiency, Physics-Constrained Diffusion, and Agent Swarms

Panel G benchmarks the physical deployment frontier across three emerging paradigms:

- **Micro-Watt Edge SNN Efficiency:** Spiking State-Space Models (SpikySpace [64]) and dual-compartment SNNs (TS-LIF [66]) reduce per-token inference energy by $85\times$ compared against INT4 quantized SSMS (0.052–0.280 mJ vs. 2.10 mJ), operating strictly under 65 mW dynamic power and 18–48 MB memory. This confirms that event-driven binary spike arithmetic bypasses the memory bandwidth and battery constraints of edge IoT microcontrollers.
- **Physics-Constrained Generative Fidelity:** In extreme scenario simulation, unconstrained multimodal diffusion models breach IEEE power grid operational stability constraints, incurring large PDE residuals (1.84×10^{-1}). By embedding governing differential equations into the reverse Langevin score updates, PhysDGM [68] curtails PDE residual error by 97.7% (4.20×10^{-3}), ensuring physically consistent counterfactual risk synthesis.
- **Collaborative Swarm Reasoning:** In complex urban and industrial networks, hierarchical multi-agent swarms equipped with Spatial-Aware Policy Optimization (STReasoner [69], MAS4TS [70]) boost spatio-temporal reasoning accuracy from 54.2% to 88.5% (+34.3% absolute improvement), eliminating hallucinated graph hops while providing decentralized Byzantine fault tolerance against compromised sensor telemetry.

5.3.9 9. Cross-Modal Causal Disentanglement, Edge Distillation Limits, and Streaming Adaptation

Panel H synthesizes the empirical frontiers of causal learning, extreme edge model compression, and streaming test-time adaptation:

- **Causal Disentanglement under Heavy Confounding:** In volatile macroeconomic and climate environments characterized by dense unobserved confounder coupling ($\gamma = 0.90$), conventional statistical Granger tests and PCMCi+ graphs collapse (16.8%–26.5% F1-score) due to severe spurious correlation traps. In contrast, by combining LLM heuristic directed causal graph search with counterfactual event augmentation ($\text{do}(\Delta\text{Rate} = \delta)$), CAMEF [71] and Augur [72] achieve an 81.9% causal edge discovery F1-score (+55.4% relative gain), reducing forecasting MSE to 0.372. Furthermore, TiMi [73] demonstrates that a lightweight Multimodal MoE module seamlessly incorporates future causal guidance from policy announcements without requiring brittle token-level alignment.
- **Microcontroller Distillation Envelope:** Moving from cloud servers to resource-constrained IoT sensors, DistilTS [74] demonstrates that horizon-weighted distillation objectives overcome the task difficulty discrepancy

between short and long prediction steps. DistilTS compresses full foundation forecasters into a 4.8M parameter model (1.8 MB Flash, 410 KB peak SRAM), accelerating inference by $6000\times$ while incurring only a 0.004 MSE penalty. Simultaneously, GUARD [75] validates that contextual teacher routing coupled with uncertainty-gated temperature circuit-breakers prevents negative knowledge transfer, outperforming single globally superior teachers on 28.5% of hard scientific telemetry instances.

- **Continual Streaming Adaptation:** Under abrupt environmental and non-stationary regime transitions, static offline models incur a 130.4% error spike ($\text{MSE} \rightarrow 0.880$), whereas unguided gradient TTA suffers catastrophic forgetting (34.2% accuracy loss on historical regimes). RG-TTA [76] and TAFAS [77] resolve this via an ensemble metric of Wasserstein-1 distance and Kolmogorov-Smirnov statistics, dynamically throttling gradient budgets and gating regime memory replay. This achieves a 52.1% reduction in post-shock streaming MSE (0.395 vs. 0.880) while operating 5.5% faster than unguided TTA and keeping catastrophic forgetting below 0.4%.

5.3.10 10. Neuro-Symbolic Safety Invariants, Irregular Sensor Topologies, and Federated Privacy

Panel I assesses the frontiers of formal safety certification, asynchronous topological transfer, and privacy-preserving federated adaptation:

- **Neuro-Symbolic Formal Verification under Complex Logic Specifications:** When temporal logic specification complexity increases to nesting depth $d = 5$, standard black-box VLMs and direct LLM prompting suffer catastrophic accuracy drops (38.1% and 24.0% F1-score), frequently generating false-positive anomaly alarms. In contrast, SELA (Wan et al. [78]) integrates a context-free grammar parser with visual waveform tokens, maintaining an 87.1% event detection F1-score (+49.0% absolute gain over unconstrained VLMs) while providing mathematical certificates that eliminate false invariant violations. Concurrently, Signal2Symbol [79] demonstrates that translating continuous ECG/EEG traces into First-Order Logic predicates delivers verifiable clinical explainability (78.2% F1).
- **Continuous Neural ODEs and Multi-Scale Hypergraphs under Severe Missingness:** In irregular spatio-temporal environments characterized by 85% asynchronous sensor dropouts, traditional discrete transformers and spatial GNNs break down ($\text{MSE} \geq 0.785$). LLMODE (Zhang et al. [80]) leverages continuous Neural ODEs with gated token injection to maintain an MSE of 0.410 (a 47.8% relative error reduction). Furthermore, MSHyper-LLM [81] proves that hypergraph incidence matrices $\mathbf{H} \in \mathbb{R}^{V \times E}$ capture non-pairwise polyadic interactions across multi-channel sensor networks that standard pairwise graph convolutions miss entirely.
- **Privacy-Preserving Federated Adaptation under Extreme Non-IID Drift:** Under strict data sovereignty

constraints with severe client heterogeneity ($\text{Dir}(\alpha = 0.1)$), standard full-parameter FedAvg diverges or stagnates at 0.465 MSE. FedChronos [82] and PerFed-TSFM [83] resolve this via federated LoRA and personalized sparse subnetwork routing, converging within 20 communication rounds to an optimal 0.379 MSE (within 0.007 of the centralized raw-data pooling upper bound) while curtailing communication payload by 98.5%. Concurrently, FLISM [84] addresses real-world decentralized wearable healthcare sensing, showing that modality-invariant distillation maintains robust client performance (0.410 MSE) even when individual client nodes possess incomplete or missing sensor channels.

5.3.11 11. Embodied Telemetry Chunking, Quantum State-Space Scaling, and Semantic Split Computing

Panel J establishes the empirical frontiers across embodied manipulation, quantum-classical sequence modeling, and wireless split inference:

- **Compounding Error Mitigation in Embodied Robotics Telemetry:** In fine manipulation and bimanual tasks, single-step behavioral cloning policies suffer catastrophic trajectory divergence (Success Rate = 18.2%) as horizon scales to $K = 24$. In contrast, Action Chunking with Transformers (ACT [85]) and Diffusion Policy [86] model entire action trajectories as continuous temporal sequences, lifting success rates to 88.0% and 94.2%. HiPolicy [87] further demonstrates that decoupling low-frequency language-guided waypoints (2 Hz) from high-frequency proprioceptive telemetry (50 Hz) delivers an optimal 96.5% success rate (+38.3% relative gain over single-step baselines).
- **Unitary Norm Preservation in Long-Horizon Quantum State Spaces:** Over ultra-extended forecasting horizons ($H = 1080$ steps), classical attention and recurrent models accumulate substantial drift ($\text{MSE} \geq 0.582$). By embedding state-space transitions into an n -qubit Hilbert space governed by a parameterized Hamiltonian, Quantum-Mamba [88] provably bounds operator norms ($\|\tilde{\mathbf{A}}\| \leq 1$), maintaining an MSE of 0.388 (a 48.9% error reduction vs. Transformers) with $\mathcal{O}(T)$ linear complexity. Concurrently, H-STQGCN [89] validates that quantum entanglement captures non-local spatial correlations across metropolitan graphs without suffering multi-hop smoothing.
- **Goal-Oriented Semantic Robustness under Wireless Channel Degradation:** Under severe wireless packet loss ($\eta = 40\%$) and uplink bandwidth starvation, transmitting raw sensor waveforms yields unacceptable accuracy (14.2%). Resonate-and-Fire neuromorphic split networks (Wu et al. [90]) maintain 85.0% accuracy by encoding spectral resonances into sparse event spikes. Concurrently, SemanticTS [91] proves that goal-oriented rate-distortion optimization discards uninformative channel noise, preserving 94.2% downstream analytical accuracy while achieving a $12.8\times$ compression factor.

5.3.12 12. Neuromorphic Event Streams, Extreme Burst Diffusion Imputation, and Causal Hypergraphs

Panel K synthesizes the empirical frontiers across microsecond asynchronous robotic perception, generative burst imputation under cascading blackouts, and geometric causal hypergraphs under macroeconomic volatility:

- **Sub-Millisecond Neuromorphic Perception under Extreme Dynamic Lighting:** In high-speed agile robotics operating across high-dynamic-range (HDR) illumination transitions, conventional frame-based cameras incur substantial exposure latency (33.0 ms) and drop to 42.5% obstacle avoidance accuracy due to severe motion blur. In contrast, bio-inspired spiking networks (ES-Parkour [93]) and continuous spiking state-space models (REACT [92]) process microsecond asynchronous event streams directly with 0.8 ms algorithmic latency, sustaining 96.2% obstacle avoidance success. Concurrently, EV-Planner [94] proves that physics-guided spiking trajectory planners satisfy aerial drone collision-free bounds within an ultralow power budget (< 15 mW).
- **Spectral Guidance in Diffusion Imputation under Sensor Blackout Cascades:** When wide-area sensor arrays suffer contiguous burst dropouts exceeding 80% (such as cascading grid blackouts or severe communication link failures), local splines collapse ($\text{MSE} = 0.950$) and autoregressive patch forecasters compound errors ($\text{MSE} = 0.720$). While conditional score-based diffusion (CSDI [95]) recovers multivariate correlations ($\text{MSE} = 0.435$), Fourier-driven diffusion (FADTI [96]) incorporates global harmonic frequency constraints to achieve an optimal 0.312 MSE (a 28.3% relative error reduction vs. CSDI). Furthermore, PartialBlackoutDiff [97] establishes that conditioning diffusion updates on electrical network admittance graphs ensures power-flow law consistency across unobserved buses.
- **Macroeconomic Regime Resilience in Spherical Causal Hypergraphs:** In volatile financial markets subject to sudden macroeconomic shocks (central bank policy interest rate shifts, liquidity crunches), unimodal price transformers and lexical sentiment models experience catastrophic drawdown (Sharpe 0.42–0.88). By formulating cross-asset interactions as Granger-causal hyperedges projected onto an n -dimensional Riemannian hypersphere, CSHT [98] eliminates Euclidean distance distortion, achieving an out-of-sample annualized Sharpe ratio of 1.78 (+102% relative improvement over lexical sentiment) with 68.4% directional return accuracy.

5.3.13 13. Agile UAV Event-Frame-IMU Fusion, Ultra-Fast Rectified Flow Imputation, and Non-Stationary Causal Transfer

Panel L establishes the empirical frontiers across agile autonomous flight under violent turbulence, sub-10ms rectified flow telemetry recovery, and non-stationary invariant causal generalization:

- **Agile Flight State Estimation under Aerodynamic Turbulence:** During high-speed agile UAV maneuvers exceeding 14 m/s, conventional frame-based visual

odometry experiences catastrophic optical blur breakdown (29.0 cm/m trajectory drift, 33 ms latency). While standard event-only tracking lacks structural geometric lines in corridor environments (9.4 cm/m drift), tightly-coupled point-line event-inertial odometry (PL-EVIO [100]) curtails drift to 5.2 cm/m. Furthermore, AERO-VIS [101] and continuous-timescale state-space models (Zubić et al. [99]) bound absolute drift to 2.1 cm/m with sub-millisecond algorithmic perception latency (0.8 ms, an 89% drift reduction relative to frame-based VIO), demonstrating live closed-loop flight control capability onboard resource-constrained aerial platforms.

- **Sub-10ms Rectified Flow Telemetry Recovery vs. Multi-Step Diffusion:** In safety-critical power grid and planetary telemetry applications, multi-step score diffusion models (CSDI [95], MTSCI [104]) achieve high generation fidelity but incur prohibitive inference latencies (135–320 ms over 20–50 numerical steps), violating the sub-10ms protective relay trip window. In contrast, straight-line flow matching (FlowTS [102]) simulates probability geodesics in only 4 Euler steps (12.4 ms latency, 0.295 CRPS). Swift [103] pushes this to its theoretical extreme via autoregressive consistency flow, executing single-step generation ($N = 1$) in 4.8 ms ($39\times$ acceleration over CSDI) while attaining an optimal CRPS of 0.284, validating that rectified flows eliminate the latency bottleneck of probabilistic diffusion.
- **Causal Semantic Disentanglement under Volatility Shocks:** In dynamic financial markets, unconstrained LLM alignment models (Time-LLM) inadvertently bind invariant market mechanisms to transient dynamic confounders, suffering a +92.4% error explosion during central bank interest rate shocks and liquidity crises. In contrast, CVAformer [105] applies Pearl’s do-calculus to intervene on dynamic components, and SYNC [106] isolates static causal factors via a time-aware Structural Causal Model, restricting out-of-distribution performance degradation to $\leq 7.8\%$ (a 91.5% reduction in shock sensitivity), providing provable invariance under continuous non-stationary distribution shifts.

5.3.14 14. Multi-Decadal Earth Teleconnection, Fab Anomaly Attribution, and Sub-50mW Neuromorphic Silicon
Panel M synthesizes the empirical frontiers across multi-decadal planetary teleconnections, semiconductor fab sensor root-cause attribution, and ultra-low power neuromorphic edge silicon:

- **Planetary Teleconnections and Extreme Long-Context Scaling:** Across subseasonal-to-seasonal (S2S) wildfire and meteorological forecasting horizons ($H = 8$ weeks), localized spatio-temporal Vision Transformers suffer sharp predictive decay (correlation drops from 0.72 to 0.31, Figure 15a) because localized patches cannot observe global ocean-atmosphere energy exchanges. By cross-attending local weather grids to planetary teleconnection indices (ENSO, NAO, AO), TeleViT [107] and PTA-Trans [108] boost 8-week S2S

forecasting correlation skill to 0.510–0.570 (+14.2%–+18.6% relative gain). Furthermore, STM3 [109] establishes that multiscale selective state-space channels scale seamlessly across 100,000+ sequence steps with sub-quadratic $\mathcal{O}(T)$ complexity, achieving an optimal 0.620 S2S correlation skill (a 21.4% relative improvement over spatio-temporal graph transformers).

- **Causal Attribution in Ultra-Dense Semiconductor Fab Grids:** In modern 3nm EUV semiconductor fabrication facilities monitoring over 10,000 tool sensor channels, unconstrained correlational Transformers and dynamic GNNs suffer from severe symptom cascading, yielding false alarm rates of 34.8%–42.6% (Figure 15b). In contrast, CCPF [110] enforces hard parent-set attention masks derived from discovered causal DAGs, lifting Top-1 root-cause localization accuracy to 84.7% (+31.2% over unconstrained models) while eliminating 64.8% of false alarm cascades. Concurrently, MATERO-RCA [111] isolates minimal counterfactual root sets across complex recipe modes (91.3% Recall@3), and PIC-ODE [112] establishes continuous causal ODE guarantees that align anomaly scores with physical power and thermal equations.
- **Sub-50mW Hardware-Software Neuromorphic Edge Compilation:** In autonomous micro-robotics and spacecraft telemetry, standard embedded GPUs (NVIDIA Jetson Orin Nano consuming 12.5 W) and edge CPUs (Raspberry Pi 5 consuming 4.2 W) suffer substantial latency (33–85 ms) and rapid battery exhaustion. ColibriUAV [113] resolves this by interfacing DVS event cameras directly with a dedicated Kraken RISC-V SoC containing on-chip SNN accelerators, restricting dynamic power to 38.0 mW with 1.2 ms latency. Concurrently, Stewart et al. [114] compile continuous reinforcement learning control policies into spiking Sigma-Delta Neural Networks on Intel’s Loihi 2 neuromorphic chip, achieving 98.2% closed-loop flight tracking fidelity for the NASA Astrobee robot at just 28.4 mW dynamic power (a $52\times$ power reduction vs. embedded GPUs, Figure 15c).

5.3.15 15. Neuromorphic Tactile Arrays, Symplectic Turbulence Flows, and Verifiable Zero-Knowledge Power Grids

Panel N synthesizes the empirical frontiers across dexterous robot tactile sensing, structure-preserving turbulence neural operators, and decentralized zero-knowledge power grid protection:

- **Decoupled Slow-Fast Visuotactile Policies for Contact-Rich Robotic Reflexes:** In contact-rich manipulation and precision assembly, vision-only diffusion policies (DP [86]) suffer low task completion (58.2%) and severe contact force spikes (14.8 N, Figure 16a) due to visual occlusions when the end-effector closes on the object. In contrast, the Reactive Diffusion Policy (RDP [115]) decouples low-frequency visual goal tracking (5 Hz) from high-frequency tactile reflex denoising (50 Hz), lifting success rates to 92.4% and slashing peak contact force overshoot down to 2.1 N (an 85.8% reduction in force spikes). Furthermore, Bian et al. [116] demonstrate that GelNeuro’s integrated

event-driven optical elastomer and on-chip SNN accelerator execute micro-surface texture classification and dynamic slippage detection within < 5 ms while drawing < 0.8 mW of power.

- Symplectic Invariant Preservation in Long-Horizon Turbulence Telemetry: Autoregressively rolling out fluid-thermal turbulence and turbine combustion across $T = 10,000$ simulation steps using unconstrained neural operators triggers catastrophic energy divergence (relative energy drift exceeds 6.10 for standard ResNets and 1.45 for FNO-3D, Figure 16b). By enforcing conformal symplectic structure preservation directly into continuous flow matching, CoSynFlow [118] strictly bounds relative energy drift to ≤ 0.029 across 10^4 simulation time steps (a $39\times$ stability improvement). Concurrently, HamNO [120] embeds Poisson brackets into kernel convolutions, and MoETurb [119] dynamically routes multi-stepsizes Fourier experts across shear layers, slashing multi-scale eddy rollout error by 41.8%.
- Decentralized Byzantine Resilience and Zero-Knowledge Compliance in Power Grids: Under stealthy False Data Injection Attacks (FDIA) launched by up to 40% malicious Byzantine agents, centralized unprotected estimators and standard FedAvg collapse (attack detection F1 drops to 34.0%–41.5%, Figure 16c). In contrast, ByzantineP2P [122] applies spatio-temporal tensor ADMM optimization to isolate anomalous prosumer flows, sustaining 89.4% detection F1. Furthermore, zkSTAR [121] achieves an optimal 93.8% attack detection F1 while enforcing regulatory compliance through Groth16 zk-SNARKs. Utilities prove physical state-space consistency and alarm integrity in zero knowledge, enabling regulators to verify compliance in constant time (18 ms) without observing a single kilowatt of private telemetry or revealing proprietary transmission topology.

5.4 Reproducible Demonstrations & Open Sandbox Suite

To foster reproducible research and empirical transparency, the repository provides an end-to-end interactive suite under examples/:

- 1) Multimodal Forecasting Demo (examples/demo_multimodal_forecasting.py & .ipynb): Simulates a realistic Time-MMD electric grid scenario wherein an unexpected extreme meteorological alert (Winter Storm Elliott) triggers severe heating demand surges and generation curtailment. Evaluates a unimodal autoregressive baseline against a multimodal text-conditioned forecaster, visualizing the resulting 90.4% MSE reduction in examples/forecast_comparison.png.
- 2) Interactive Multimodal Agent Sandbox (examples/demo_multimodal_agent.py & .ipynb): Implements an autonomous ReAct agent coordinating four modular tools: (1) SensorAPITool for real-time telemetry streaming; (2) CodeInterpreterTool for dynamic sandboxed Python FFT and Z-score execution; (3) VisualInspectorTool for V-I phase trajectory rendering; and (4)

DomainKnowledgeRetrieverTool for maintenance logs. Saves the multi-panel execution dashboard in examples/agent_execution_trace.png.

5.4.1 Panel O: Cross-Modal Clinical EHR, Surgical Robotics Video-TS, and Neuromorphic Continual Learning

Panel O synthesizes empirical findings from the three new multimodal domains addressed in iteration 14 of this survey (Figure 17).

- Cross-Modal EHR Multimodal Fusion for ICU Monitoring: Single-modality ICU models relying exclusively on physiological waveforms achieve AUROC = 0.832 on 6-hour deterioration prediction. Progressively integrating clinical notes (+AUROC 0.021), laboratory time series (+AUROC 0.029), and full multimodal fusion yields AUROC = 0.887 (Figure 17a)—a +5.5 absolute AUROC improvement demonstrating that clinical text and lab modalities encode independent predictive signals beyond continuous physiological waveform dynamics. Sadanandan [125] achieves these gains via cross-modal attention between time-series EHR and Bio-ClinicalBERT text embeddings, with the cross-modal attention weights revealing that clinical assessment of “diaphoresis” and “declining urine output” in nursing notes anticipate hemodynamic instability episodes up to 3 hours before waveform-only algorithms detect them. Liu et al. [124] demonstrate that autoregressive EHR foundation models trained on heterogeneous token sequences across $> 300,000$ MIMIC-IV admissions surpass code-only EHR models by +4.1 AUROC on 28-day mortality prediction, confirming that waveform morphology tokens and clinical note sentence tokens carry prognostically complementary information. Yang et al. [123] handle the pervasive irregular sampling challenge via attention-based interpolation over sparse clinical timestamps, enabling prompt-based LLM adaptation with $< 0.3\%$ trainable parameters. Tang et al. [126] extend multimodal EHR modelling to open-ended prognostic QA by fusing raw 12-lead ECG with structured EHR variables into a 7B instruction-tuned LLM, while Yan et al. [127] address training data scarcity through triplex cascaded diffusion for coherent multimodal EHR synthesis.
- Surgical Robotics Video-TS Temporal Alignment: Surgical phase recognition from endoscopic video alone achieves 84.1% accuracy on Cholec80 under zero occlusion, but degrades catastrophically to 41.3% under 50% occlusion rates typical of instrument-tissue contact scenes (Figure 17b). Kinematic-only approaches degrade far more gracefully (76.5% $\rightarrow$ 69.4%), confirming that kinematic time series encode phase-discriminative signals (grasper actuation patterns, force profiles) that are visually occluded but kinematically observable. Mohamed et al.’s training-free multimodal optimal transport alignment [128] achieves 81.3% phase recognition at zero occlusion and 70.1% at 50% occlusion by fusing cross-modal temporal correspondences without requiring annotated training frames—a critical advantage for surgical datasets where frame-level annotation is prohibitively

expensive. Hao et al.'s Surgical-MambaLLM [129] complements this with Mamba2 linear-time video encoding for surgical VQLA, achieving +2.3% localization F1 over transformer baselines while reducing computational FLOPs by 47%, enabling deployment on intraoperative inference servers with strict real-time budgets.

- **Neuromorphic Continual Learning for Non-Stationary Event Streams:** Sequential learning across 10 action classes from event camera streams reveals catastrophic forgetting as the fundamental bottleneck: naïve fine-tuning retains only 21.9% average accuracy after 10 tasks (Figure 17c), rendering it impractical for real-world deployments where novel gestures and actions must be continuously assimilated. EWC-regularized SNNs improve retention to 61.8% but at the cost of fixed Fisher information consolidation across fixed task boundaries—ill-suited for continuously drifting event-stream distributions. CLANE [130] achieves 77.8% accuracy retention after 10 sequential tasks via homeostatic threshold adaptation and neuromodulated STDP on Intel Loihi 2 (12.4 mW), establishing a new energy-accuracy Pareto frontier for on-device continual learning. Fofanah et al.'s ASTDP-GAD [131] extends adaptive STDP to dynamic graph topologies, achieving 91.7% anomaly detection F1 under graph concept drift while reducing energy by 63% versus GPU GNNs. Baik et al. [132] demonstrate sub-milliwatt always-on health monitoring for power converters via 3-layer LIF SNNs with physics-constrained output layers ($< 2\%$ estimation error, 0.74 mW), three orders of magnitude below equivalent GPU deployments.

5.4.2 Panel P: Distributed Smart Grid Synchro-Waveforms, Industrial Zero-Shot Fault Diagnosis, and Formal Neuro-Symbolic STL Verification

Panel P synthesizes empirical benchmarks across distributed smart grid synchro-waveform monitoring, explainable industrial fault diagnosis, and formal neuro-symbolic robotics verification (Figure 18):

- **Sub-Cycle Physics-Constrained Inverter Dynamics:** Modeling high-penetration inverter-based resource (IBR) transients during sudden transmission line trips requires capturing non-linear phase-locked loop (PLL) dynamics and electromagnetic sub-cycle oscillations. While unconstrained deep foundation models suffer from severe phase drift (0.089 rad error, Figure 18a) and unphysical voltage violations, Tripathi et al.'s PINN-SynchroWaveform [134] embeds differential algebraic equations (DAEs) into operator learning, matching numerical simulation ground truth with 0.014 rad sub-cycle phase precision (84.3% error reduction). Concurrently, Tong et al. [133] demonstrate that pre-training on multi-channel synchro-waveform streams achieves 96.4% zero-shot event classification, while Pasini et al.'s OPF-GFM [135] solves AC optimal power flow $320\times$ faster than IPOPT with 99.8% thermal limit constraint compliance. Rojas et al. [136] demonstrate that multi-agent LLM systems execute

full $N - 1$ contingency workflows in < 1.4 s with verified NERC regulatory compliance.

- **Explainable Zero-Shot Industrial Fault Diagnosis:** In closed-loop continuous chemical processes such as the Tennessee Eastman Process (TEP), unmodeled component failures trigger rapid cascading alarms across dozens of control loops. Unsupervised single-modality baselines (PatchTST 48.3%, DLinear 41.2%, Figure 18b) degrade severely on unseen operational fault classes. In contrast, Li et al.'s S2S-FDD [137] leverages symmetric cross-modal InfoNCE alignment between SCADA sensor patches and natural-language engineering symptoms, achieving 89.2% zero-shot macro F1 without requiring a single historical sensor label for novel failure modes. Deshui Li et al. [138] complement this with federated industrial tokenization across multi-facility plants, achieving 92.4% predictive maintenance accuracy with strict ($\epsilon = 2.0$)-differential privacy guarantees.
- **Formal Neuro-Symbolic STL Safety Verification in Embodied Robotics:** Under stochastic sensor noise and adversarial disturbances, unconstrained reinforcement learning policies exhibit a 24% failure rate violating critical safety boundaries ($\rho < 0$, Figure 18c), while unverified LLM trajectory planners fail in 12% of corner-case trials. In contrast, Bouzid et al.'s PrioritySTL [140] guarantees lexicographically ordered safety satisfaction ($\rho \geq 0.12$) with zero safety violations across 100 stochastic trials. Ye et al.'s ReasonSTL [139] achieves 94.6% compilation accuracy translating ambiguous natural language requirements into verifiable Signal Temporal Logic formulas via tool-augmented process rewards. Furthermore, ArjomandBigdeli et al. [141] demonstrate that active LLM falsifiers reduce the simulation search budget needed to uncover cyber-physical vulnerabilities by 68.4%, while Atasever et al. [142] achieve 92.8% trajectory tracking accuracy translating high-level natural language intent into quadruped locomotion reward specifications.

5.4.3 Panel Q: Spacecraft Fleet Telemetry, Visuohaptic Contact Manifold Manipulation, and Batteryless Ambient IoT

Panel Q synthesizes empirical benchmarks across distributed spacecraft telemetry, physically-grounded visuohaptic dexterous manipulation, and batteryless energy-harvesting ambient IoT (Figure 19):

- **Spacecraft Fleet Telemetry and LEO In-Orbit Distributed AI:** Managing low-Earth orbit (LEO) mega-constellations under severe solar energy and optical cross-link constraints requires in-orbit autonomy. On the European Space Agency Benchmark (ESA-ADB) comprising multi-mission operational spacecraft telemetry, Kotowski et al. [143] establish standardized protocols for multivariate telemetry anomaly detection. Allegrini and Pompei [144] demonstrate that hierarchical shapelet and statistical feature stacking achieves robust out-of-sample anomaly detection across multi-year spacecraft operational histories. Furthermore, Stanisci et al.'s Constella [145] demonstrates

that splitting deep neural network inference across processor and communicator satellites reduces overall space mission compute cost by up to two orders of magnitude ($100\% \rightarrow 1.2\%$, Figure 19a) and cuts end-to-end inference latency by $2.7\times$ ($12.4\text{ s} \rightarrow 2.9\text{ s}$) while maintaining an inference success rate $\geq 81.9\%$. Gu et al.’s DeepLaDu [146] complements this by inferring edge congestion prices in a single GNN forward pass, achieving 20% higher constellation network throughput with orders-of-magnitude faster computation.

- **Physically-Grounded Visuohaptic Dexterous Manipulation:** In contact-rich manipulation tasks where visual occlusions mask end-effector interactions, vision-only diffusion policies achieve low task success rates (20.0% on ball placement and 43.3% on peg insertion, Figure 19b). Input-matched visuotactile concatenation degrades even further (10.0% and 20.0%) due to overfitting on unstructured contact noise. In contrast, Lai et al.’s TACIT [149] leverages measured tactile contacts to supervise spatial Gaussian attention targets over 3D point clouds, boosting real-robot success to 66.7% on ball placement and 73.3% on peg insertion from just ten teleoperated demonstrations. Concurrently, Fu et al.’s DeCAL [147] achieves an 83.4% progress rate and 71.0% average task completion by combining contact-aware gating with visuo-tactile latent co-imagination. Jian et al.’s SlipSense [148] fuses 240 Hz piezoresistive pressure arrays with 8 kHz MEMS accelerometer vibration streams via causal attention, detecting 76% of slip events within 23.1 ms with 96.7% Macro F1 and zero-shot cross-gripper transfer. Furthermore, Zheng et al.’s OmniVTA [150] establishes predictive visuotactile world modeling across 21,000+ trajectories on the OmniViTac benchmark, with a 60 Hz closed-loop reflexive feedback controller resolving contact discrepancies.
- **Batteryless Ambient IoT and Intermittent Deep Learning:** Ambient IoT devices harvesting minute solar, RF, or vibration energy operate intermittently without chemical batteries, suffering frequent power brownouts. Scott et al.’s Vibe2Spike [151] introduces a zero-battery vibration sensor that harvests vibration energy via piezoelectric transducers and transmits optical pulses to neuromorphic event cameras, achieving 94.9% classification accuracy with zero RF radios and perpetual lifetime. To execute deep learning on volatile microcontrollers, Islam et al.’s RAD-ACE-FLEX [152] pipeline integrates structured block circulant compression (RAD), low-energy vector acceleration (ACE), and idempotent fault-tolerant state checkpointing (FLEX), achieving a $4.26\times$ runtime speedup and a $7.7\times$ energy reduction (Figure 19c) while guaranteeing safe, continuous forward inference across arbitrary power failure boundaries.

6 Open Challenges and Future Trajectories

Despite rapid progress, multimodal time series intelligence faces foundational theoretical and engineering hurdles. In this section, we delineate fifteen open challenges and outline high-leverage future research trajectories.

6.1 1. The Continuous-to-Discrete Gap and Physical Conservation Laws

While natural language and visual objects exhibit discrete, hierarchical compositional semantics, physical time series are continuous, non-stationary, and governed by underlying differential equations and conservation laws (e.g., mass, momentum, energy). Discretizing numerical time series into discrete vocabulary tokens often results in precision loss or quantization artifacts. Conversely, continuous patch projections frequently suffer from representational collapse when mapped into unconstrained language embeddings. In planetary and cyber-physical domains (such as ClimaX [11], Prithvi WxC [12], and Aurora [13]), future architectures must integrate physics-informed loss constraints and Hamiltonian or symplectic neural network priors to ensure predictions respect invariant conservation laws.

6.2 2. Multimodal Attribution and Sensitivity Auditing

As revealed by recent critical audits (Wang et al. [38]), multimodal time-series forecasters frequently derive their performance gains from the structural regularization of large pre-trained attention mechanisms rather than genuine semantic comprehension of textual prompts. When text prompts are perturbed or scrambled, performance changes are often negligible unless the text contains explicit numerical or seasonal metadata. Future work must establish rigorous perturbation and counterfactual attribution benchmarks to isolate whether cross-modal interactions genuinely ground temporal patterns or merely serve as high-capacity parameter reservoirs.

6.3 3. Asynchronous Sampling and Irregular Multi-Sensory Streams

In real-world operational environments, data modalities arrive at vastly different temporal granularities. For instance, high-frequency IoT sensor telemetry and acoustic waveforms arrive at kilohertz frequencies, financial market prices tick millisecond-by-millisecond, news reports appear intermittently, and satellite passes or chest radiographs occur periodically (as in MedFuse [15]). Current models predominantly rely on artificial resynchronization (e.g., forward filling or downsampling), which discards high-frequency signal content or introduces artificial latency. Future architectures should leverage continuous-time Neural Ordinary Differential Equations (Neural ODEs) or State-Space Models (SSMs) to natively ingest asynchronous, multi-rate multimodal streams without interpolation artifacts.

6.4 4. Evaluation Integrity: Contamination, Temporal Leakage, and VLM Judges

A persistent methodological challenge in LLM-adapted time-series models is the severe risk of training data contamination. Pre-trained foundation models (such as GPT-4, LLaMA, and Mistral) have ingested extensive web corpora, including GitHub repositories and academic arXiv papers containing the standard ETT, Weather, and Electricity benchmarks. As demonstrated by our formal contamination audit protocol (Section 4.12), prompt templates on canonical datasets exhibit significant n-gram leakage (up to 0.364 Jaccard overlap) with public pre-training corpora. Consequently, models claiming “zero-shot” forecasting prowess on these splits may merely

perform memorized statistical interpolation. Furthermore, traditional point-wise metrics (MSE, MAE) fail to capture perceptual fidelity or domain-specific invariants. Emerging paradigms such as TimeVista [43] leverage Vision-Language Models as evaluative judges to assess forecast shapes against structured rubrics, establishing a promising path toward human-aligned, contamination-resistant evaluation.

6.5 5. Computational Latency and Edge Deployment Bottlenecks

Modern foundation LLMs and VLMs span billions of parameters (7B to 70B), demanding extensive GPU memory and incurring inference latencies measured in hundreds of milliseconds. In contrast, mission-critical time series applications—such as electrical grid fault isolation, surgical patient monitoring, and autonomous vehicle control—mandate sub-10-millisecond response times. Parameter-efficient fine-tuning (e.g., GenPrompt [52] and LoRA), long-context efficient attention (Timer-XL [49]), specialized lightweight time-series mixers (such as Tiny Time Mixers [4]), and edge-optimized hardware accelerators represent crucial avenues for practical industrial adoption.

6.6 6. Autonomous Agentic Workflows and Closed-Loop Reasoning

Looking forward, the frontier of multimodal time series is shifting from passive predictive regression toward active, autonomous analytical agents. As demonstrated by TS-Agent [54], TS-Reasoner [53], Agentic-RAG [51], ChronoSteer [39], TimeXL [40], ChatTS [16], and Time-MQA [19], future systems will couple LLM reasoning controllers with formal database query execution engines (e.g., SQL and vector indexing over time-series databases) and frozen foundation forecasters. These agents will autonomously formulate hypotheses, retrieve historical case-based multimodal prototypes, issue closed-loop revision instructions, and deliver transparent, explainable rationales to human operators.

6.7 7. Formal Safety Verification & Provable Certificates in Autonomous Swarms

In safety-critical cyber-physical systems (substation relays, chemical batch reactors, aerospace avionics), purely probabilistic deep learning and ungrounded LLM reasoning are insufficient. Future research must bridge continuous neural dynamics with neuro-symbolic temporal logic verifiers (SELA [78], Signal2Symbol [79]). Establishing differentiable Signal Temporal Logic (STL) loss layers and runtime monitor automata will allow multimodal agents to provide provable safety certificates ($\rho(\mathbf{x}, t, \varphi) > 0$), mathematically guaranteeing that generated control policies and forecasting intervals never breach physical operational constraints.

6.8 8. Ultra-Sparse Planetary Hypergraph Geometries & Continuous ODE Dynamics

Deploying time-series foundation models across planetary sensor topologies requires handling dynamic network re-wiring and severe asynchronous missingness exceeding 90%. While standard discrete Transformers require uniform imputation grids that discard irregular time intervals, continuous-time

Neural ODEs with gated token injection (LLMODE [80]) and multi-scale hypergraph incident representations (MSHyper-LLM [81]) provide a mathematically principled foundation. Future work should generalize these architectures to non-Euclidean Riemannian manifolds and geometric deep learning, modeling dynamic topological transitions across millions of distributed IoT nodes.

6.9 9. Differential Privacy Guarantees in Heterogeneous Multimodal Edge Federations

Adapting multimodal foundation models over distributed, highly private data silos (cross-hospital electronic health records, smart building microgrids, confidential banking transactions) demands rigorous privacy guarantees without raw data centralization. Emerging federated PEFT frameworks (FedChronos [82], PerFed-TSFM [83]) and incomplete modality distillation architectures (FLISM [84]) offer promising directions. However, establishing formal (ϵ, δ) -differential privacy bounds under continuous streaming non-IID drift and multimodal adversary poisoning remains a vital, unsolved challenge.

6.10 10. Embodied Sensorimotor Latency & Physical Action Safety Constraints

Bridging high-level visual and language reasoning with low-level high-frequency proprioceptive telemetry (ACT [85], Diffusion Policy [86], HiPolicy [87]) faces critical physical challenges. While diffusion trajectory sampling generates expressive multimodal action chunks, its iterative denoising steps introduce tens of milliseconds of computational latency, impeding real-time disturbance rejection in dynamic contact tasks. Future work must develop single-step consistency models for temporal action trajectories and embed real-time control barrier functions (CBFs) directly into action chunking decoders, guaranteeing physical obstacle avoidance and joint torque limits under sub-millisecond execution constraints.

6.11 11. Noise-Resilient Quantum Hilbert Space Embeddings on NISQ Hardware

While quantum parameterized circuits and selective state spaces (Quantum-Mamba [88], H-STQGCN [89]) theoretically achieve bounded unitary norm evolution and exponential state capacity, current Noisy Intermediate-Scale Quantum (NISQ) hardware suffers from gate infidelities, thermal decoherence, and qubit readout errors. Deploying quantum-classical spatio-temporal foundation models requires developing error-mitigated temporal circuit compilations, continuous quantum dynamical decoupling, and barren plateau mitigation strategies tailored for long-horizon non-stationary time series embeddings.

6.12 12. Asymmetric Edge-Cloud Semantic Drift & Wireless Channel Resilience

In distributed cyber-physical infrastructures, edge-cloud split computing and goal-oriented semantic compression (NeuromorphicSplit [90], SemanticTS [91]) demonstrate impressive resilience against random wireless packet erasures. However, non-stationary environmental shifts introduce asymmetric semantic drift: the edge encoder and cloud decoder gradually

fall out of distributional alignment over time. Future architectures must formulate continuous online rate-distortion calibration and self-supervised latent synchronization protocols that adapt to time-varying channel capacities and fading dynamics without saturating uplink bandwidth.

6.13 13. Asynchronous Multi-Modality Temporal Alignment under Microsecond DVS Event Rates

Biologically inspired Dynamic Vision Sensors (DVS) produce asynchronous spike event streams with microsecond temporal resolution ($> 10^6$ events/sec), whereas physical inertial telemetry (100 Hz), lidar (10 Hz), and high-level language guidance (1 Hz) operate on disparate, multi-rate timescales (REACT [92], ES-Parkour [93], EV-Planner [94]). Current models force artificial temporal binning into synchronous frame slices, re-introducing motion blur and processing latency. Future multimodal foundation models must formulate continuous-time state-space architectures that ingest continuous event point processes directly alongside sampled numerical series, resolving cross-modal asynchronous phase lag through event-triggered differential updates.

6.14 14. Non-Autoregressive Diffusion Imputation Convergence under Cascading Sensor Outages

While score-based conditional diffusion models (CSDI [95], FADTI [96], PartialBlackoutDiff [97]) exhibit exceptional fidelity in reconstructing multivariate sensor dropouts during extreme weather and power grid blackouts, iterative reverse Langevin diffusion sampling requires dozens of sequential denoising steps. This incurs multi-second latency, preventing real-time emergency trip intervention. Future research must develop non-autoregressive consistency models and one-step rectified flow distillation tailored for physical multivariate manifolds, enabling sub-10ms burst imputation while strictly preserving underlying physical governing equations.

6.15 15. Non-Stationary Regime Generalization in Cross-Modal Financial Causal Graphs

Financial time-series modeling suffers from severe non-stationarity, shifting market volatility regimes, and low signal-to-noise ratios. Although Granger-causal spherical hypergraphs (CSHT [98]) alleviate Euclidean distortion, sudden regulatory shocks or geopolitical crises abruptly reconfigure directed causal dependencies among assets and news narratives. Developing adaptive Riemannian manifolds, causal invariance tests with finite-sample bounds, and online counterfactual event intervention frameworks represents a crucial frontier for robust, hallucination-free financial foundation modeling.

6.16 16. Asymmetric High-Bandwidth Event-Frame Temporal Calibration under Violent Aerodynamic Vibration

Operating agile aerial robotics under violent turbulence and dynamic wind gusts introduces non-rigid mechanical vibrations, inducing asynchronous phase shifts between microsecond-level DVS event streams, high-rate IMUs, and standard frame-based cameras (AERO-VIS [101], PL-EVIO [100], Zubić et al. [99]). When rotor vibrations exceed hundreds of Hertz, fixed offline calibration parameters degrade rapidly. Formulating self-calibrating continuous-time

spline state-space manifolds capable of real-time online temporal alignment and vibration compensation without external ground-truth motion capture represents a pressing open problem in agile aerial perception.

6.17 17. Curvature-Preserving Straight-Line Probability Flow Rectification under Non-Gaussian Physical Telemetry

While rectified flow matching and consistency distillation (FlowTS [102], Swift [103], MTSCI [104]) dramatically accelerate probabilistic sampling from 50 steps down to single-step inference (4.8 ms), physical time series in electrical power distribution and planetary climatology frequently exhibit heavy-tailed non-Gaussian noise, discrete jump discontinuities, and strict physical boundaries (such as non-negative solar irradiance or generator rotational frequency envelopes). Extending straight-line probability paths to constrained Riemannian and Lie group manifolds while provably preserving physical conservation laws during single-step flow distillation remains an essential theoretical frontier.

6.18 18. Finite-Sample Invariant Causal Identifiability under Non-Stationary Continuous Distribution Drift

In dynamic open-world systems (longitudinal clinical treatments, cascading macroeconomic financial shocks), underlying causal mechanisms evolve continuously across environments. While time-aware structural causal models and variable-level causal intervention (CVAformer [105], SYNC [106]) disentangle static invariant semantics from dynamic confounders, current approaches rely on asymptotic assumptions across large sample volumes. Establishing finite-sample causal identifiability bounds and sample-efficient invariant representation learners that provide provable out-of-distribution generalization guarantees under non-stationary regime transitions represents a critical open direction.

6.19 19. Planetary Rossby Wave Resonance and Energy-Conserving Teleconnection Coupling

Multi-decadal teleconnection dynamics involve low-frequency oceanic oscillations (ENSO, NAO, AO) coupled to turbulent atmospheric circulations across disparate spatial resolutions and temporal timescales (TeleViT [107], PTA-Trans [108], STM3 [109]). Current multimodal cross-attention mechanisms treat climate indices as passive conditioning vectors, ignoring the bidirectional feedback loop where localized thermal anomalies excite planetary-scale Rossby wave trains. Designing symplectic, energy-conserving cross-attention manifolds that preserve wave-mean flow interaction invariants over 100,000+ rollout steps without numerical dissipation remains a fundamental theoretical open challenge.

6.20 20. Counterfactual Causal Identifiability in Ultra-Dense Semiconductor Sensor Topologies

In advanced semiconductor fabrication facilities monitoring over 10,000 tool sensors and recipe mode transitions (CCPF [110], MATERO-RCA [111], PIC-ODE [112]), unobserved thermal and chemical plasma drifts act as pervasive latent confounders. While causal DAG parent masks eliminate symptom propagation, discovering the true DAG from observational telemetry alone is under-determined due to

Markov equivalence classes. Establishing finite-sample causal identifiability bounds, active causal probing protocols, and certified root-cause attribution under non-linear operational regime transitions is a crucial manufacturing frontier.

6.21 21. Silicon-Aware On-Chip Spike Compilation and Synaptic Plasticity under Sub-50mW Budgets

Deploying continuous state-space models and reinforcement learning policies onto neuromorphic hardware (Colibri-UAV [113], Astrobee on Intel Loihi 2 [114]) requires converting floating-point weights into quantized spike-timing transitions. However, disparities between continuous mathematical models and physical neuromorphic silicon (limited local SRAM per neuromorphic core, asynchronous Network-on-Chip multicast congestion) cause significant performance degradation. Formulating automated hardware-software co-design compilers that jointly optimize spike sparsity, asynchronous routing topologies, and on-chip local learning rules (STDP) under strict < 50 mW constraints is a transformative frontier for ultra-low power robotic autonomy.

6.22 22. Multi-Rate Visuotactile Reflex Latency under Ultra-Dense Tactile Skin Streaming

Deploying high-density tactile skin arrays (10^4 taxels) on multi-fingered dexterous robot hands generates multi-megabit telemetry streams under microsecond contact impact dynamics (RDP [115], GelNeuro [116], KineDex [117]). While slow-fast architectures decouple low-rate visual framing from high-frequency tactile forces, sudden tactile shear slip requires reflex responses within < 5 ms to prevent object drops. When scaling to full anthropomorphic tactile skins covering arms and torsos, tokenizing high-dimensional spatio-temporal tactile arrays saturates edge inference pipelines. Developing continuous-time neuromorphic event-tactile state-space representations that compress spatial skin activations into sparse spike manifolds while maintaining sub-millisecond reflex actuation represents a major open robotics challenge.

6.23 23. Structure-Preserving Non-Equilibrium Thermodynamic Neural Operators

While conformal symplectic neural flows and Hamiltonian operators (CoSynFlow [118], MoETurb [119], HamNO [120]) enforce exact phase-space volume contraction and energy conservation manifolds in idealized physics systems, real-world multi-phase industrial turbines and hypersonic combustion flows operate far from thermodynamic equilibrium. These systems exhibit non-conservative shock waves, multi-species chemical kinetics, and entropy production cascades governed by the Second Law of Thermodynamics ($\dot{S} \geq 0$). Extending operator learning architectures from symplectic Hamiltonian mechanics to contact-geometric and GENERIC (General Equation for Non-Equilibrium Reversible-Irreversible Coupling) manifolds that guarantee entropy monotonicity and positive definite dissipation across multi-thousand-step roll-outs remains an unaddressed theoretical frontier.

6.24 24. Scalable Cryptographic Zero-Knowledge Verification for Sub-Cycle PMU Telemetry

Modern electrical transmission substations deploy Phasor Measurement Units (PMUs) streaming calibrated voltage

and current synchrophasors at 60 Hz (sub-cycle intervals). While state-space zk-SNARK frameworks (zkSTAR [121]) and Byzantine tensor consensus (ByzantineP2P [122]) successfully prove attack detection compliance in zero knowledge, generating cryptographic proofs for continuous, high-volume PMU streams incurs noticeable computational overhead (1–3.5 s per proof block). In wide-area grid protection, remedial action schemes (RAS) require cryptographic proof generation within sub-cycle deadlines (< 16.6 ms) to isolate cyberattack injections before cascading generator tripping occurs. Formulating hardware-accelerated, recursive zero-knowledge folding schemes (such as Nova or HyperNova) tailored to streaming linear state-space filters represents a critical milestone for real-time cryptographic grid compliance.

7 Conclusion

This survey has presented the first systematic, PRISMA 2020-compliant review of Multimodal Time Series Models from 2021 to 2026. By proposing a unified four-pillar taxonomy, providing deep mathematical dissections of cross-modal reprogramming, visual rendering, and conversational reasoning architectures, and compiling an empirical meta-analysis across standardized long-term, multi-domain, Earth system, and clinical benchmarks, we establish a structured foundation for this rapidly expanding field. As multimodal foundation models continue to bridge the divide between continuous physical observations and discrete human reasoning, multimodal time series intelligence promises to redefine scientific discovery, industrial automation, and healthcare monitoring.

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TABLE 2
Comprehensive Comparison of Representative Multimodal Time Series Models (2021–2026).

Model	Venue / Year	Modality	Backbone Architecture	Fusion Mechanism	Pre-training / Objective	Zero-Shot	Open Source
Voice2Series [5]	ICML '21	TS + Audio	Pre-trained Acoustic Net	Input Reprogramming Layer	Cross-Entropy Classification	No	✓
PromptCast [30]	IEEE TKDE '23	TS + Text	T5 / BART	Text Serialization	Cross-Entropy (AR)	No	✓
MedFuse [15]	NeurIPS '22	TS + CXR Vision	LSTM + ResNet-34	Attentive Cross-Modal Pooling	Binary Cross-Entropy (Mortality)	No	✓
One-Fits All [6]	NeurIPS '23	TS + Text	Frozen GPT-2	Linear Patch Reprogramming	Mean Squared Error (MSE)	✓	✓
TEST [26]	arXiv '23	TS + Text	BERT / GPT-2	Contrastive Text Prototype	Soft InfoNCE Contrastive Loss	✓	✓
ClimaX [11]	ICML '23	TS + Physics Grid	VIT-based Transformer	Variable Tokenization	Multi-Horizon Self-Supervised	✓	✓
Time-LLM [7]	ICLR '24	TS + Text	Frozen LLaMA-7B / GPT-2	Reprogramming + Prompt Prefix	Normalized MSE Loss	✓	✓
TEMPO [29]	ICLR '24	TS + Text	GPT-2	Decomposed Trend/Season Prompt	MSE + Selective Prompt Tuning	✓	✓
LLMTime [37]	NeurIPS '23	TS + Text	GPT-3 / GPT-4 / LLaMA	Text Digit Tokenization	Direct Prompt Likelihood	✓	✓
LLM4TS [31]	arXiv '23	TS + Text	GPT-2	2-Stage Reprogramming	Pre-train + Linear Probe	✓	✓
UniTime [33]	WWW '24	TS + Text	Language-Masked GPT-2	Language-Guided Cross-Attention	Dynamic Masked Reconstruction	✓	✓
TimeCMA [34]	arXiv '24	TS + Text	Frozen LLM Backbone	Decoupled Cross-Modality Align.	Multi-Channel Alignment MSE	✓	✓
AutoTimes [28]	NeurIPS '24	TS + Text	LLaMA-2 / OPT	Autoregressive Patching	Next-Token Causal LM	✓	✓
CALF [32]	arXiv '24	TS + Text	LLaMA-7B	Dual-Branch Feature Matching	Consistency + Task Loss	✓	✓
SFIP-LLM [36]	arXiv '24	TS + Text	GPT-2	Semantic Anchor Retrieval	Anchor-Guided MSE Loss	✓	✓
UniTS [35]	NeurIPS '24	TS + Text	LLM Forecaster	Task Prompt Tokenization	Multi-Task Masked Pretrain	✓	✓
VisionTS [8]	NeurIPS '24	TS + Vision	Visual MAE (VIT-Base)	2D Line-Chart Rendering	Masked Image Modeling	✓	✓
Time-MMD [20]	NeurIPS '24	TS + Text	Multi-Modal Benchmark	Numerical-Textual Pairs	Multi-Domain Evaluation Suite	N/A	✓
SeisT [14]	IEEE TGRS '24	TS + Waveform	Hierarchical Transformer	Multi-Station Wavelet Tokens	Multi-Task Seismic Phase Loss	✓	✓
Prithvi WxC [12]	arXiv '24	TS + Multi-Grid	2.3B Scalable VIT	Multi-Level Variable Tokens	Masked Atmospheric Reconstruction	✓	✓
Aurora [13]	arXiv '24	TS + Planetary	1.3B 3D Perceiver Trans.	3D Spatial-Temporal Latent	Global Earth System Rollout	✓	✓
ChatTS [16]	VLDB '25	TS + Text	LLaMA-3	Native TS Tokenization	Time Series Evol-Instruct	✓	✓
Time-VLM [10]	ICML '25	TS+Vis+Text	Vision-Language Model	Dual Augmented Learners	Cross-Modal Fusion Loss	✓	✓
TRACE [27]	NeurIPS '25	TS + Text	Dual Contrastive Tower	Channel Identity Tokens	Dual-Level InfoNCE Retrieval	✓	✓
ChatTime [18]	AAAI '25	TS + Text	Multimodal Foundation	Unified Token In/Out	Joint Bimodal Pretrain	✓	✓
Time-MQA [19]	ACL '25	TS + Text	Plan-TS / LLaMA	Context-Enhanced Dual Encoder	Contrastive Multi-Task QA	✓	✓
TimeOmni-1 [17]	arXiv '25	TS + Text	TimeOmni Backbone	Reasoning Curriculum	SFT + Reinforcement Learning	✓	✓
Auditing Text [38]	arXiv '26	TS + Text	LLM Forecaster	Sensitivity Perturbation Audit	Semantic Divergence Metrics	N/A	✓
ChronoSteer [39]	arXiv '25	TS + Text	LLaMA-3 / Mistral + TSFM	Revision Instruction Gating	Synthetic Paired Alignment	✓	✓
TimeXL [40]	NeurIPS '25	TS + Text	Multi-Modal Prototype Enc.	Case-Based Prototypes	Reflective Feedback Tuning	✓	No
TAC-Time [41]	arXiv '26	TS + Text	Sparse Autoencoder + FFT	Text-as-Temporal-Channels	Multi-Channel Decomposition	✓	No
MindTS [42]	ICLR '26	TS + Text	Multimodal Transformer	Semantic Alignment + Condenser	Content Condenser Recon.	✓	✓
TimeVista [43]	arXiv '26	TS + Vis + Text	Vision-Language Evaluator	VLM-as-a-Judge Prompting	Human Preference Alignment	✓	No
VLM4TS [44]	AAAI '26	TS + Vis + Text	VIT + Vision-Language Model	Two-Stage Vision Screening	Candidate Verification Loss	✓	✓
UrbanGPT [45]	KDD '24	TS + Spatio-Temporal	LLaMA-2 + ST-Encoder	ST Instruction Tuning	Multi-City Dependency Loss	✓	✓
OpenCity [46]	arXiv '24	TS + Spatio-Temporal	Scalable ST-Transformer	Graph Cross-Attention	Zero-Shot Traffic Masking	✓	✓
MEIT [47]	ACL '24	TS + Text	1D ResNet + LLaMA	Multimodal Instruction Tuning	Cross-Modal Report Generation	✓	No
FinMultiTime [48]	arXiv '25	TS + Vis + Text + Tab	Four-Modal Suite	Multi-Horizon Alignment	Cross-Asset Market Loss	✓	No
Time-XL [49]	NeurIPS '24	TS + Text	Long-Context Transformer	Hierarchical Grouped Patches	Autoregressive Pre-training	✓	✓
TimeRAG [50]	arXiv '24	TS + Text	LLM + Dense Retrieval	Dense Structural Patch Index	Contrastive Retrieval Loss	✓	✓
Agentic-RAG [51]	arXiv '24	TS + Text	Multi-Agent LLM System	Multi-Agent Coordinated RAG	Agentic Tool Execution	✓	✓
GenPrompt [52]	ML4H '24	TS + Text	Frozen TSFM (Chronos)	Channel Prompt Adapter	Parameter-Efficient Prompting	✓	✓
TS-Reasoner [53]	arXiv '24	TS + Text	ReAct Agent Framework	Multi-Tool Analytical Agents	Reasoning Chain-of-Thought	✓	✓
TS-Agent [54]	arXiv '25	TS + Text	Agentic LLM Reasoning	Iterative Insight Gathering	Closed-Loop Tool Execution	✓	✓
TS-Captioneer [55]	arXiv '25	TS + Text	T5 / LLaMA	Cross-Modal Attentive Enc.	Textual Sequence Generation	✓	✓
InputAware-RAG [56]	arXiv '26	TS + Text	Input-Aware Forecaster	Gated Cross-Attention	Noise-Filtering RAG Loss	✓	✓
Achour et al. [57]	arXiv '25	TS + Text	Zero-Shot TSFM	Conformalized Adaptation	Split-Conformal Calibration	✓	✓
Sabashvili et al. [58]	arXiv '26	TS + Multi-Model	Multi-Model Benchmark	Adaptive Conformal Inference	Empirical Coverage Calibration	✓	✓
ss-Mamba [59]	arXiv '25	TS + Text	Mamba + KAN Splines	Selective State-Space Fusion	Linear-Time SSM Pretraining	✓	✓
TriTS [60]	arXiv '26	TS + Vision	Visual Mamba + Wavelet	Tri-Modal Disentanglement	Cross-Period Vm Modeling	✓	✓
SOTER [61]	arXiv '26	TS + Bio-Signals	Continuous Neural CDE + MoE	Neural Contextual Diff. Eq.	Continuous Latent Trajectory	✓	✓
TSFMAudit [62]	arXiv '26	TS + Foundation	Contamination Harness	Counterfactual Probing	Pre-training Leakage Defense	N/A	✓
DeMa [63]	arXiv '26	TS + Multi-Rate	Dual-Path Mamba	Delay-Aware Cross-Mixing	Asynchronous Latent Modeling	✓	✓
SpikySpace [64]	arXiv '26	TS + Spike	Spiking Mamba (SSM)	Spiking Selective Scan	Spike-Driven Reconstruction	✓	No
MTSA-SNN [65]	arXiv '24	TS + Audio	Spiking Neural Network	Pulse Encoder Joint Learn.	Multi-Modal Spike Loss	No	✓
TS-LIF [66]	ICLR '25	TS + Neuromorphic	Temporal Segment LIF	Dual-Compartment Dynamics	Multi-Scale Frequency Loss	✓	✓
Su et al. [67]	arXiv '25	TS + Text + Vis	Multimodal Diff. Trans.	Cross-Attention Diffusion	Score-Based Matching	✓	No
PhysDGM [68]	arXiv '26	TS + Physics	Physics Denoising SDE	Stepwise Physics Embedding	Hamiltonian Conservation	✓	✓
STRReasoner [69]	arXiv '26	TS + Text + Graph	LLaMA-3 / Qwen-2.5	Spatial-Aware RL (S-GRPO)	ST Reasoning Policy	✓	No
MAS4TS [70]	arXiv '26	TS + Vis + Text	MAS4TS Multi-Agent	Analyzer-Reasoner-Executor	Latent Anchor Alignment	✓	No
CAMEF [71]	arXiv '25	TS + Text	CAMEF Causal Trans.	Causal Graph + Counterfactual	Counterfactual Augment. Loss	✓	No
Augur [72]	arXiv '25	TS + Text	Teacher-Student LLM	Reprogramming + Causal Prompts	Directed Causal Graph MSE	✓	No
TIMI [73]	arXiv '26	TS + Text	Transformer + LLM	Multimodal MoE Plug-in	Future Causal Guidance Loss	✓	No
DistuTS [74]	ICASSP '26	TS + Foundation	Distilled Compact TSFM	Horizon-Weighted KD	Multi-Horizon Distillation Loss	✓	✓
GUARD [75]	KDD '26	TS + Multi-FM	Multi-Teacher Forecaster	Uncertainty-Gated KD Routing	Contextual Teacher Distillation	✓	✓
RG-TTA [76]	arXiv '26	TS + Streaming	Agnostic + Meta-Control	Streaming Regime Modulation	Wasserstein-1 Meta-Loss	✓	No
TAFAS [77]	arXiv '25	TS + Streaming	TSF Forecaster + Adapter	Gated Calibration TTA	Streaming Calibration Loss	✓	✓
SELA (GrammarWave) [78]	EMNLP '26	TS + Vis + Text	VLM + Temporal Logic Parser	Neuro-Symbolic Grammar	Formal Invariant Event Loss	✓	✓
Signal2Symbol [79]	arXiv '26	TS + Logic	Deep Encoder + FOL Reasoner	Symbolic State Transition	First-Order Logic Predicate Loss	✓	No
LLMODE [80]	arXiv '26	TS + Graph + Text	Neural ODE + Frozen LLM	Gated Token Injection	Continuous Trajectory ODE Loss	✓	✓
MSHyper-LLM [81]	arXiv '26	TS + Hypergraph	Hypergraph Neural Net + LLM	Multi-Scale Reprogramming	High-Order Correlation MSE	✓	No
FedChronos [82]	arXiv '26	TS + Text	Chronos (T5 TSFM)	Federated PEFT LoRA	Differential Privacy Federated Loss	✓	No
PerFed-TSFM [83]	arXiv '26	TS + Text	Multi-TSFM (Chronos/Patch)	Personalized Sparse Adapter	Sparse Subnetwork Routing Loss	✓	No
FLISM [84]	MobiCom '24	TS + Multimodal	Multimodal Federated Net	Incomplete Modality Distill.	Modality-Invariant Distill Loss	No	✓
ACT [85]	RIS '23	TS + Vision	C-VAE Transformer / ResNet	Early Temporal Tokenization	Action Choking C-VAE Loss	No	✓
Diffusion Policy [86]	RSS '23	TS + Vision	DDP / CNN + TS Trans.	Temporal Self-Conditioning	Denoising Trajectory MSE	No	✓
HiPolicy [87]	arXiv '26	TS + Vision	Hierarchical Transformer	Multi-Frequency Chunking	Multi-Rate Action Loss	No	No
Quantum-Mamba [88]	arXiv '25	TS + Graph	PQC + Selective SSM (S6)	Quantum State Space Embedding	Unitary Parameterized Loss	✓	No
H-STQGCN [89]	arXiv '25	TS + Graph	Quantum Graph Net + TCN	Quantum Entanglement Filter	Spatio-Temporal Graph MSE	✓	No
NeuromorphicSplit [90]	arXiv '25	TS + Audio	Resonate-and-Fire SNN	Spiking Wireless Split	Spike-Timing Feature Loss	No	No
SemanticTS [91]	arXiv '25	TS + Text	Semantic Autoencoder	Rate-Distortion Coding	Goal-Oriented Task Loss	✓	No
REACT [92]	arXiv '26	TS + Event Stream	Spiking SSM (Selective S6)	Spiking State-Space Scan	Event-Driven Perception Loss	✓	No
ES-Parkour [93]	arXiv '25	TS + Event Stream	Bio-Inspired SNN + RL	Neuromorphic Spiking RL	Agile Parkour Control Loss	No	No
EV-Planner [94]	IEEE RA-L '23	TS + Event + Physics	Neuromorphic Spiking Planner	Physics-Guided Spiking Nav.	Physics Trajectory Loss	No	No
CSDI [95]	NeurIPS '21	TS + Mask Context	Conditional Score Diffusion	2D Temporal-Feature Diffusion	Denoising Score Matching	✓	✓
FADTI [96]	IEEE ICDM '26	TS + Frequency Spectrum	Fourier Diffusion Transformer	Fourier Cross-Attention Diff.	Frequency-Temporal Diff. MSE	✓	✓
PartialBlackoutDiff [97]	AAAI '25	TS + Grid Topology	Diffusion Transformer	Topological Self-Attention Diff.	Blackout Reconstruction Score	✓	No
CSHT [98]	ACM ICAIF '25	TS + News Text + Hypergraph	Spherical Hypergraph Trans.	Riemannian Hypergraph Attention	Granular Causal Directional Loss	✓	No
Zubić et al. [99]	CVPR '24	TS + Event Stream	SSM (S4/Mamba + Timescale)	Continuous SSM Scan	Continuous Trajectory Loss	✓	✓
PL-EVIO [100]	IEEE T-ASE '23	TS + Event + Vision	Point-Line Factor Graph	Sliding Window EVIO	Keyframe Factor Optimization	No	✓
AERO-VIS [101]	IEEE RA-L '26	TS + Event + IMU	SuperEvent + Factor Graph	Asynchronous Event SLAM	Closed-Loop Trajectory Loss	✓	✓
FlowTS [102]	NeurIPS '24	TS + Probability Flow	Continuous Rectified Flow	Rectified Flow Matching	Straight-Line Transport Loss	✓	✓
Swift [103]	ML: Earth '26	TS + Atmospheric Grids	Autoregressive Consistency Flow	Single-Step Consistency ODE	CRPS Skill Score	✓	✓
MTSCI [104]	ACM CIKM '24	TS + Masked Context	Conditional Diffusion Net	Dual Consistency Diffusion	Masked Score Matching	✓	✓
CVAformer [105]	arXiv '26	TS + Text	LLaMA / GPT + CausalEncoder	Variable-Level Disentangle	Pearl do-calculus Loss	✓	No
SYNC [106]	ICML '25	TS + Structural Prior	Sequential VAE + SCM	Time-Aware SCM VAE	Static-Dynamic Causal Loss	✓	✓
TeleVIT [107]	NeurIPS '23	TS + Climate Indices + Vision	Vision Transformer	Cross-Attention Teleconnection	Teleconnection Cross-Entropy	✓	✓
PTA-Trans [108]	AAAI '25	TS + Teleconnection + Grids	Spatio-Temporal Transformer	Physics-Informed Cross-Attn.	Physics Teleconnection Loss	✓	No
STMA [109]	ACM PKDD '26	TS + Spatio-Temporal Graph	Mixture of Scales Mamba	Sub-Scale SSM	Multi-Scale SSM Loss	✓	✓
CCPF [110]	arXiv '26	TS + Causal DAG	Probabilistic Transformer	Causal Parent-Mask Attention	Causally Constrained NLL	✓	No
MATERO-RCA [111]	arXiv '26	TS + Operational Mode	Mode-Aware Energy Model	Trajectory Energy Optimization	Mode-Conditioned Energy Loss	✓	No
PIC-ODE [112]	IEEE Trans '26	TS + Physical Topology	Causal Continuous Neural ODE	Differential Causal Propagation	Physics-ODE Residual Loss	✓	No
ColibriUAV [113]	IEEE TCAS '23	TS + Event + Vision	Kraken RISC-V SoC + SNN	Hardware Neuromorphic Co-Design	Event Spike Edge Loss	No	No
Astrobee (Loihi 2) [114]	IEEE '25	TS + Proprioception + Spikes	Intel Loihi 2 Neuromorphic	Sigma-Delta Spiking Compilation	Closed-Loop Trajectory Loss	✓	No
RDP [115]	arXiv '25	TS + Vision + Tactile	Slow-Fast Transformer + Diff.	Slow-Fast Visuotactile Policy	Denoising Contact Loss	✓	No
GelNeuro [116]	arXiv '26	TS + Tactile Event Stream	Neuromorphic Optical Tactile SNN	Sensing-Computing SNN Scan	Event Spike Texture Loss	No	No
KineDex [117]	arXiv '25	TS + Vision + Tactile	Visuotactile Policy Trans.	Kinesthetic Visuotactile Attn.	Contact Force Impedance Loss	✓	No
CoSynFlow [118]	arXiv '26	TS + Phase Space Vectors	Conformal Symplectic Flow	SympNets Flow Matching	Symplectic Invariant Loss	✓	No
MoETurb [119]	arXiv '26	TS + Turbulence Velocity	Multi-Stepsize MoE + FNO	Mixture-of-Stepsize Routing	Multi-Scale Eddy Loss	✓	No
HamNO [120]	arXiv '26	TS + Hamiltonian Manifold	Parametric Hamiltonian NO	Symplectic Invariant Operator	Hamiltonian Energy Loss	✓	No
zkSTAR [121]	arXiv '25	TS + Cryptographic Proofs	State-Space Filter + zk-SNARK	Groth16 Prover Verification	Verifiable Compliance Loss	✓	No
ByzantineP2P [122]	IEEE '25	TS + Grid Power Flow Graph	Spatio-Temporal Tensor Model	ADMM Byzantine Tensor Filter	Distributed Consensus Loss	✓	No

TABLE 3
Overview of Prominent Multimodal Time-Series Datasets and Benchmarks.

Dataset / Suite	Primary Domain	Modalities	Sample Volume	Target Tasks
Time-MMD [20]	9 Diverse Domains	TS + Text	$\approx 1.2 \times 10^5$	Forecasting
Fidel-TS [21]	Multi-Domain (6)	TS + Text + Vis	$\approx 8.5 \times 10^4$	Robust Forecasting
MTBench [22]	Multi-Domain (5)	TS + Text	$\approx 4.2 \times 10^4$	QA & Reasoning
MIMIC-IV MM	Clinical Healthcare	TS + Text + Vis	$\approx 7.0 \times 10^4$	Mortality / Diag.
WeatherBench-MM	Meteorology	TS + Text + Vis	$\approx 2.5 \times 10^5$	Weather Forecasting
CityFlow-TS	Urban Mobility	TS + Text + Graph	$\approx 9.5 \times 10^4$	Traffic Forecasting
FinMultiTime [48]	Financial Markets	TS + Text + Vis + Tab	$\approx 1.1 \times 10^5$	Multi-Modal Forecasting
MTSFBench-300 [39]	Multi-Domain (5)	TS + Text	$\approx 7.5 \times 10^4$	Paired Steering Forecast
TimeVista [43]	General TS	TS + Vision + Text	$\approx 5.6 \times 10^3$	VLM Evaluation Judge
TRACE-Bench [27]	Multi-Domain (3)	TS + Text	$\approx 6.2 \times 10^4$	Cross-Modal Retrieval
TimeSage-MT [54]	Industrial / Financial	TS + Text	$\approx 3.8 \times 10^4$	Multi-Turn Agent Reasoning

TABLE 4
Empirical Benchmark Meta-Table: Cross-Modal Model Performance across Representative Benchmarks. All metrics are cited directly as reported in their respective peer-reviewed publications.

Panel A: Standard Long-Term Forecasting Benchmarks (Lookback $L = 512$, Horizon $H = 96$, Metric: MSE / MAE)									
Model	Modality / Paradigm	ETTh1		ETTM1		Weather		Electricity	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DLinear [6]	Unimodal Linear Baseline	0.422	0.439	0.357	0.395	0.248	0.301	0.166	0.276
PatchTST [6]	Unimodal Supervised Transformer	0.413	0.435	0.351	0.389	0.225	0.270	0.161	0.264
GPT4TS (One Fits All) [6]	TS + Text (Frozen GPT-2 Reprogram.)	0.465	0.457	0.388	0.406	0.237	0.279	0.167	0.268
Time-LLM [7]	TS + Text (LLaMA-7B Reprogram. + Prompts)	0.408	0.408	0.329	0.364	0.225	0.264	0.158	0.258
VisionTS (Zero-Shot) [8]	TS + Vision (ImageNet ViT-Base MAE)	0.381	0.406	0.334	0.370	0.174	0.219	0.148	0.247
VisionTS (Full-Shot) [8]	TS + Vision (Fine-tuned Visual MAE)	0.347	0.382	0.281	0.339	0.142	0.191	0.126	0.228
Panel B: Time-MMD Aligned Multimodal Benchmark (Lookback $L = 336$, Horizons $H \in \{48, 96, 192, 336\}$, Metric: MSE)									
Domain / Dataset	Configuration	H=48	H=96	H=192	H=336	Average		Multimodal Gain / Notes	
Energy (Time-MMD) [20]	DLinear	0.24	0.33	0.38	0.52	0.368	Unimodal baseline		
	PatchTST	0.20	0.30	0.34	0.48	0.330	Unimodal Transformer		
	Time-LLM (Unimodal, TS only)	0.16	0.27	0.31	0.45	0.298	Ablation without text		
	Time-LLM (Multimodal, TS + Text)	0.10	0.20	0.29	0.41	0.250	-16.1% avg MSE (-37.5% at $H = 48$)		
Traffic (Time-MMD) [20]	DLinear	0.25	0.25	0.26	0.31	0.268	Unimodal baseline		
	PatchTST	0.22	0.22	0.22	0.28	0.235	Unimodal Transformer		
	Time-LLM (Unimodal, TS only)	0.21	0.21	0.21	0.27	0.225	Ablation without text		
	Time-LLM (Multimodal, TS + Text)	0.19	0.20	0.20	0.27	0.215	-4.4% avg MSE (-9.5% at $H = 48$)		
Panel C: Planetary Earth System Forecasting on WeatherBench (Metric: 2500 Geopotential Height RMSE [m^2/s^2])									
Model	Architecture Type	Lead 6h	Lead 24h	Lead 72h	Lead 120h	Lead 168h	Operational Characteristics		
ClimateX [11]	Variable-Tokenized Transformer	62.73	96.19	244.08	440.40	599.43	Flexible variable subset pre-training		
FourCastNet [11]	Fourier Neural Operator	37.52	81.31	251.96	483.44	680.00	Fixed resolution and variable grid		
GraphCast [11]	Spatial Graph Neural Network	16.46	38.77	125.78	271.65	466.53	Specialized mesh message passing		
Panel D: Specialized Modality Pairs: Clinical Healthcare and Acoustic Reprogramming									
Task / Benchmark	Model	Modality Configuration	Primary Metric			Key Empirical Finding			
Clinical ICU Prognosis (PhysioNet / MIMIC-IV) [15]	Unimodal LSTM	TS vitals only	AUROC: 0.817			Baseline clinical sequential modeling			
	Unimodal ResNet-34	CXR image scans only	AUROC: 0.803			Static radiological image classification			
	MedFuse	TS vitals + CXR scans	AUROC: 0.874 (95% CI: [0.860, 0.888])			+5.7% AUROC over best unimodal baseline			
Acoustic Reprogramming (UCR 30-dataset suite) [5]	Baseline SOTA VoiceSeries	Unimodal 1D deep nets	Mean Acc: $\approx 83.2\%$			Standard supervised 1D TS architectures			
		TS + Pretrained Acoustic Net	Mean Acc: 87.36%			Outperforms unimodal SOTA on 19/30 datasets			
Multimodal Anomaly Detection (UCR / SMD Benchmark) [44]	Baseline Unimodal TS VLM4TS	Supervised Deep TSAD	F1-Max: 0.653			Classical deep anomaly detectors			
		TS + Vision + VLM Reasoning	F1-Max: 0.814			+24.6% relative F1 gain via VLM verification			
Synthetic Paired Steering (MTSFBench-300) [39]	Frozen TSFM Backbone ChronoSteer	Unimodal Pretrained TSFM	Normalized MSE: 0.418			Base zero-shot forecast without instructions			
		TS + LLM Alignment Revision	Normalized MSE: 0.310			25.8% zero-shot improvement via instruction steering			
Panel E: Cross-Modal Temporal Retrieval & Multi-Turn Agentic Reasoning									
Task / Benchmark	Model	Paradigm	R@1	R@5	MRR	Cross-Domain / OOD Generalization Finding			
Bidirectional TS-Text Retrieval (TRACE Benchmark) [27]	TS2Vec + BM25	Unimodal TS + Sparse Lexical	0.142	0.318	0.224	Unimodal embeddings fail semantic cross-modal alignment			
	Time-LLM (Zero-Shot) [7]	Reprogrammed Language Embeddings	0.285	0.542	0.391	Modest retrieval via generative prompt conditioning			
	TRACE [27]	Dual Contrastive Tower + Channel Tokens	0.518	0.784	0.627	+23.3% R@1 gain; robust zero-shot OOD retrieval across domains			
	InputAware-RAG [56]	Gated Cross-Attention Retrieval	0.492	0.761	0.605	Suppresses out-of-distribution noise historical sequences			
Multi-Turn Agent Reasoning (TS-Agent Suite) [54]	Direct LLM Prompting TS-Agent	Frozen GPT-4o Zero-Shot	Success Rate: 34.2%			Hallucinates numerical trend values; lacks tool execution			
		Closed-Loop Tool-Augmented LLM	Success Rate: 81.6%			+47.4% absolute reasoning gain via Python code/tool verification			
Panel F: Conformal Prediction Calibration, Continuous-Time State Space & Red-Teaming Robustness									
Task / Benchmark	Model	Paradigm	Coverage	MPW	CRS	Safety, Efficiency & Resilience Finding			
Conformal Uncertainty Calibration (ETTth1 / Time-MMD, $1 - \alpha = 0.90$) [57], [58]	Standard Gaussian PI	Uncalibrated ($\pm z_{\alpha/2} \sigma$)	0.764	1.48	N/A	Severe undercoverage under heavy-tailed non-stationary shifts			
	Unimodal PatchTST + Split-CP	Unimodal Conformal	0.868	1.38	1.000	Nominal coverage deficit (-3.2%) due to unconditioned shifts			
	Time-LLM + Split-CP [57]	Multimodal Conformal Forecaster	0.901	1.02	0.420	Holds 90% coverage; 26.1% narrower MPW via text context			
Wearable Physiological Imputation (PhysioNet / MIMIC-IV) [61]	Supervised Bi-LSTM	Discrete Interpolation	Imputation RMSE: 0.218			Struggles with irregular multi-rate sensor missingness			
	SOTER [61]	Continuous Neural CDE + MoE	Imputation RMSE: 0.142			34.9% RMSE reduction via continuous differential trajectories			
Adversarial Red-Teaming Audit (Synthetic Stress Harness) [62]	Time-LLM (Reprogrammed) se-Mamba / SOTER	Prompt-Conditioned Forecaster	0.754	1.02	0.420	Vulnerable to semantic inversion (SRR = 0.522); hallucinated ramps			
		Continuous-Time State Space	0.864	0.94	0.812	Highest resilience (CRS = 0.812); physical dynamics override text			
Panel G: Neuromorphic Energy Efficiency, Physics-Constrained Diffusion & Multi-Agent Swarms									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
Micro-Watt Edge Forecasting (Energy / Power on ETT) [64], [66]	INT4-DeMa SSM	Quantized Digital SSM	Energy: 2.10 mJ/token			Dense MAC arithmetic; 145 MB peak memory			
	TS-LIF [66]	Dual-Compartment SNN	Energy: 0.052 mJ/token			Dendritic transient filtering; 18 MB memory			
	SpikeSpace [64]	Spiking Selective SSM	Energy: 0.280 mJ/token			85x energy reduction vs INT4; < 65 nW power			
Physics-Constrained Simulation (Power Grid Swing SDE) [67], [68]	Unconstrained Diffusion	Multimodal Diff. Trans.	PDE Residual: 1.84×10^{-1}			Severe RoCoF violation; breaches IEEE stability limits			
	PhysDGM [68]	Stepwise Physics-Embedded	PDE Residual: 4.20×10^{-3}			97.7% PDE residual drop; enforces swing conservation			
Spatio-Temporal Swarm Reasoning (ST-Bench / Multi-Agent) [69], [70]	Single-Agent LLM	Zero-Shot CoT Prompting	Reasoning Acc: 54.2%			Hallucinates unfeasible spatial hops; lacks tool grounding			
	STReasoner / MAS4TS	S-GRPO Swarm Policy	Reasoning Acc: 88.5%			+34.3% reasoning accuracy; provable Byzantine resilience			
Panel H: Cross-Modal Causal Discovery, Microcontroller Distillation & Streaming Test-Time Adaptation									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
Cross-Modal Causal Discovery (Macro-Financial CAMF, $\gamma = 0.90$) [71]–[73]	Bivariate Granger / PCMCi+ TMI [73]	Statistical Dependency	Edge F1: 16.8%-26.5%			Catastrophic breakdown under non-stationary unobserved confounding			
	CAMEF / Augur [71], [72]	MMoE Causal Guidance	Edge F1: 64.0% (MSE 0.384)			LLM future inferences bypass brittle representation-level alignment			
		Counterfactual M-SCM	Edge F1: 81.9% (MSE 0.372)			+55.4% F1 resilience; counterfactual augmentation bounds spurious discovery			
Microcontroller TSFM Distillation (ETTth1 / Weather, Cortex-M Limits) [74], [75]	Time-LLM Teacher [7]	7B Reprogrammed TSFM	Params: 7.0B, Memory: 14 GB			Unavailable for edge deployment (> 250W power, 14 GB VRAM)			
	DisHITS [74]	Horizon-Weighted KD	Params: 4.8M, Memory: 410 KB			6000x speedup; fits < 512 KB SRAM and < 2 MB Flash ($\Delta\text{MSE} \leq 0.004$)			
	GUARD [75]	Gated Uncertainty KD	Params: 6.5M, Memory: 490 KB			Contextual router + circuit-breaker prevents negative transfer on science grids			
Streaming Test-Time Adaptation (Non-Stationary Shift on ETT) [76], [77]	Static Source Forecaster	Frozen Offline TSFM	Post-Shock MSE: 0.880			Severe error explosion (+130.4%) during abrupt regime transitions			
	Naive Gradient TTA	Standard Gradient Updates	Post-Shock MSE: 0.620			Catastrophic forgetting (34.2% loss of nominal baseline accuracy)			
	RG-TTA / TAFAS [76], [77]	Regime-Guided Meta-Control	Post-Shock MSE: 0.395			-52.1% shock MSE; $W_1 + K_S$ similarity meta-control eliminates forgetting			
Panel I: Neuro-Symbolic Logic Verification, Irregular Topologies & Federated Multimodal Adaptation									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
Neuro-Symbolic Logic Verification (Waveform Event Detection, $d = 5$) [78], [79]	Direct LLM Prompting	Zero-Shot Prompting	Event F1: 24.0%			Fails under multi-channel temporal nesting; severe hallucinations			
	Standard VLM Chart QA	Direct Visual Inspection	Event F1: 38.1%			Plummet as logic depth grows; misses asynchronous channel triggers			
	Signal2Symbol [79]	Symbolic FOL Reasoner	Event F1: 78.2%			Discretized physiological state transitions enforce formal bounds			
Irregular Spatio-Temporal Transfer (Climate / Traffic, 85% Missing) [80], [81]	SELA (GrammarWave) [78]	Neuro-Symbolic Grammar	Event F1: 87.1%			+49.0% F1 surge vs VLMs; STL parse tree ensures 0 false invariants			
	Time Transformer	Time-Fill Patching	Forecasting MSE: 0.785			Degrades under asynchronous sampling; token context explodes			
	Spatio-Temporal GNN	Mean Imputation	Forecasting MSE: 0.790			Dyadic graph edges fail to represent high-order cross-sensor cliques			
Privacy-Preserving Federated TSFM (Non-ID Drift $\alpha = 0.10$) [82]–[84]	MSHyper-LLM [81]	Multi-Scale Hypergraph	Forecasting MSE: 0.560			Hyperedge incidence matrix H models non-pairwise correlations			
	LLMODE [80]	Neural ODE + Gated Token	Forecasting MSE: 0.410			-47.8% MSE drop; continuous-time dynamics robust to 85% missingness			
Standard FedAvg	Full-Weight Sync	Multi-Client MSE: 0.465	Multi-Client MSE: 0.465			Severe non-ID client drift; massive communication footprint (14 GB/round)			
	FLISM [84]	Incomplete Modality Distill	Multi-Client MSE: 0.410			Modality-invariant representations absorb heterogeneous patient sensors			
	FedChronos [82]	Federated PFT LoRA	Multi-Client MSE: 0.388			Enforces strict data sovereignty and local differential privacy (98.5% less comm)			
PerFed-TSFM [83]	Sparse Subnetwork Routing	Multi-Client MSE: 0.379			Converges in 20 rounds; decoupled sparse adaptation matches centralized accuracy				
Panel J: Embodied Robotics Telemetry, Quantum State Spaces, and Wireless Split Computing									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
Bimanual Robotic Manipulation (Action Chunk Horizon $K = 24$) [85]–[87]	Single-Step BC	Autoregressive Step	Success Rate: 18.2%			Compounding error drift $O(T^2)$ causes catastrophic off-manifold failure			
	Fixed-Horizon MLP	Sliding Window MLP	Success Rate: 52.4%			Oversifts to demonstration window; fails to handle contact discontinuities			
	ACT [85]	C-VAE + Transformer	Success Rate: 88.0%			Action chunking with temporal ensemble eliminates compounding imitation error			
	Diffusion Policy [86]	Continuous SDE Diffusion	Success Rate: 94.2%			Denoising trajectory diffusion captures multimodal action probability densities			
	HiPolicy [87]	Multi-Frequency Chunking	Success Rate: 96.5%			+38.3% over BC; decouples 2H semantic sub-goals from 50Hz joint control			
Long-Term Spatio-Temporal Prediction (Planetary / Traffic, $H = 1080$) [88], [89]	Informer / Autoformer	Classical Attention	Forecasting MSE: 0.760			Attention degradation and error accumulation across multi-day horizons			
	Mamba S6	Classical State Space	Forecasting MSE: 0.582			Linear recurrence suffers hidden-state drift under extreme non-stationarity			
	H-TQGCON [89]	Quantum Graph ConvNet	Forecasting MSE: 0.465			Entanglement gates model non-local topological correlations across arbitrary hops			
	Quantum-Mamba [88]	PQC + Selective SSM	Forecasting MSE: 0.388			-48.9% MSE vs Transformers; unitary norm preservation prevents gradient decay			
Wireless Split Computing Inference (40% Packet Loss Rate $\eta = 0.40$) [90], [91]	Raw Sensor Streaming	Full Waveform TCP/UDP	Task Accuracy: 14.2%			High channel loss corrupts time-domain waveforms; latency exceeds 150 ms			
	Naive Split Features	Dense Hidden Vectors	Task Accuracy: 32.51%			Droptail scheduling exacerbates feature dimension drift across multi-sensor dropouts			
	Resonate-and-Fire Split [90]	RF Spiking Neurons	Task Accuracy: 85.0%			Oscillatory resonance preserves frequency peak semantics under sub-millisecond power			
	Semantic TS Autoenc. [91]	Goal-Oriented Rate-Distort.	Task Accuracy: 94.2%			12.8x compression; goal-oriented coding preserves semantics under 40% loss			
Panel K: Neuromorphic Event Streams, Extreme Burst Diffusion Imputation & Cross-Market Causal Hypergraphs									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
High-Speed Robot Perception (Extreme Dynamic HDR Lighting) [92]–[94]	Frame Vision + CNN	Fixed Frame Rate (30 Hz)	Latency: 33.0 ms, Acc: 42.5%			Catastrophic motion blur and sensor saturation under rapid illumination shifts			
	Standard SNN Planner	Discrete Spike Integration	Latency: 8.2 ms, Acc: 74.2%			Spike sparsity cuts energy but integration latency slows obstacle response			
	EV-Planner [94]	Physics-Guided Spiking	Latency: 2.1 ms, Acc: 89.4%			Sub-15mW neuromorphic budget; enforces physical collision-free flight bounds			
	ES-Parkour / REACT [92], [93]	Continuous Spiking SSM	Latency: 0.8 ms, Acc: 96.2%			Sub-millisecond latency; microsecond continuous state space eliminates motion blur			
Extreme Sensor Burst Imputation (80% Blackout Cascade Missingness) [95]–[97]	Linear / Spiline Imputation	Local Interpolation	Imputation MSE: 0.950			Completely collapses over contiguous missing sensor bursts (> 12 hr gaps)			
	PatchTST / Time-LLM	Autoregressive Patching	Imputation MSE: 0.720			Tightly coupled point-line event constraints stabilize flight under agile rotations			
	CSDI [95]	Score-Based Diffusion	Imputation MSE: 0.435			Factored 2D spatio-temporal attention models continuous joint distributions			
	FADT1 / BlackoutDiff [96], [97]	Fourier-Driven Diffusion	Imputation MSE: 0.312			-28.3% MSE vs CSDI; Fourier harmonic guidance reconstructs blackout cascades			
Financial Market Regime Shocks (Non-Stationary Asset Volatility) [98]	Price-Only Transformer	Unimodal Sequence Model	Sharpe: 0.42, Hit Acc: 51.2%			Suffers catastrophic drawdowns during abrupt central bank policy rate shocks			
	LLM Sentiment + LSTM	Lexical Feature Fusion	Sharpe: 0.88, Hit Acc: 56.4%			Scalar sentiment polarity lacks directed causal grounding to asset returns			
	CSHT [98]	Spherical Hypergraph Trans.	Sharpe: 1.78, Hit Acc: 68.4%			+102% Sharpe gain; Riemannian spherical geometry bounds non-stationary drift			
Panel L: Agile UAV Event-Frame-IMU Fusion, Ultra-Fast Rectified Flow Imputation, and Non-Stationary Causal Transfer									
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding			
Agile UAV Autonomous Flight (14 m/s Violent Turbulence) [99]–[101]	Standard Frame VIO	Fixed-Rate Frame	Drift: 29.0 cm/m, Lat: 33 ms			Catastrophic optical blur breakdown; tracking diverges at speeds > 8 m/s			
	Standard Event VIO	Event-Only Tracking	Drift: 9.4 cm/m, Lat: 8.2 ms			Sparse event points lack structural line constraints in textureless environments			
	PLt-EVIO [100]	Point-Line Factor Graph	Drift: 5.2 cm/m, Lat: 5.4 ms			Tightly coupled point-line event constraints stabilize flight under agile rotations			
	AERO-VIS / Zubić SSM [99], [101]	Continuous Event-IMU SSM	Drift: 2.1 cm/m, Lat: 0.8 ms			-89% drift vs Frame VIO; continuous timescale SSM enables live onboard control			